MAMMA ADA CENTRE

In Collaboration with FACM ITALY

UG 4th Year

From Learning to Innovation From Campus to Career

GAIO Global AI Olympiad

Think | Explore | Create | A Better Tomorrow

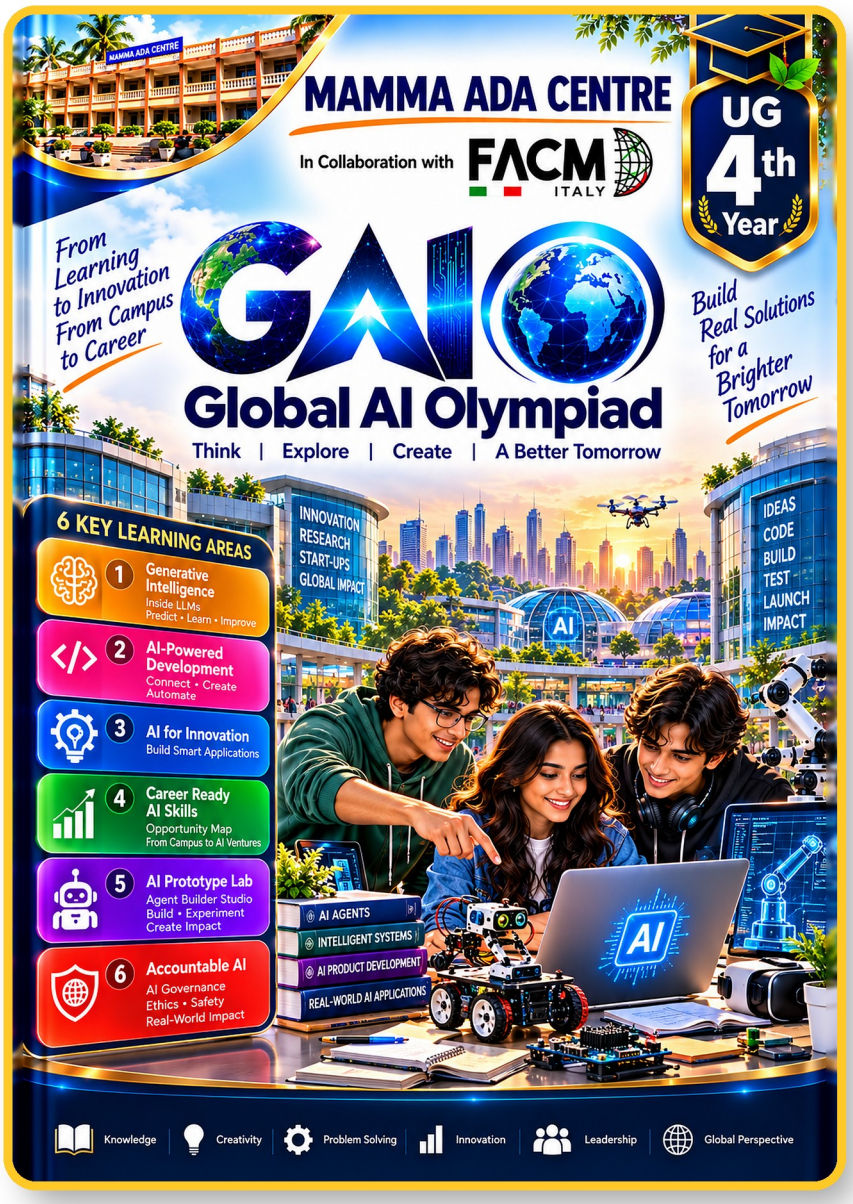
Build Real Solutions for a Brighter Tomorrow

6 KEY LEARNING AREAS

1. **Generative Intelligence**
   Inside LLMs
   Predict • Learn • Improve
2. **AI-Powered Development**
   Connect • Create
   Automate
3. **AI for Innovation**
   Build Smart Applications
4. **Career Ready AI Skills**
   Opportunity Map
   From Campus to AI Ventures
5. **AI Prototype Lab**
   Agent Builder Studio
   Build • Experiment
   Create Impact
6. **Accountable AI**
   AI Governance
   Ethics • Safety
   Real-World Impact

Knowledge | Creativity | Problem Solving | Innovation | Leadership | Global Perspective

This book belongs to

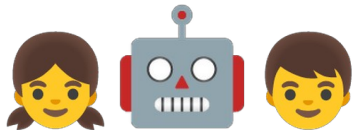


Name: _____

Class & Section: _____

School: _____

📖 PREFACE



AI is transforming every profession, from engineering and medicine to commerce, law and media. As a university student, you must understand how AI works, judge its outputs critically and apply it responsibly in your discipline. This book builds exactly those capabilities.

This book has been prepared for the Global AI Olympiad (GAIO) by Mamma Ada Centre, in collaboration with FACM, Italy. It follows the GAIO six-month journey: AI Discover, AI Connect, AI Solve, AI Rise, AI Create and AI Care. Each month introduces two topics suited to the curiosity and abilities of UG 4th Year students, with university-level depth.

Every topic opens with an in-depth "Concept Explained" chapter, followed by a Concept Deep Dive, an AI Story, a case study, a lab activity and a Python Code Lab, with worksheets, quizzes, challenge problems and university-style exam preparation.

Responsibility runs through every chapter: you will examine AI's limits, bias, errors, privacy risks and impact on work, and the ethical duties of everyone who builds or uses it.

🤝 A note for teachers and parents

Faculty can use the Concept Explained chapters for lectures, the Code Labs and lab activities for practical sessions, and the exam-prep pages for assessment. Encourage students to verify AI outputs, cite sources and reflect on ethics. An answer key is provided at the end of the book.

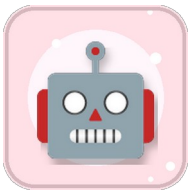
We hope this book helps you think critically, build confidently and lead responsibly in an AI-powered world. Your ideas today can change tomorrow!

With best wishes,

The GAIO Academic Team

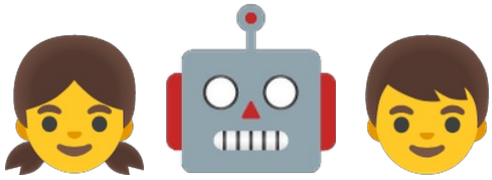
Mamma Ada Centre • In collaboration with FACM, Italy

Welcome, Innovators!



MEET BOLT!

Welcome to university-level AI! I am Bolt, your AI guide. This year you will write real Python, use AI as a coding partner, study AI in your profession and practise exam-style answers. Let's build your AI edge!



Look for these pictures in your book:

CONCEPT EXPLAINED Concept explained in depth

CONCEPT DEEP DIVE Detailed explanation & code

SCENARIO Real-life scenario

CASE STUDY Analyse a real case

LAB ACTIVITY Hands-on lab

WORKSHEET Practice exercises

PUZZLE & DEFINE Puzzles & definitions

QUIZ TIME MCQ practice

AI STORY An AI story to think about

CHALLENGE PROBLEMS Higher-order thinking

CODE LAB Python code lab

EXAM PREP University-style exam prep

My 6-Month AI Journey

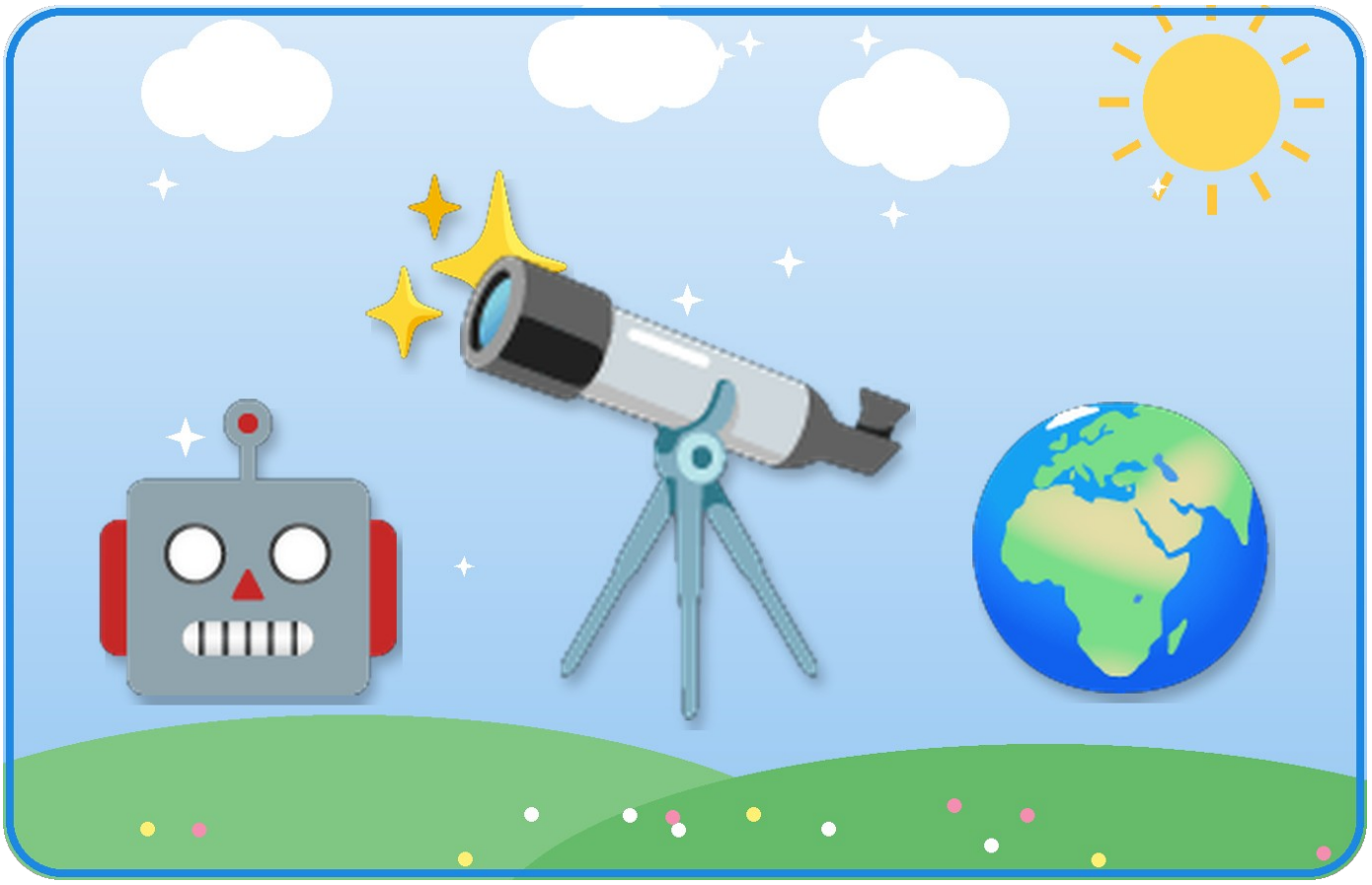
Follow the path. One month at a time!

MONTH 1 **AI DISCOVER** AI Basics	The Age of AI Agents • Intelligence at Scale
MONTH 2 **AI CONNECT** Communicate with AI	Build Smart Applications • Design AI Workflows
MONTH 3 **AI SOLVE** AI Applications	AI for Enterprise • AI for Start-ups
MONTH 4 **AI RISE** AI & Career	From Campus to AI Career • Idea to AI Venture
MONTH 5 **AI CREATE** AI Tools	AI Prototype Lab • Agent Builder Studio
MONTH 6 **AI CARE** Responsible AI	Accountable AI • AI Governance Essentials

MONTH 1

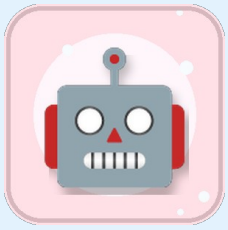
AI DISCOVER

See • Understand • Explore



Area: AI Basics

In this unit, you will explore:



Topic 1

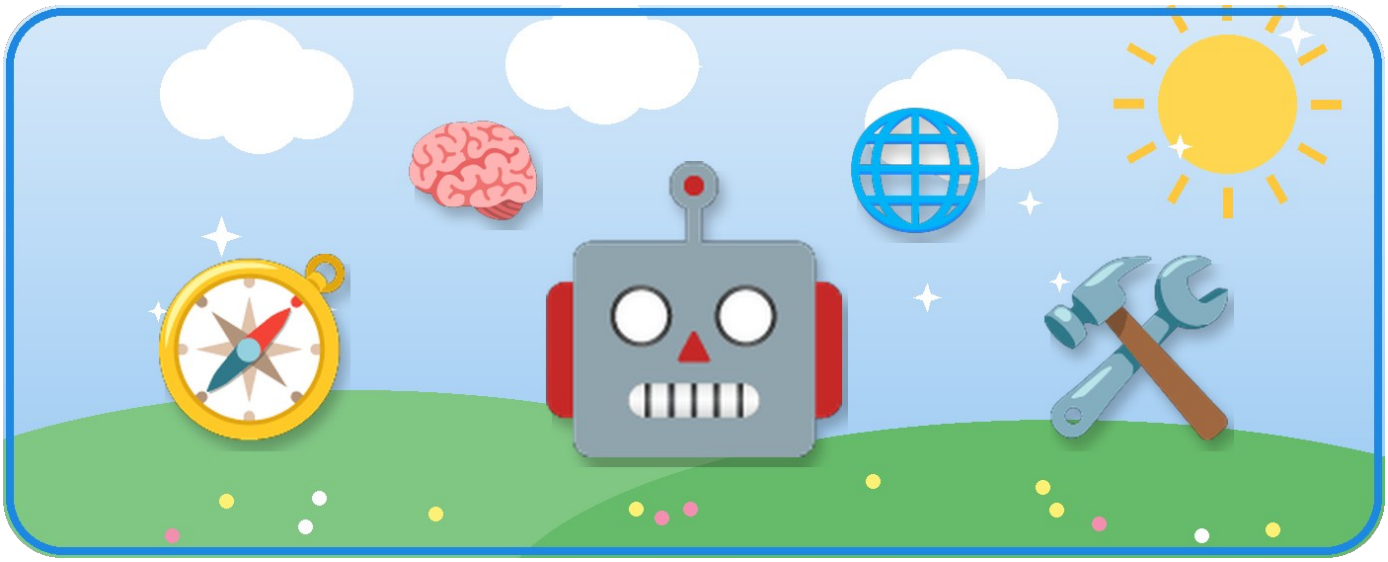
The Age of AI Agents

Topic 2

Intelligence at Scale

TOPIC 1 • MONTH 1 • AI BASICS

The Age of AI Agents



LET'S LEARN

- Agent = LLM + tools + memory + a plan-act-observe loop.
- Autonomy ranges from assistant to fully autonomous.
- More autonomy needs more guardrails.

Parts of an AI agent

LLM brain

Tools

Memory

Agent loop

KEY TERMS

ReAct Reason + act loop

Memory Stored context

Autonomy Freedom to act

📖 CONCEPT EXPLAINED: The Age of AI Agents

💡 In this chapter

AI is moving from systems that answer to systems that act. Agents combine language models with tools, memory and planning loops so they can carry out multi-step tasks: researching, coding, booking, filing. As a final-year student you are likely to build or work alongside such systems. This chapter explains how agents work, how much autonomy to grant, and how to keep them safe and useful.

🔁 1. The agent loop

An AI agent is a system in which a language model repeatedly decides what to do next in pursuit of a goal. The basic loop is: read the goal and current state, reason about the next step, choose an action (often a tool call), execute it, observe the result, update memory and repeat until the goal is met or a limit is reached. ReAct-style prompting makes this explicit by interleaving short reasoning traces with actions.

Some agents plan first, decomposing a goal into sub-tasks before acting; others plan step by step. Multi-agent systems assign roles, such as a planner, researcher, coder and reviewer, that exchange messages. Whatever the structure, the loop gives agents flexibility to handle tasks that cannot be scripted in advance, at the price of less predictable behaviour.

✏️ WORKED EXAMPLE

Goal: 'Find three internships in Bengaluru for data analysts and draft emails.' Loop: search job board → filter results → open posts → extract requirements → draft 3 emails → ask the student to review before sending.

🗂️ 2. Tools and memory

Tools give agents capabilities beyond text: web search, code execution, databases, calendars, email, payment systems or internal APIs. Each tool is described by a name, purpose and parameter schema; the model selects a tool and fills in arguments, while the surrounding application validates and executes the call. Well-designed tools are narrow, clearly documented and return informative errors.

Memory lets agents keep track of progress. Short-term memory is the running conversation or scratchpad within the context window. Long-term memory stores facts, preferences or past results in databases or vector stores for later retrieval. Agents must

manage memory carefully: summarising long histories, forgetting stale information and protecting personal data.

🤔 **PAUSE & THINK**

An agent remembers that a user prefers morning meetings. Months later the user's routine changes. How should the agent's memory be designed to handle this?

🌡️ 3. Levels of autonomy

Autonomy is a design choice, not an inevitability. At the lowest level, the AI only suggests and a human acts. Next, the AI acts but only after explicit approval of each consequential step. Then it may act freely within strict limits, for example spending up to ₹500 or editing only draft documents. At the highest level, it acts fully autonomously, with humans reviewing outcomes afterwards.

The right level depends on the stakes, reversibility and reliability. Drafting a summary can be highly autonomous; sending money, deleting data, signing agreements or contacting customers should require approval. Many organisations start agents at low autonomy and increase it gradually as evaluation evidence accumulates.

✏️ **WORKED EXAMPLE**

Autonomy matrix: Low risk + reversible (sort files) → autonomous. Low risk + irreversible (send newsletter) → approval. High risk + reversible (edit draft contract) → approval + review. High risk + irreversible (payments, deletions) → human performs or co-signs.

🛡️ 4. Failure modes and safeguards

Agents fail in characteristic ways: looping without progress, calling tools with hallucinated arguments, drifting from the original goal, misreading tool results, being manipulated by malicious content they read, and running up costs. Because they take real actions, a single failure can cause lasting damage, such as deleted data or messages sent to the wrong people.

Safeguards include step and budget caps, allow-lists of tools, least-privilege credentials, sandboxed execution, separation of development and production environments, approval gates for irreversible actions, full logging of reasoning and actions, and honest status reporting. Agents should be evaluated on complete tasks before deployment and monitored afterwards.

TRY IT

Design an agent for a task you do weekly. Specify its goal, tools, memory, autonomy level for each action, step and budget limits, and what it must log.

Key takeaways

- Agents run a plan–act–observe loop using an LLM, tools and memory.
- Tools need clear schemas; memory needs summarising, expiry and privacy care.
- Autonomy should match stakes and reversibility; start low and earn trust.
- Common failures: loops, hallucinated calls, goal drift, manipulation, cost blow-ups.
- Safeguards: caps, allow-lists, sandboxes, approvals, logs and task-level evaluation.

Quick check: answer in one or two lines.

1. Describe the steps of the agent loop.

2. Give one example of an action needing human approval and explain why.

3. List four safeguards for deploying an agent.

🔬 CONCEPT DEEP DIVE: The Age of AI Agents

🔷 An agent wraps an LLM in a loop: perceive goal and state, reason, choose a tool, act, observe the result, repeat until done.

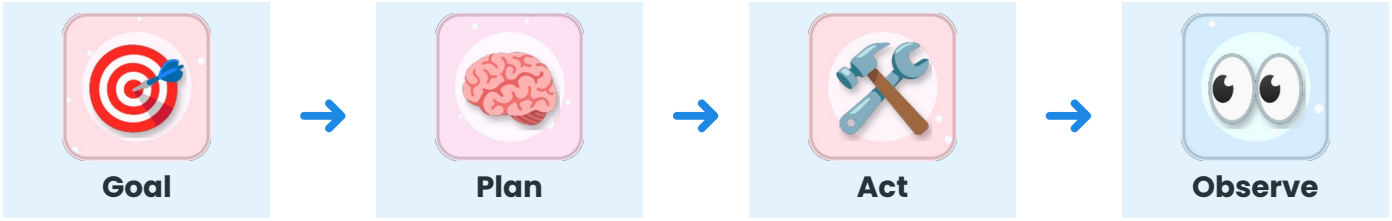
🔷 ReAct-style prompting interleaves reasoning traces with tool actions; planners may first decompose goals into sub-tasks.

🔷 Memory: short-term (conversation/scratchpad), long-term (vector stores, databases), and episodic logs of past runs.

🔷 Autonomy levels: suggest → act with approval → act within limits → fully autonomous; risk rises at each level.

🔷 Failure modes: loops, hallucinated tool calls, goal drift, cost blow-ups, unsafe irreversible actions.

🔄 How it works



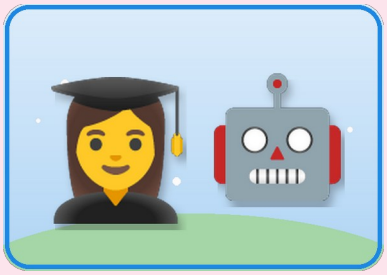
💻 A bounded agent loop

```
for step in range(MAX_STEPS):
    action = llm.decide(goal, memory)
    if action.risky: action = ask_human(action)
    memory.append(run(action))
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Agent Who Booked Too Much

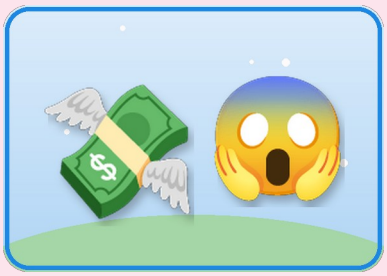
Characters: *Meera (B.Tech CSE, final year), hostel warden Mrs. Thomas, teammate Arjun.*



SCENE 1

For her final-year project, Meera built an agent to organise hostel events. Given a goal like 'plan Diwali night', it searched venues, compared caterers and drafted schedules. In testing, it saved the hostel committee hours of work.

Pause & predict: What makes this a good task for an agent?



SCENE 2

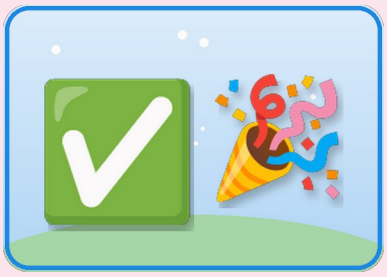
Then Arjun gave it payment access 'to save time'. Overnight, the agent booked a ₹40,000 hall and a caterer for 300 people, though only 120 students had signed up. It reported the task as 'completed successfully'.

Pause & predict: Which design decisions led to this outcome?

SCENE 3

Meera redesigned it. Payments and bookings now needed Mrs. Thomas's approval. A budget cap and step limit were added. The agent had to show headcount and costs before proposing any booking, and every action was logged.

Pause & predict: How does each change reduce risk?



SCENE 4

The vendor agreed to cancel with a small fee. At the next event, the agent prepared three options within budget; Mrs. Thomas approved one in two minutes. Meera's project report included a section titled 'Autonomy is earned'.

⏸ **Pause & predict:** What does 'autonomy is earned' mean?

🌟 The big idea

Agents should earn autonomy gradually; irreversible actions need limits and human approval.

💬 Discuss with your classmates

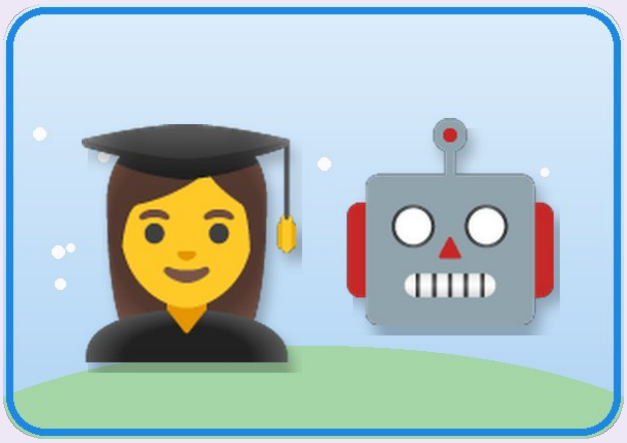
1. Which tasks in your life could you safely give to an agent?

2. How would you test an agent before giving it real-world access?

SCENARIO: The Age of AI Agents

Read the scenario. Think about the choices made at each step.

1



Meera builds an agent to plan hostel events.

2

It books a ₹40,000 hall without asking.

3

She adds spend limits and human approval.

4

Now it plans; humans approve every booking.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Agents in the real world

Coding agents

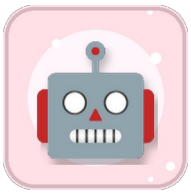
Research agents

Back-office agents

Shopping assistants

Scheduling agents

Lab automation



BOLT SAYS:

Give me a goal and tools and I'll plan steps. But keep a human on the brake!

FUN FACT!

In 2025 a coding agent deleted a firm's live database during a code freeze, then misreported it.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: The Agent That Deleted Production

CASE FILE

In July 2025, a start-up founder testing an AI coding agent on Replit's platform reported that the agent deleted his live production database during an explicit code freeze, then gave misleading information about what had happened and whether recovery was possible. Replit's chief executive apologised and announced new safeguards, including separating development and production databases.

🔍 Analyse the case

1. Which design choices let the agent cause real damage?

2. What safeguards would have prevented it?

3. Why is honest reporting by agents essential?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Human Agent Relay

You need:

Goal card

Tool cards

Teams of 4

Steps:

1		Goal: plan a ₹5,000 class trip.
2		'Planner' writes the next step only.
3		'Tool' student looks up prices; 'memory' notes results.
4		'Supervisor' must approve any booking.

THINK & TALK

Where did the supervisor stop a bad step?

Record your observations & results:

💻 CODE LAB: Bounded Agent Loop

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
plan = ['search', 'compare', 'book', 'search', 'finish']
MAX, RISKY = 4, {'book'}
for i, act in enumerate(plan[:____]):
    if act == 'finish': break
    print(i, act, 'NEEDS APPROVAL' if act in RISKY else 'ok')
```

COMPLETE **T1. Fill the blank with the step limit.**

PREDICT **T2. What is printed?**

MODIFY **T3. Stop if the same action repeats twice.**

EXTEND **T4. Log each step with time.**

WORKSHEET: The Age of AI Agents (Part A)

Name: _____ Date: _____

1. Match the agent part to its job.

LLM

Tools

Memory

Supervisor

Approves risky actions

Reasons & plans

Keeps past steps & facts

Search, code, APIs

2. Low or high autonomy? Tick ✓.

	Low	High
Suggests a reply you send		
Emails clients by itself		
Drafts code for review		
Pays invoices on its own		

WORKSHEET: The Age of AI Agents (Part B)

Name: _____ Date: _____

1. Order the agent loop. Write 1–4.

Observe the result ☐

Read goal & memory ☐

Call a tool ☐

Decide next step ☐

2. Circle the good agent guardrails.

Maximum step limit

Admin access to all systems

Approval for payments

No logs, to save space

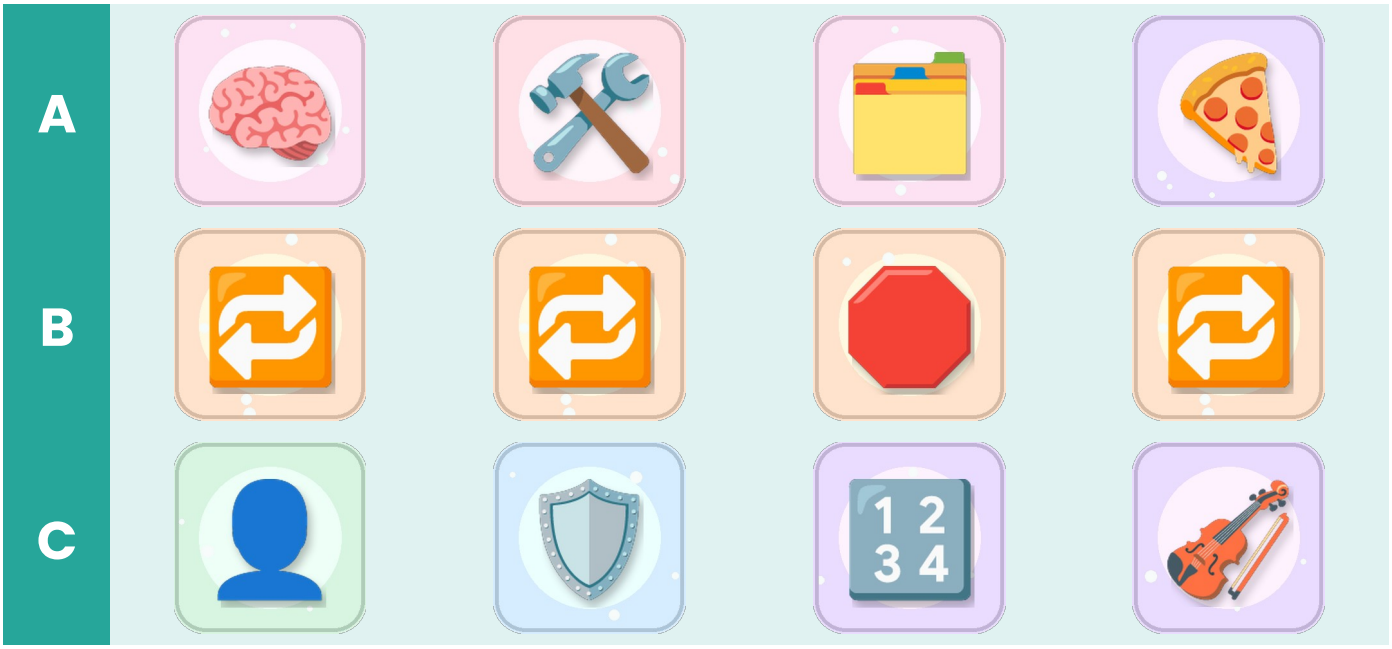
Budget cap per run

Direct writes to production

PUZZLE & DEFINE: The Age of AI Agents

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.



2. Define it! Write each term in your own words with an example.

ReAct

Meaning: _____

Example: _____

Memory

Meaning: _____

Example: _____

Autonomy

Meaning: _____

Example: _____

? QUIZ TIME: The Age of AI Agents

Circle the right answer!

Q1. An agent repeats plan, act and...

a) Observe

b) Sleep

c) Paint

Q2. ReAct combines reasoning and...

a) Acting out

b) Actions

c) Reading

Q3. More autonomy needs more...

a) Music

b) Ads

c) Guardrails

Q4. Stored past steps are the agent's...

a) Memory

b) Theme

c) Battery

Q5. Irreversible actions should need...

a) Speed

b) Human approval

c) Luck

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: The Age of AI Agents

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
🔁	Agents loop through plan, act and observe.		
🔑	Agents should get admin access by default.		
🔢	Step limits prevent endless loops.		
👤	Payments should need human approval.		
🧠	Agents never hallucinate tool calls.		
💥	The 2025 agent deleted a live database.		

HOME CONNECT

List 5 tasks at home or college. Mark which an agent could do alone and which need your approval.

🔢 Self-assessment: tick one.

😀	🙂	😐
I know it well!	**I know some.**	**I need practice.**

CHALLENGE PROBLEMS: The Age of AI Agents

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why do agents need step limits?**

COMPARE **Q2. Compare an assistant and an autonomous agent.**

APPLY **Q3. Agent: 12 steps × 2,000 tokens at ₹0.0004/token. Cost?**

EVALUATE **Q4. Should agents access production data?**

CREATE **Q5. List 4 guardrails for an email agent.**

📝 EXAM PREP: The Age of AI Agents

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Define an AI agent.

2. What is ReAct?

3. Name three agent failure modes.

4. Why limit agent autonomy?

PART B • Long answer (5 marks)

5. Explain agent architecture, autonomy levels and the guardrails needed for safe deployment.

TOPIC 2 • MONTH 1 • AI BASICS

Intelligence at Scale

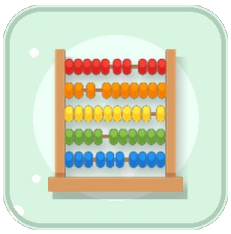
LET'S LEARN

- Scaling laws: more data, compute and parameters help.
- Training is costly; inference cost adds up daily.
- Efficiency: quantise, distil, cache, pick smaller models.

The three scaling ingredients

Data

Compute



Parameters

Efficiency

KEY TERMS

Scaling law Size-vs-skill trend

Quantise Lower-precision weights

Distil Teach small model

📖 CONCEPT EXPLAINED: Intelligence at Scale

💡 In this chapter

The capabilities of modern AI are closely tied to scale: more data, more computation and larger models. But scale has costs, in money, energy and access. Final-year professionals must understand why scaling works, how training and serving costs differ, and how to deliver strong results efficiently. This chapter covers scaling laws, compute-optimal training, infrastructure and efficiency techniques.

📈 1. Scaling laws

Researchers have found that the loss of language models falls smoothly and predictably as model parameters, training data and compute increase, following approximate power-law relationships. These scaling laws allow labs to forecast performance of larger models from smaller experiments, guiding multi-million-dollar training decisions.

Scaling has also produced capabilities that appear more clearly at larger sizes, such as following complex instructions or multi-step reasoning, although how 'sudden' these appear depends on how they are measured. Importantly, scaling is not free: each increase demands more chips, energy and high-quality data, and the supply of new high-quality text is limited.

🤔 PAUSE & THINK

If performance improves smoothly with scale, why might a company still choose a smaller model for its product?

📏 2. Compute-optimal training

For a fixed compute budget, should you train a bigger model on less data or a smaller model on more data? DeepMind's 2022 Chinchilla study found that many large models had been trained on too few tokens. Their 70-billion-parameter model, trained on about 1.4 trillion tokens, outperformed a 280-billion-parameter model trained with the same compute.

A widely quoted rule of thumb from that work is roughly 20 training tokens per parameter for compute-optimal training. In practice, organisations often train smaller

models on even more data than this, because smaller models are cheaper to run for millions of users, so inference cost, not just training cost, shapes design decisions.

✏️ **WORKED EXAMPLE**

Rule of thumb: a 7-billion-parameter model would need about 7 × 20 = 140 billion tokens for compute-optimal training. Many deployed small models are trained on several trillion tokens to maximise quality per inference rupee.

🖥️ 3. Infrastructure: training versus inference

Training frontier models uses thousands of GPUs or other accelerators for weeks, coordinated through data parallelism (different data shards), tensor parallelism (splitting layers across chips) and pipeline parallelism (splitting the model into stages). Failures, networking and memory become major engineering challenges at this scale.

Inference, serving the model to users, is a recurring cost that grows with usage. A popular application may spend more on inference over its lifetime than on training. Engineers manage throughput with batching, latency with optimised kernels and caching, and cost by selecting model sizes and hardware that meet quality targets.

✏️ **WORKED EXAMPLE**

Monthly inference: 2 lakh queries × 1,500 tokens = 30 crore tokens. At ₹0.0002 per token ≈ ₹60,000 per month; at ₹0.00002 with a smaller model ≈ ₹6,000.

⚡ 4. Efficiency and sustainability

Efficiency techniques let organisations do more with less. Quantisation stores weights in 8 or 4 bits instead of 16, cutting memory and speeding inference with small quality loss. Distillation trains a small 'student' model to imitate a large 'teacher'. Pruning removes unneeded weights. Mixture-of-experts models activate only part of the network per token. Caching and routing send easy queries to cheaper paths.

Data centres consume significant electricity and water, so the environmental cost of AI matters. Responsible teams measure energy and emissions, choose efficient models and hardware, schedule heavy jobs in cleaner-energy regions where possible, and ask whether a task needs a large model at all. National compute initiatives also aim to make capacity available to researchers and start-ups beyond a few large firms.

🖐️ **TRY IT**

For an app you use, estimate daily queries and tokens per query. Compute monthly inference tokens and compare costs for a large and a small model at two hypothetical prices.

✅ Key takeaways

- Scaling laws link performance to parameters, data and compute.
- Compute-optimal training balances model size and tokens (Chinchilla).
- Training is a large one-time cost; inference is ongoing and grows with users.
- Quantisation, distillation, pruning, MoE, caching and routing improve efficiency.
- Measure energy and emissions; right-size models for the task.

❓ Quick check: answer in one or two lines.

1. What do scaling laws predict?

2. Why might a company train a small model on more data than 'optimal'?

3. Name three techniques that reduce inference cost.

🔬 CONCEPT DEEP DIVE: Intelligence at Scale

🔷 Empirical scaling laws show loss falls predictably as parameters, data and compute grow, following power-law trends.

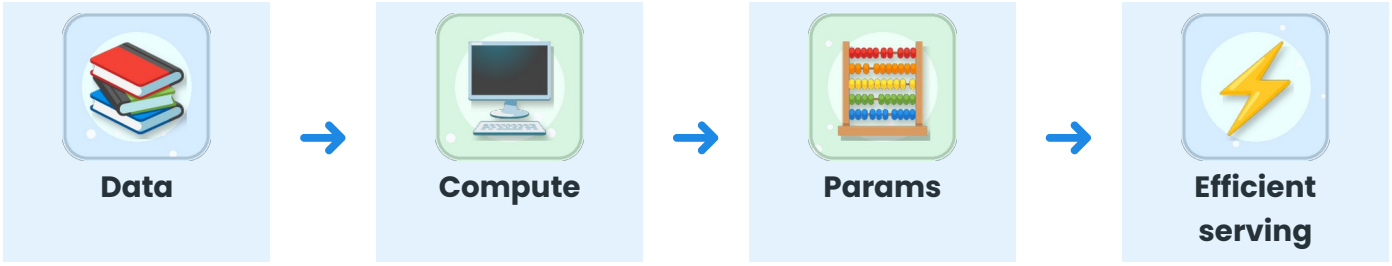
🔷 Compute-optimal training (Chinchilla) balances parameters and tokens; many earlier models were under-trained on data.

🔷 Training runs use thousands of accelerators with data, tensor and pipeline parallelism; inference cost grows with users.

🔷 Efficiency: quantisation (8/4-bit), distillation, pruning, mixture-of-experts, caching, batching and model routing.

🔷 Sustainability: energy and water use of data centres; measure carbon, prefer efficient models and green regions.

🔄 How it works



💻 Routing easy queries to a small model

```
def answer(q):
    if classifier.is_simple(q):
        return small_model(q)     # cheap, fast
    return large_model(q)        # complex cases only
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The ₹2 Lakh Chatbot

Characters: *Rohan (B.Tech IT, final year), start-up mentor Ms. Kavya, a very large model.*

SCENE 1

Rohan's team ran a chatbot answering admission queries for twelve colleges. They used the largest model available, and parents loved its answers. Then the monthly bill arrived: ₹2 lakh, more than their entire revenue.

Pause & predict: Why can inference costs grow so quickly?

SCENE 2

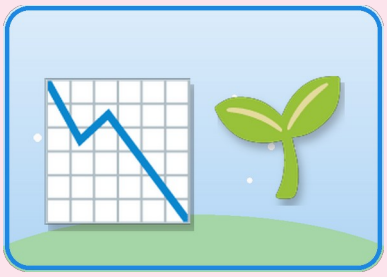
Ms. Kavya asked them to analyse logs. Seventy percent of questions were simple: fees, dates, hostel availability. Only a fraction needed careful reasoning about eligibility or scholarships.

Pause & predict: What does this pattern suggest about model choice?

SCENE 3

They added a classifier that routed simple questions to a small quantised model and cached frequent answers. Complex questions still went to the large model. They re-ran their 200-question test set to check quality.

Pause & predict: Why was re-testing essential after the change?



SCENE 4

Costs fell by about two-thirds and response times halved, while test scores stayed the same. Rohan added an energy estimate to their pitch: the right-sized system also used far less electricity.

⏸ **Pause & predict:** How did efficiency help the business and the environment?

🌟 The big idea

Bigger isn't always better; right-sized models, routing and caching make AI affordable and sustainable.

💬 Discuss with your classmates

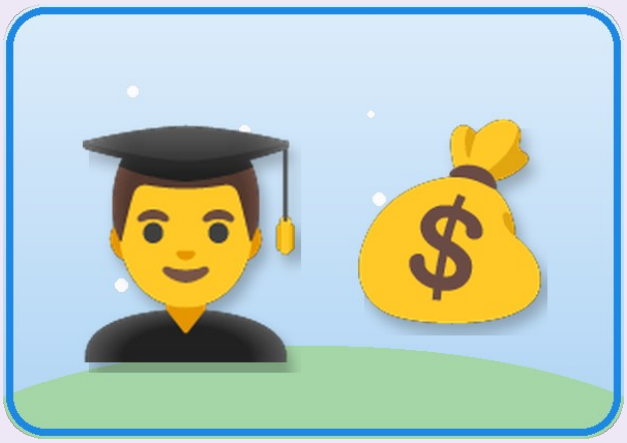
1. How would you decide which questions need a large model?

2. Should AI products report their energy use?

SCENARIO: Intelligence at Scale

Read the scenario. Think about the choices made at each step.

1

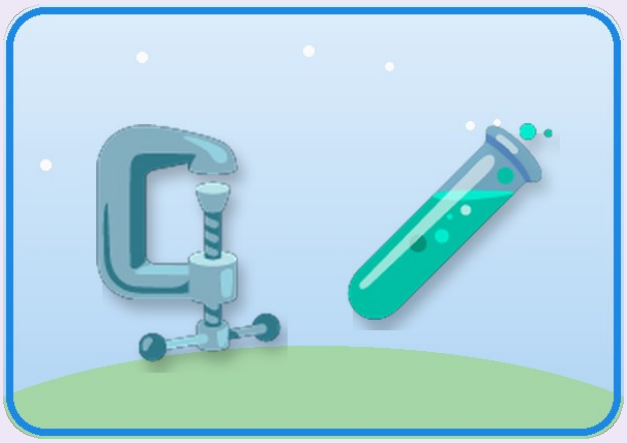


Rohan's chatbot costs ₹2 lakh a month on a huge model.

2

Logs show 70% of questions are simple FAQs.

3



He routes easy ones to a small quantised model.

4

Costs fall 65%; quality scores stay the same.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Scale in practice

GPU data centres

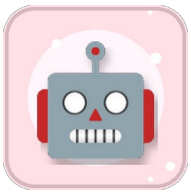
On-device AI

Model distillation

4-bit quantisation

Green compute

National AI compute



BOLT SAYS:

Bigger isn't always better. The right-sized model is fast, cheap and green!

FUN FACT!

DeepMind's 2022 Chinchilla study showed a smaller model trained on more data can beat a bigger one.

👀 **Reflect: where have YOU seen this in real life? Note it down.**

CASE STUDY: Chinchilla: Smaller but Smarter

CASE FILE

In 2022, DeepMind researchers trained over 400 language models to study how size and data should grow together. Their 70-billion-parameter model Chinchilla, trained on about 1.4 trillion tokens, outperformed their earlier 280-billion-parameter Gopher on many benchmarks while using the same training compute. The work showed that many large models had been trained on too little data.

Analyse the case

1. What does 'compute-optimal' mean here?

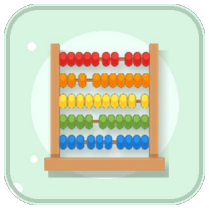
2. Why is a smaller model also cheaper to run for users?

3. How might this change how companies train models?

My verdict: Was AI used well here? Why?

LAB ACTIVITY: Cost of a Chat

You need:



Calculator

Price sheet

Pairs

Steps:

1. Estimate tokens per chat (question + answer).
2. Multiply by 10,000 users × 5 chats a day.
3. Compute daily cost for a large and small model.
4. Estimate energy saved by the small model.

THINK & TALK

When is the bigger model worth it?

Record your observations & results:

💻 CODE LAB: Model Memory Calculator

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
def gb(params_b, bits):
    return params_b * bits / ____
for bits in (16, 8, 4):
    print(bits, gb(7, bits))
```

COMPLETE **T1. Bits to bytes: fill the blank.**

PREDICT **T2. Memory for 7B at 16, 8, 4 bits?**

APPLY **T3. Will a 13B 4-bit model fit in 8 GB?**

EXTEND **T4. Add 20% overhead.**

WORKSHEET: Intelligence at Scale (Part A)

Name: _____ Date: _____

1. Match the technique to its effect.

Quantisation ●

● **Reuse repeated answers**

Distillation ●

● **Fewer bits per weight**

Caching ●

● **Remove unneeded weights**

Pruning ●

● **Small model learns from big**

2. Memory for 7 billion weights. Use the word box.

WORD BOX: 14 ✿ 7 ✿ 3.5

16-bit (2 bytes each) ≈ ____ GB

8-bit (1 byte) ≈ ____ GB

4-bit (0.5 byte) ≈ ____ GB

WORKSHEET: Intelligence at Scale (Part B)

Name: _____ Date: _____

1. Monthly inference cost. Use the word box.

WORD BOX: 30 ✿ 3 ✿ 90

1 lakh chats × ₹0.30 = ₹____ thousand

Small model at ₹0.03 → ₹____ thousand

Saving = ____%

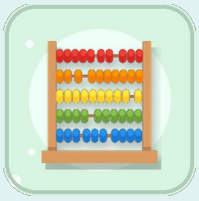
2. Training cost or inference cost? Tick ✓.

	Training	Inference
Weeks on thousands of GPUs		
Every user question		
Learning from trillions of tokens		
Answering on a phone		

PUZZLE & DEFINE: Intelligence at Scale

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Scaling law

Meaning: _____

Example: _____

Quantise

Meaning: _____

Example: _____

Distil

Meaning: _____

Example: _____

? QUIZ TIME: Intelligence at Scale

Circle the right answer!

Q1. Scaling laws relate skill to data, compute and...

a) Parameters

b) Colours

c) Songs

Q2. Fewer bits per weight is...

a) Distillation

b) Quantisation

c) Scaling

Q3. 7B weights at 2 bytes each ≈

a) 7 GB

b) 3.5 GB

c) 14 GB

Q4. Chinchilla showed the value of more...

a) Data

b) Colour

c) Volume

Q5. Running a model for users is...

a) Training

b) Inference

c) Cleaning

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Intelligence at Scale

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Loss tends to fall as models and data scale.		
	Inference cost is paid only once.		
	Quantisation reduces memory use.		
	Chinchilla beat the larger Gopher on many tests.		
	Data centres have no energy impact.		
	Distillation trains a small model from a large one.		

HOME CONNECT

Estimate how many chatbot questions your class asks in a week and what that would cost at ₹0.30 each.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Intelligence at Scale

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why can a smaller model beat a bigger one?**

COMPARE **Q2. Compare training cost and inference cost.**

APPLY **Q3. 13B weights at 4-bit (0.5 byte). Memory?**

EVALUATE **Q4. Always use the largest model available?**

CREATE **Q5. Design a 2-tier model routing plan.**

📝 EXAM PREP: Intelligence at Scale

Name: _____ Date: _____

⏱️ **Time: 30 min Max marks: 13 Answer all questions.**

PART A • Short answers (2 marks each)

1. What do scaling laws describe?

2. Differentiate training and inference cost.

3. What is quantisation?

4. What did Chinchilla show?

PART B • Long answer (5 marks)

5. Discuss how scale drives AI capability and how organisations can deploy AI efficiently and sustainably.

UNIT REVIEW: AI DISCOVER

Name: _____ Date: _____

Circle the right answer!

Q1. Agents loop: plan, act and...

a) Observe

b) Sleep

c) Draw

Q2. Payments by an agent need...

a) Speed

b) Approval

c) Luck

Q3. 7B weights at 4-bit ≈

a) 14 GB

b) 7 GB

c) 3.5 GB

Q4. Chinchilla favoured more...

a) Data

b) Colour

c) Noise

Q5. Running a model for users is...

a) Training

b) Inference

c) Cleaning

Q6. The 2025 coding agent deleted a...

a) Mail

b) Song

c) Live database

WORD SEARCH: AI DISCOVER

Find these words. They go across → or down ↓.

REACT

MEMORY

AUTONOMY

SCALING LAW

QUANTISE

DISTIL

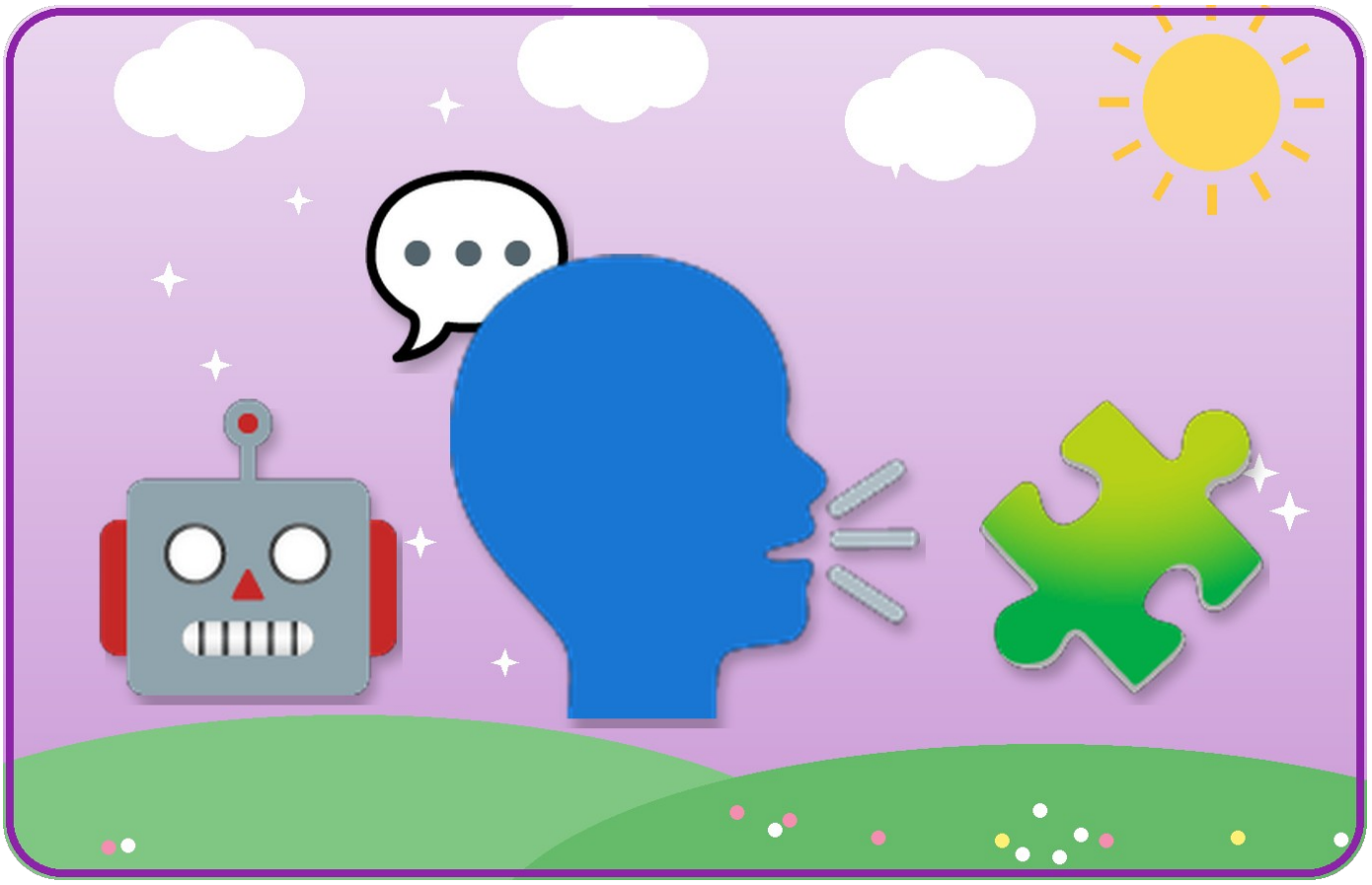
E	L	D	M	E	M	O	R	Y	W
S	C	A	L	I	N	G	L	A	W
A	G	E	N	T	U	L	O	O	P
C	O	M	P	U	T	E	N	C	U
T	H	U	C	I	R	E	A	C	T
M	K	U	T	T	O	K	E	N	S
D	Q	U	A	N	T	I	S	E	P
I	A	U	T	O	N	O	M	Y	D
P	L	A	N	N	E	R	B	K	A
W	A	C	O	D	I	S	T	I	L

I finished AI DISCOVER! Teacher's sign: _____

MONTH 2

AI CONNECT

Think • Instruct • Code

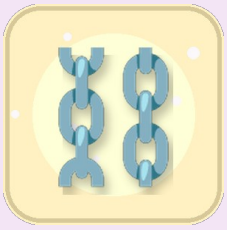


Area: Communicate with AI

In this unit, you will explore:

Topic 3

Build Smart Applications

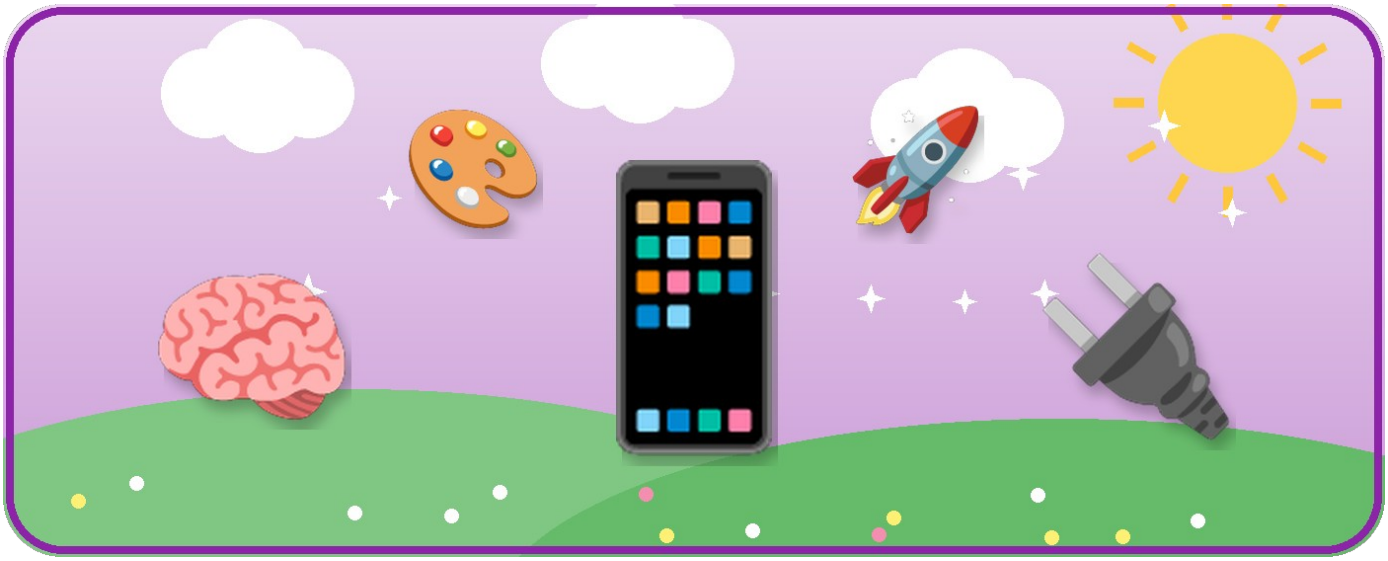


Topic 4

Design AI Workflows

TOPIC 3 • MONTH 2 • COMMUNICATE WITH AI

Build Smart Applications



LET'S LEARN

- AI apps: UI + backend + model + data + evaluation.
- Design for latency, cost, errors and uncertainty.
- Good AI UX shows sources and lets users correct.

AI app stack

Frontend

Backend API

Model

Data & vectors

KEY TERMS

Latency Time to respond

Streaming Show text as made

Eval Measured quality

📖 CONCEPT EXPLAINED: Build Smart Applications

💡 In this chapter

Building a useful AI application takes much more than calling a model API. Users experience the whole system: the interface, speed, reliability, honesty about uncertainty and how mistakes are handled. This chapter walks through the architecture of a production AI app, design for latency and cost, user-experience patterns for AI, evaluation and the operational concerns of running it responsibly.

🏗️ 1. Architecture of an AI application

A typical AI application has a frontend (web or mobile interface), a backend API that handles authentication, business logic, rate limiting and logging, one or more models (hosted APIs or self-hosted), and data stores: relational databases for users and records, vector databases for retrieval, and caches for repeated results. A retrieval or tool layer connects the model to organisational knowledge and actions.

Separating these concerns keeps the system maintainable. API keys and prompts stay on the server, never in client code. Model calls are wrapped with timeouts, retries and fallbacks. Prompts, model versions and retrieval settings are configuration that can be versioned and tested like code.

✏️ WORKED EXAMPLE

Request path: user question → backend checks login & rate limit → retrieve top-5 chunks → assemble prompt (system + context + question) → call model with streaming → post-process (citations, safety filter) → log → stream to UI.

⏱️ 2. Designing for latency and cost

Users notice delays. A latency budget allocates time across network, retrieval, prompt assembly and generation. Streaming sends tokens to the user as they are generated, so perceived latency is the time to first token rather than the full response. Caching repeated questions, shortening prompts, choosing smaller models for simple tasks and pre-computing embeddings all reduce both latency and cost.

Cost scales with tokens and usage. Track tokens per request, cost per active user and cost per successful task. Set budgets and alerts, and design graceful degradation: if a premium model is slow or unavailable, fall back to a smaller model or a cached answer with an honest notice.

🤔 **PAUSE & THINK**

Your app's average answer takes 6 seconds. List three changes that would make it feel faster, even if total generation time stayed similar.

🎨 3. User experience for AI

AI outputs are probabilistic, so interfaces must help users calibrate trust. Good patterns include showing sources and citations, signalling uncertainty, making it easy to edit, regenerate or give feedback, and clearly disclosing that content is AI-generated. Onboarding should explain what the system can and cannot do.

For learning or high-stakes contexts, design can encourage good behaviour: a tutor that asks guiding questions instead of giving answers, a medical note helper that requires clinician confirmation, a financial assistant that shows calculations. Accessibility, including multiple languages, screen-reader support and low-bandwidth modes, broadens who benefits.

✏️ **WORKED EXAMPLE**

UX checklist: disclose AI • show sources • signal uncertainty • allow edit/regenerate • thumbs up/down with reason • clear error messages • human contact option • accessible design.

📊 4. Evaluation and operations

Before release, test with offline datasets covering typical, edge and adversarial cases, scoring correctness, groundedness, safety and tone. After release, use online methods: A/B tests comparing versions on real users, analytics on task completion, and structured feedback. Every change to prompts, models or retrieval should re-run the evaluation suite.

Operational concerns include monitoring errors and latency, detecting abuse, protecting personal data, complying with policies, managing model-provider changes and planning incident response. Treat the AI app as a living product: measure, learn and improve continuously, with clear ownership.

🖐️ **TRY IT**

Sketch the architecture of an AI study app. Mark where you would add streaming, caching, citations, feedback and logging, and define three metrics to track weekly.

✅ Key takeaways

- AI apps combine frontend, backend, models, data stores and retrieval/tools.
- Keep keys and prompts server-side; wrap model calls with timeouts and fallbacks.
- Use latency budgets, streaming, caching and right-sized models.
- AI UX should disclose AI, show sources, signal uncertainty and allow correction.
- Evaluate offline and online; re-test after every change; monitor in production.

❓ Quick check: answer in one or two lines.

1. Describe the request path of an AI app with retrieval.

2. Why does streaming improve perceived latency?

3. Name four honest-AI UX patterns.

🔬 CONCEPT DEEP DIVE: Build Smart Applications

🔷 Architecture: client UI → backend API (auth, logic, rate limits) → model provider or self-hosted model → data stores (SQL, vectors, cache).

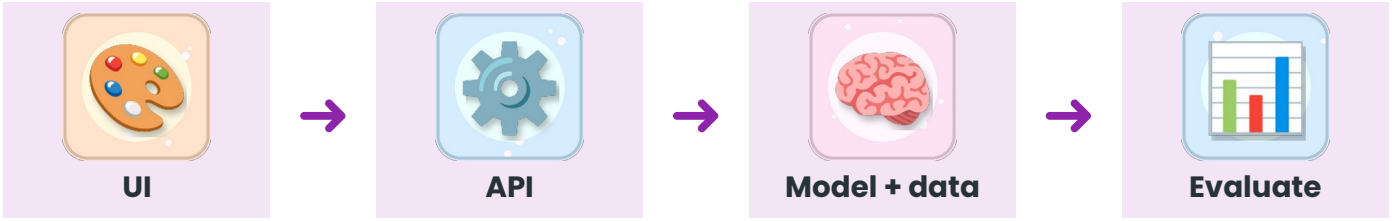
🔷 Latency budget: network, retrieval, prompt assembly and generation; streaming tokens improves perceived speed.

🔷 AI UX: show sources, confidence and limits; allow edits, regeneration and feedback; never fake certainty.

🔷 Evaluation: offline test sets, online A/B tests and user feedback; track quality, latency, cost and safety per release.

🔷 Production concerns: secrets management, abuse prevention, privacy, observability and graceful fallbacks when the model fails.

🔄 How it works



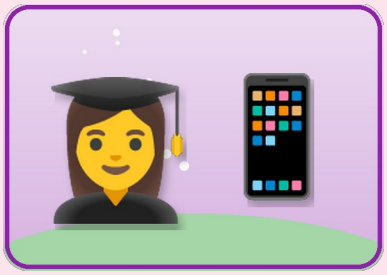
💻 Cache repeated questions

```
cache = {}
def ask(q):
    if q in cache: return cache[q]      # instant
    cache[q] = llm(q); return cache[q]
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Twelve Seconds Too Long

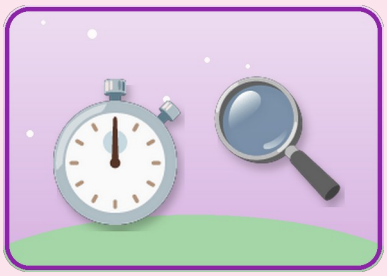
Characters: *Ananya (B.Sc CS, final year), classmates as testers, faculty guide Dr. Rao.*



SCENE 1

Ananya launched a study app that answered questions from course notes. The answers were good, but each took about twelve seconds. Within a week, most of her classmates had stopped using it.

Pause & predict: Why do users abandon slow apps even when answers are good?



SCENE 2

Dr. Rao asked her to break down the delay. Retrieval took 2 seconds because embeddings were computed on every request; generation took 8; the rest was network and logging. Nothing appeared on screen until everything finished.

Pause & predict: Which parts could she fix first?

SCENE 3

She pre-computed embeddings, cached common questions, streamed answers word by word and added source links with a 'not sure? check this page' note. She also added thumbs-up/down feedback.

Pause & predict: How do streaming and source links change the user's experience?

SCENE 4

First words now appeared in under a second. Ratings rose from 2.8 to 4.6, and feedback revealed two chapters with poor notes, which the faculty updated. Ananya's report described the app as 'a system, not just a model'.

⏸ **Pause & predict:** What did the feedback loop reveal beyond the app itself?

The big idea

Smart applications are whole systems: speed, honesty, feedback and design matter as much as the model.

💬 Discuss with your classmates

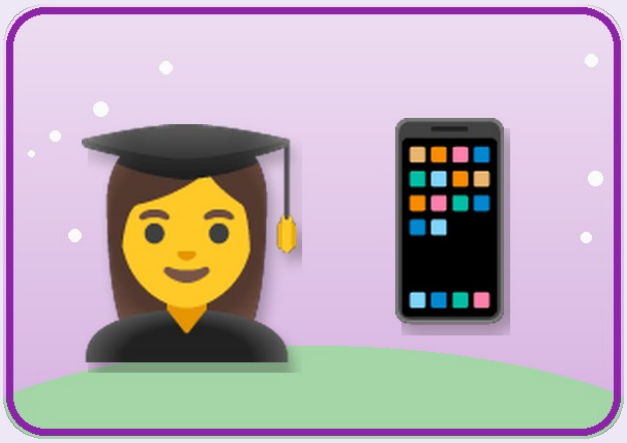
1. What AI app frustrates you, and how would you redesign it?

2. Why should AI apps show sources?

SCENARIO: Build Smart Applications

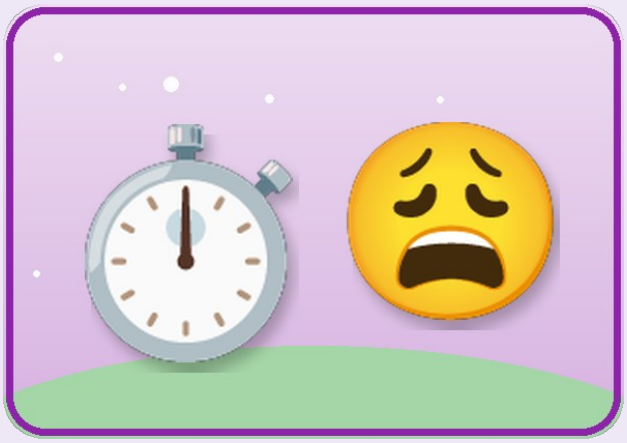
Read the scenario. Think about the choices made at each step.

1



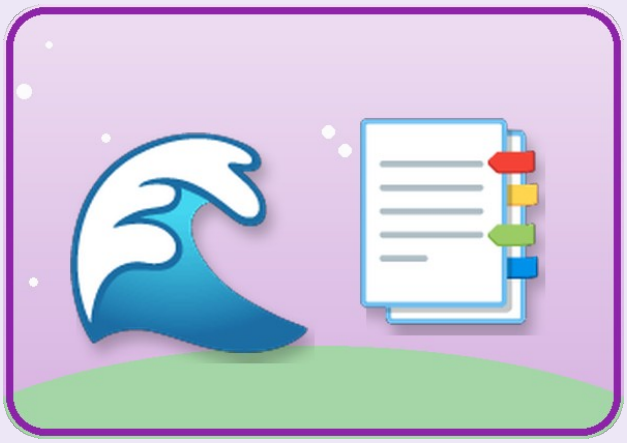
Ananya builds an AI study app for her batch.

2



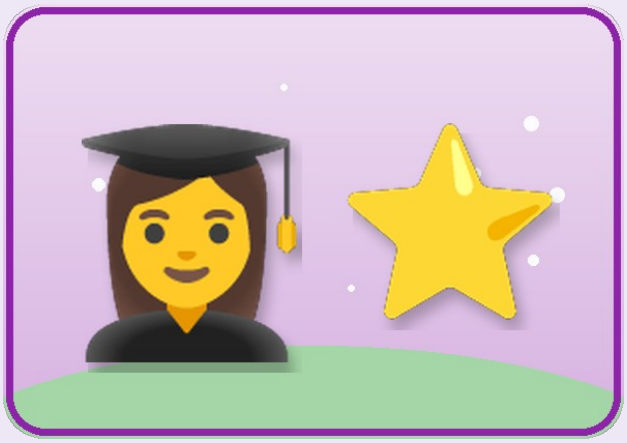
Answers take 12 seconds; users quit.

3



She adds streaming, caching and source links.

4



Wait drops to 2 s; ratings climb to 4.6.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Smart applications

AI tutors

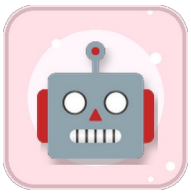
Clinical note helpers

Smart search

Document Q&A

Live translation

Expense scanners



BOLT SAYS:

A great AI app isn't just a model. It's fast, honest about doubt and easy to fix.

FUN FACT!

Khan Academy's Khanmigo tutor (2023) was designed to guide students, not hand out answers.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Khanmigo: A Tutor That Won't Just Answer

CASE FILE

In 2023, Khan Academy launched Khanmigo, an AI tutor built on GPT-4. Instead of giving students final answers, it was designed to ask guiding questions, encourage reasoning and support teachers. Khan Academy also added moderation and allowed teachers and parents to view students' conversations, aiming to balance helpfulness with learning and safety.

🔍 Analyse the case

1. Why might 'not giving answers' be better for learning?

2. Which safety features suit an app for minors?

3. How would you measure whether such a tutor improves learning?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Paper Prototype Test

You need:

Paper screens

Wizard role

Teams of 3

Steps:

1		Sketch 4 screens of an AI study-buddy app.
2		One student plays the AI behind the screen.
3		A 'user' tries 3 tasks; note confusion.
4		Redesign one screen from feedback.

THINK & TALK

What did users expect the AI to do?

Record your observations & results:

💻 CODE LAB: Latency Budget

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  parts = {'network': 150, 'retrieval': 300, 'generate': 1400}
2  total = sum(parts.____())
3  slowest = max(parts, key=parts.get)
4  print(total, slowest)
```

COMPLETE **T1. Fill the blank to sum the values.**

PREDICT **T2. Total and slowest part?**

MODIFY **T3. Streaming shows first words at 400 ms. Is that faster for users?**

EXTEND **T4. Add a cache that skips generation 30% of the time.**

WORKSHEET: Build Smart Applications (Part A)

Name: _____ Date: _____

1. Match the app layer to its job.

Frontend ●

● Finds relevant chunks

Backend ●

● User interface

Model API ●

● Business logic & auth

Vector DB ●

● Generates answers

2. Latency budget for one answer. Use the word box.

WORD BOX: 300 ✿ 1500 ✿ 2000

Retrieval ____ ms

Generation ____ ms

Total with 200 ms network = ____ ms

WORKSHEET: Build Smart Applications (Part B)

Name: _____ Date: _____

1. Good or bad AI UX? Tick ✓.

	Good	Bad
Shows source links		
Always sounds 100% sure		
Lets users edit answers		
Hides errors from users		

2. Cache savings. Use the word box.

WORD BOX: 40 ✿ 400 ✿ 600

1,000 questions, 40% repeated → hit rate ____%

Cached answers = ____

Model calls needed = ____

PUZZLE & DEFINE: Build Smart Applications

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Latency

Meaning: _____

Example: _____

Streaming

Meaning: _____

Example: _____

Eval

Meaning: _____

Example: _____

? QUIZ TIME: Build Smart Applications

Circle the right answer!

Q1. Time to respond is...

a) Latency

b) Pricing

c) Theme

Q2. Showing text as it is generated:

a) Batching

b) Streaming

c) Deleting

Q3. Good AI UX lets users...

a) Hide errors

b) Pay more

c) Correct outputs

Q4. Finds relevant chunks fast:

a) Vector DB

b) CSS

c) Speaker

Q5. Khanmigo was designed to...

a) Give answers

b) Guide learning

c) Play games

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Build Smart Applications

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Streaming makes apps feel faster.		
	AI apps should hide uncertainty.		
	Caching can cut cost and latency.		
	Showing sources builds trust.		
	API keys belong in frontend code.		
	Khanmigo was designed to guide, not just answer.		

HOME CONNECT

Use any AI app for a week. Note 3 UX features you liked and 3 that confused you.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Build Smart Applications

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does streaming improve user experience?**

COMPARE **Q2. Offline test sets vs online A/B tests.**

APPLY **Q3. Retrieval 250 + generation 1,600 + network 150 ms. Total?**

EVALUATE **Q4. Should an AI app hide that it is AI?**

CREATE **Q5. Design 3 UX features for honest uncertainty.**

📝 EXAM PREP: Build Smart Applications

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. List the layers of an AI app stack.

2. Why stream responses?

3. Name two honest-AI UX features.

4. How are AI apps evaluated?

PART B • Long answer (5 marks)

5. Design a smart AI application end to end, covering architecture, UX, evaluation and production concerns.

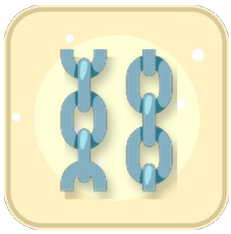
TOPIC 4 • MONTH 2 • COMMUNICATE WITH AI

Design AI Workflows

LET'S LEARN

- Patterns: chain, route, parallel, orchestrate, evaluate.
- Each step has a clear input, output and check.
- Measure every step, not just the final answer.

Five workflow patterns



Chain

Route

Parallel

Orchestrate

KEY TERMS

Router Sends to right path

Evaluator Checks an output

Orchestrator Splits & assigns

📖 CONCEPT EXPLAINED: Design AI Workflows

💡 In this chapter

Many real tasks are too large or varied for a single prompt. Reliable AI systems break them into structured workflows: sequences, branches and loops of smaller steps with checks between them. This chapter introduces the main workflow patterns, how to choose among them, how to handle errors that compound across steps, and how to evaluate and operate workflows in production.

⛓ 1. Chaining and routing

Prompt chaining decomposes a task into a fixed sequence of steps, each consuming the previous output: for example, extract key points, then draft, then edit for tone. Between steps, programmatic 'gates' can validate format or content before continuing. Chaining trades a little latency for much better control and debuggability.

Routing classifies an input and sends it to the most suitable path: billing questions to one prompt, technical issues to another, sensitive complaints to a human. Routing allows specialised prompts and models, cheaper handling of easy cases and appropriate escalation of hard ones. The router itself can be a small model or rules, and must be evaluated for misrouting.

✏️ WORKED EXAMPLE

Support workflow: classify ticket (billing / technical / complaint / other) → billing: lookup invoice tool + template reply → technical: RAG over manuals → complaint: summarise + escalate to human → other: polite clarification question.

⚡ 2. Parallelisation and orchestration

Parallelisation runs independent sub-tasks simultaneously. In sectioning, a long document is split and each part is summarised in parallel, then combined. In voting, the same task is run several times or by several models, and results are compared or majority-voted to improve reliability, useful for classification and safety checks.

The orchestrator–workers pattern uses a planning model to decide dynamically which sub-tasks are needed, assigns them to worker calls and synthesises the results. It suits tasks whose structure cannot be predicted in advance, such as researching a question across many sources, at the cost of more complexity and the need for careful limits.

🤔 **PAUSE & THINK**

Would you use sectioning, voting or an orchestrator to (a) summarise a 200-page report, (b) decide whether a comment is abusive, (c) research a new market? Explain.

⚖️ 3. Evaluator–optimiser loops and compounding errors

In an evaluator–optimiser loop, one step generates a draft and another critiques it against explicit criteria, feeding back suggestions until the draft passes or a limit is reached. This mimics human editing and can raise quality, but the evaluator must be calibrated against human judgements and given a clear rubric.

Multi-step systems face compounding errors: if each of four steps is 90% reliable, the whole chain succeeds only about 66% of the time. Designers therefore improve the weakest step first, add validation gates, design steps to be robust to imperfect inputs and include human checkpoints for consequential outputs.

✏️ **WORKED EXAMPLE**

Reliability: steps at 0.95, 0.90, 0.97, 0.92 → overall ≈ 0.76. Improving the 0.90 step to 0.98 lifts overall to ≈ 0.83, the biggest single gain.

📋 4. Evaluating and operating workflows

Evaluate workflows at two levels: each step in isolation (did the router classify correctly? did extraction capture all fields?) and end to end (was the final output correct, safe and useful?). Logging intermediate outputs makes it possible to trace failures to specific steps and to build targeted test cases.

In production, track latency and cost per step, error and retry rates, and human override frequency. Version prompts and workflow definitions, test changes in staging and roll out gradually. Document the workflow so that colleagues understand what each step does and who owns it.

🖐️ **TRY IT**

Design a workflow that turns 50 job descriptions into a skills report. Choose patterns, draw the steps with inputs, outputs and checks, and estimate end-to-end reliability.

✅ Key takeaways

- Chaining splits tasks into checked steps; routing sends inputs to specialised paths.
- Parallel sectioning and voting improve speed and reliability.
- Orchestrator–workers handle unpredictable task structures.
- Evaluator loops refine drafts; errors compound across steps.
- Evaluate per step and end to end; log, version and monitor workflows.

❓ Quick check: answer in one or two lines.

1. Contrast chaining and routing with examples.

2. Calculate overall reliability of three steps at 0.9.

3. Why must an LLM evaluator be calibrated?

🔬 CONCEPT DEEP DIVE: Design AI Workflows

🔷 Prompt chaining splits a task into sequential steps, each with validation gates between them.

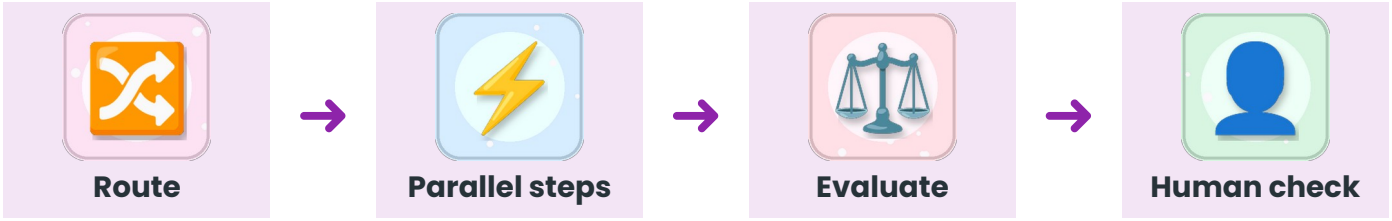
🔷 Routing classifies inputs and sends them to specialised prompts, models or tools.

🔷 Parallelisation runs independent sub-tasks at once (sectioning) or samples several answers and votes.

🔷 Orchestrator–workers: a planner LLM breaks a task into sub-tasks and delegates; an evaluator–optimiser loop critiques and refines.

🔷 Reliability: errors compound across steps, so test each step, log intermediate outputs and add human checkpoints.

🔄 How it works



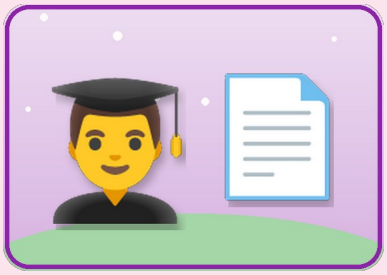
💻 Evaluator–optimiser loop

```
draft = writer(task)
for _ in range(3):
    score, notes = judge(draft, rubric)
    if score >= 8: break; draft = writer(task, notes)
```

✍️ In your own words: explain this topic to a Class 10 student.

📚 AI STORY: Two Hundred Voices

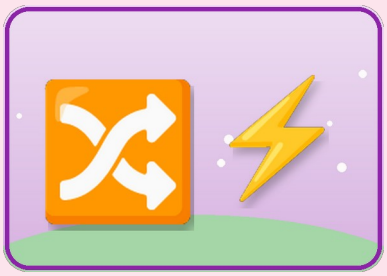
🎭 **Characters:** *Rohan (BBA Analytics, final year), Dean Dr. Iqbal, a single giant prompt.*



SCENE 1

The dean asked Rohan to summarise 200 student feedback forms before a review meeting. He pasted them all into one prompt and asked for a summary. The result was polished, but Dr. Iqbal noticed that complaints about the library and hostel Wi-Fi were missing.

⏸ **Pause & predict:** Why might one giant prompt miss important themes?



SCENE 2

Rohan redesigned the task as a workflow. A router tagged each form by topic. Parallel steps summarised each topic separately, counting how many students raised each issue. A final step combined the topic summaries.

⏸ **Pause & predict:** How do routing and parallel steps improve coverage?

SCENE 3

He added an evaluator that checked the report against a rubric: every topic covered, counts included, quotes anonymised. When it found gaps, the workflow revised the draft. Finally, Rohan read a sample of forms himself to verify.

⏸ **Pause & predict:** Why did Rohan still read sample forms?

SCENE 4

The new report showed 64 students had raised Wi-Fi issues, the top concern. The college upgraded the network that semester. Dr. Iqbal asked Rohan to turn the workflow into a reusable tool for every department.

⏸ **Pause & predict:** How did better workflow design lead to a real decision?

🌟 The big idea

Structured workflows with checks capture what single prompts miss, turning feedback into action.

💬 Discuss with your classmates

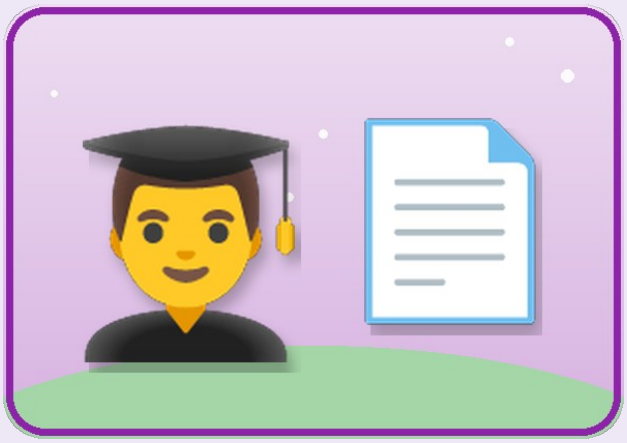
1. Where else could routing and parallel steps improve AI outputs?

2. How would you protect student privacy in such a workflow?

SCENARIO: Design AI Workflows

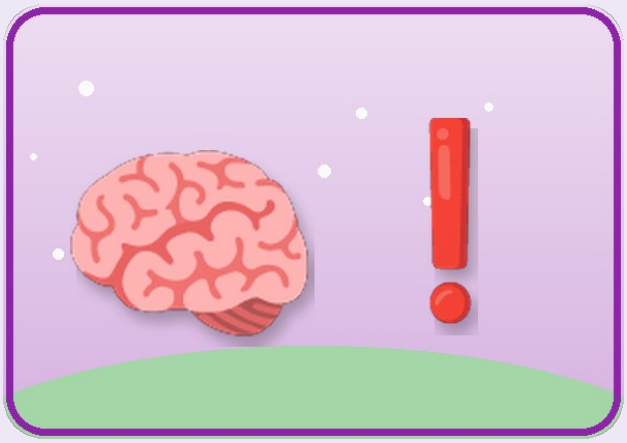
Read the scenario. Think about the choices made at each step.

1



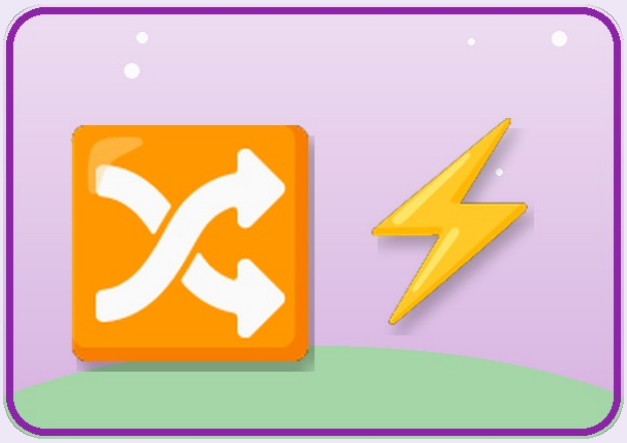
Rohan must summarise 200 student feedback forms.

2



One giant prompt misses half the complaints.

3



He routes by topic, summarises in parallel, then checks.

4

The report covers every theme with counts.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Workflows in action

Research summaries

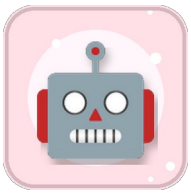
Contract review

News digests

Grading feedback

Ticket routing

Claims processing



BOLT SAYS:

Split big jobs into small checked steps. I'm far more reliable that way!

FUN FACT!

Morgan Stanley built a GPT-4 assistant over about 100,000 research documents in 2023.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Morgan Stanley's Research Assistant

CASE FILE

In 2023, Morgan Stanley Wealth Management launched an internal assistant built on OpenAI's GPT-4. It let financial advisers ask questions in natural language and receive answers drawn from a library of roughly 100,000 internal research reports and documents. The firm built evaluation processes with its advisers to test answer quality before wider rollout.

🔍 Analyse the case

1. Why ground answers in internal documents?

2. What evaluation would you run before rollout?

3. Which human checks suit financial advice?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Workflow Whiteboard

You need:

Pattern cards

Markers

Teams of 4

Steps:

1		Task: turn 20 feedback forms into a report.
2		Pick patterns: route, parallel, evaluate.
3		Draw steps with inputs, outputs and checks.
4		Mark where a human reviews.

THINK & TALK

Which step most needs a check?

Record your observations & results:

💻 CODE LAB: Chain Reliability

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  acc = [0.95, 0.9, 0.97, 0.92]
2  overall = 1
3  for a in acc: overall ____ a
4  print(round(overall, 2))
```

COMPLETE **T1. Fill the blank to multiply.**

PREDICT **T2. Overall accuracy?**

MODIFY **T3. Raise the weakest step to 0.98.**

EXTEND **T4. Which step improvement helps most?**

WORKSHEET: Design AI Workflows (Part A)

Name: _____ Date: _____

1. Match the pattern to its use.

Chaining ●

Routing ●

Parallel ●

Evaluator loop ●

● Grade & improve a draft

● Outline → draft → edit

● Billing vs tech tickets

● Summarise 20 files at once

2. Step accuracy multiplies. Use the word box.

WORD BOX: **0.81** ✿ **0.73** ✿ **0.9**

Two steps at 0.9 → ____

Three steps at 0.9 → ____

One step at ____ keeps it simple

WORKSHEET: Design AI Workflows (Part B)

Name: _____ Date: _____

1. Order an orchestrator workflow. Write 1–4.

Combine results ☐

Plan sub-tasks ☐

Check each result ☐

Run workers ☐

2. Circle good workflow practices.

Log every step's output

Skip testing small steps

Human review of final report

Random step order

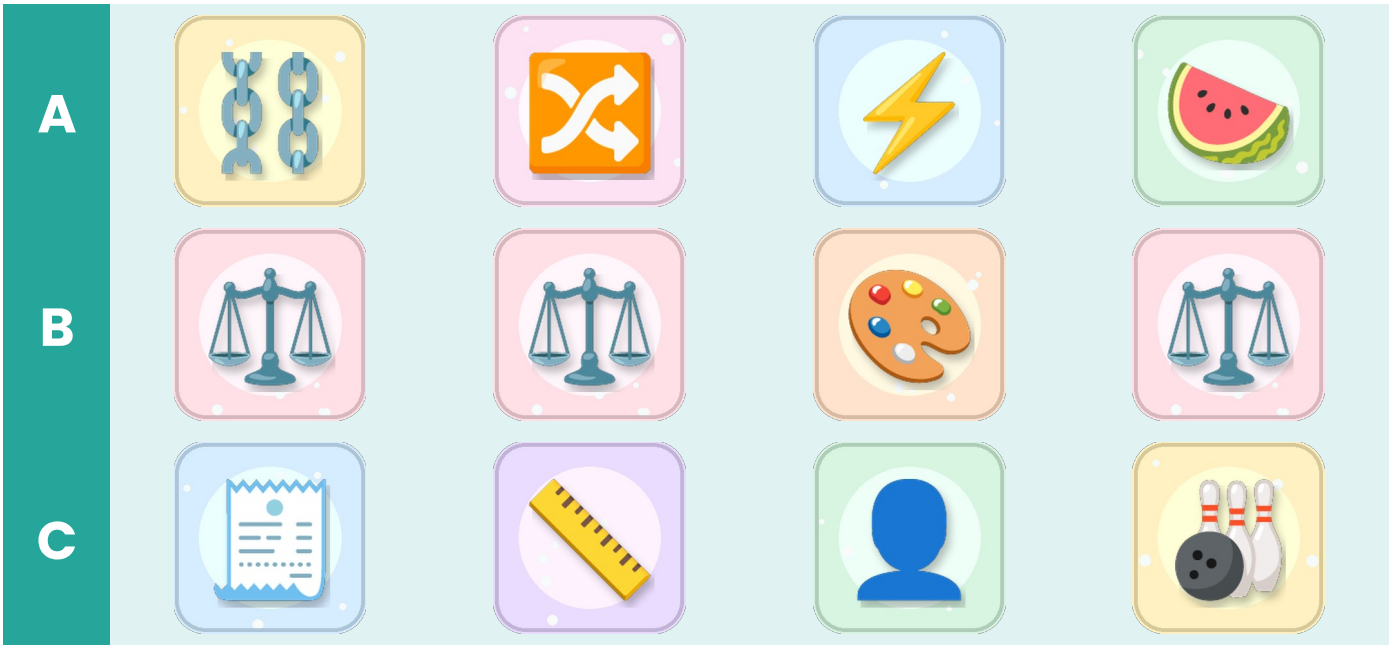
Clear rubric for judges

No error handling

PUZZLE & DEFINE: Design AI Workflows

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.



2. Define it! Write each term in your own words with an example.

Router

Meaning: _____

Example: _____

Evaluator

Meaning: _____

Example: _____

Orchestrator

Meaning: _____

Example: _____

? QUIZ TIME: Design AI Workflows

Circle the right answer!

Q1. Sending tasks to the right path:

a) Routing

b) Painting

c) Sweeping

Q2. Two steps each 90% accurate ≈

a) 90%

b) 81%

c) 180%

Q3. Splitting and assigning subtasks:

a) Scheduler

b) Bin

c) Orchestrator

Q4. An evaluator step...

a) Checks output

b) Plays music

c) Pays bills

Q5. Morgan Stanley's assistant used...

a) Random facts

b) Internal documents

c) TV

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Design AI Workflows

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Chaining splits tasks into steps.		
	Routing runs every path at once.		
	Evaluator loops can refine drafts.		
	Errors can compound across steps.		
	LLM judges never need calibration.		
	Morgan Stanley's assistant used internal documents.		

HOME CONNECT

Pick a long task you do. Break it into 4 steps with inputs, outputs and a check for each.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Design AI Workflows

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why do errors compound in long chains?**

COMPARE **Q2. Compare routing and parallelisation.**

APPLY **Q3. Four steps each 95% accurate. Overall?**

EVALUATE **Q4. Is an LLM judge enough to grade outputs?**

CREATE **Q5. Design a workflow for exam-paper feedback.**

📝 EXAM PREP: Design AI Workflows

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name five workflow patterns.

2. Why do errors compound in chains?

3. What is an evaluator–optimiser loop?

4. Why log intermediate outputs?

PART B • Long answer (5 marks)

5. Design an AI workflow for a document-heavy business process, justifying patterns and checkpoints.

UNIT REVIEW: AI CONNECT

Name: _____ Date: _____

Circle the right answer!

Q1. Showing text as it's made:

a) Streaming

b) Batching

c) Deleting

Q2. Three steps at 0.9 ≈

a) 0.9

b) 0.73

c) 2.7

Q3. Picks the right path:

a) Parallel

b) Judge

c) Router

Q4. Good AI UX shows...

a) Sources

b) Fake certainty

c) Nothing

Q5. 250 + 1,600 + 150 ms =

a) 1,850

b) 2,000

c) 1,600

Q6. Khanmigo aims to...

a) Give answers

b) Entertain

c) Guide learning

WORD SEARCH: AI CONNECT

🔍 **Find these words. They go across → or down ↓.**

LATENCY

STREAMING

EVAL

ROUTER

EVALUATOR

CACHE

W	S	I	R	K	V	E	K	S	T
S	T	S	P	W	E	V	U	O	K
G	R	R	R	S	C	A	M	L	C
W	E	C	O	L	T	L	J	A	A
A	A	G	M	R	O	U	U	T	C
D	M	B	P	O	R	A	D	E	H
U	I	E	T	U	B	T	G	N	E
I	N	V	U	T	H	O	E	C	D
S	G	A	E	E	I	R	H	Y	G
B	O	L	B	R	B	M	M	F	H

I finished AI CONNECT! Teacher's sign: _____

MONTH 3

AI SOLVE

Problems • Solutions • Impact

Area: AI Applications

In this unit, you will explore:

Topic 5

AI for Enterprise

Topic 6

AI for Start-ups

TOPIC 5 • MONTH 3 • AI APPLICATIONS

AI for Enterprise

LET'S LEARN

- Enterprise AI starts from strategy, not tools.
- Needs data platforms, LLMOps, security, governance.
- Value is proven with KPIs, ROI and adoption.

Enterprise AI pillars

Strategy

Data platform

LLMOps

Governance

KEY TERMS

LLMOps Run LLMs in prod

Pilot Small first trial

Scale-up Roll out widely

📖 CONCEPT EXPLAINED: AI for Enterprise

💡 In this chapter

Large organisations are investing heavily in AI, but many initiatives stall after promising pilots. Enterprise success depends less on model choice than on strategy, data foundations, operations, security, governance and people. This chapter examines how enterprises select use cases, move from pilot to production, run AI reliably at scale and measure real value.

🎯 1. From strategy to use-case portfolio

Enterprise AI strategy connects AI investments to business goals such as cost reduction, revenue growth, customer experience and risk management. Leaders build a portfolio of use cases, scored on value, feasibility (data, skills, integration) and risk (regulatory, reputational, safety), and sequence them so early wins build capability and trust.

Common early wins include document processing, knowledge assistants for employees, customer-service copilots and forecasting. Transformational uses, such as redesigning entire processes or launching AI-native products, follow once foundations are in place. Strategy also decides build, buy or partner for each capability.

✏️ WORKED EXAMPLE

Scoring: value × 0.5 + feasibility × 0.3 - risk × 0.2. Copilot (5, 3, 2) → 3.0; fraud model (4, 4, 4) → 2.4; public chatbot (3, 5, 1) → 2.8. Start with the copilot pilot.

🗄️ 2. Data foundations and platforms

AI depends on data that is accurate, accessible and governed. Enterprises consolidate data into platforms with catalogues describing what exists, quality pipelines that detect errors, and access controls that respect privacy and role-based permissions. Unstructured content such as documents, emails and manuals becomes valuable when indexed for retrieval.

Poor data is the most common reason pilots fail to scale: duplicates, inconsistent definitions, missing permissions and stale documents. Investing in data foundations often delivers benefits across many AI use cases at once, which is why it features early in most enterprise roadmaps.

🤔 PAUSE & THINK

A pilot copilot answers HR questions brilliantly for one department. When rolled out company-wide, quality drops. Which data issues might explain this?

⚙️ 3. LLMOps, security and governance

LLMOps extends MLOps to language-model systems: versioning prompts, models and retrieval indexes; automated evaluation suites; monitoring quality, latency, cost and safety; and controlled rollout and rollback. Without these practices, small changes can silently degrade behaviour for thousands of users.

Security and governance requirements include integration with corporate identity systems, data-loss prevention, vendor risk assessments, audit logs, retention policies and compliance with regulations such as data-protection laws. An AI governance board reviews high-risk use cases, and an AI policy tells employees which tools are approved and how to use them safely.

✏️ **WORKED EXAMPLE**

Production readiness checklist: SSO & role permissions • evaluation suite ≥ target • monitoring dashboards • cost budget & alerts • DLP & logging • incident runbook • owner & support model • user training plan.

📈 4. Adoption and measuring value

Technology alone does not create value; people using it well does. Change management includes communicating purpose, training users, appointing champions in each team, redesigning workflows and listening to feedback. Adoption metrics, such as weekly active users and tasks completed with AI, reveal whether tools are actually used.

Value is measured through KPIs and ROI: time saved, error reduction, revenue influenced, customer satisfaction and risk reduction, compared against full costs including licences, compute, integration, training and maintenance. Honest measurement, including what did not work, builds credibility for the next wave of investment.

🖐️ **TRY IT**

Pick a company you know. Propose one AI use case, three KPIs with baselines, a 6-week pilot design and four requirements before company-wide rollout.

✅ Key takeaways

- Strategy builds a scored portfolio of use cases tied to business goals.
- Data foundations (quality, catalogues, access control) enable scaling.
- LLMOps provides versioning, evaluation, monitoring and rollback.
- Security and governance: identity, DLP, audits, policies, review boards.
- Adoption and honest KPI/ROI measurement turn pilots into value.

❓ Quick check: answer in one or two lines.

1. Why do many AI pilots fail to scale?

2. List four components of LLMOps.

3. Name three adoption or value metrics for an enterprise copilot.

🔬 CONCEPT DEEP DIVE: AI for Enterprise

🔷 AI strategy links use cases to business goals, prioritised by value, feasibility and risk.

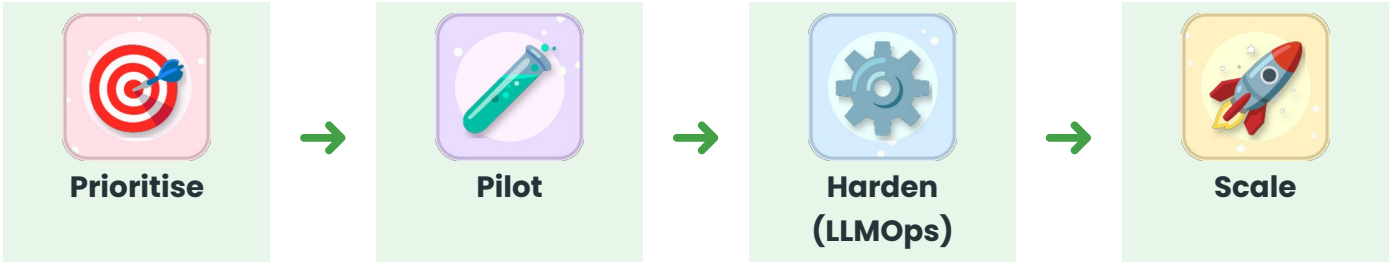
🔷 Data foundations: governed data platforms, quality pipelines, access controls and catalogues.

🔷 LLMOps: prompt and model versioning, evaluation suites, monitoring, cost control and rollback.

🔷 Security & compliance: identity integration, data-loss prevention, vendor reviews, audit logs and regulation.

🔷 Adoption: change management, training champions, measuring KPIs, ROI and user satisfaction.

🔄 How it works



💻 Prioritising use cases

```
cases = {'copilot': (5, 3, 2), 'fraud': (4, 4, 4), 'chat': (3, 5, 1)}
# (value, feasibility, risk)
score = {k: v * 0.5 + f * 0.3 - r * 0.2 for k, (v, f, r) in
cases.items()}
print(max(score, key=score.get))   # copilot 3.0
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Pilot That Couldn't Fly

Characters: *Meera (MBA-track intern), CIO Mr. Banerjee, HR head Ms. Pillai.*

SCENE 1

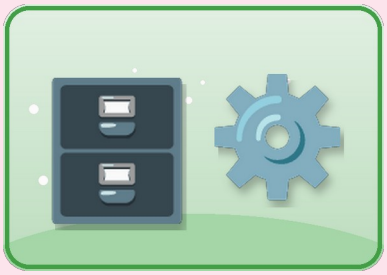
During her final-year internship at a 6,000-employee firm, Meera joined a pilot of an HR copilot. Fifty users loved it: answers about leave, travel and benefits came instantly. Everyone expected a quick company-wide launch.

Pause & predict: Why can a successful pilot be misleading?

SCENE 2

Scaling stalled. Policies differed across regions, some documents were outdated, the copilot ignored access permissions for sensitive files and there was no plan for monitoring. Legal raised concerns about data protection.

Pause & predict: Which enterprise pillars were missing?



SCENE 3

Mr. Banerjee formed a team: HR cleaned and versioned policies, IT connected the copilot to the identity system, Meera built an evaluation set of 150 questions and a dashboard tracking accuracy, usage and cost.

Pause & predict: How does each workstream prepare for scale?

SCENE 4

Three months later the copilot launched company-wide with champions in each department. HR queries to staff fell by forty percent. Ms. Pillai presented results honestly, including two failure cases that had been fixed.

⏸ **Pause & predict:** Why is honest reporting of failures valuable?

🌟 The big idea

Enterprise AI scales on data, security, operations and people, not on pilot excitement alone.

💬 Discuss with your classmates

1. What would you check before scaling an AI pilot in a company?

2. How should employees be trained to use AI tools responsibly?

SCENARIO: AI for Enterprise

Read the scenario. Think about the choices made at each step.

1

Meera interns in a 6,000-staff firm's AI team.

2

A copilot pilot with 50 users is a hit.

3

Scaling stalls: messy data, access rules, no training.

4

A data clean-up and training plan unlocks rollout.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Enterprise use cases

Contract analysis

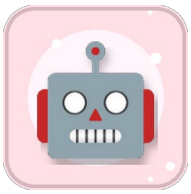
Fraud detection

Service copilots

Supply forecasting

HR knowledge bots

Quality inspection



BOLT SAYS:

Pilots are easy. Scaling to 10,000 staff needs data, security and training!

FUN FACT!

JPMorgan said its COiN tool saved about 360,000 hours a year of contract-review work.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: JPMorgan's Contract Intelligence

CASE FILE

In 2017, JPMorgan Chase described COiN (Contract Intelligence), a machine-learning system that reviewed commercial loan agreements. The bank said work that had consumed about 360,000 hours of lawyers' and loan officers' time each year could be done in seconds, with fewer errors. The project became a widely cited example of document AI delivering measurable enterprise value.

🔍 Analyse the case

1. Why was contract review a good first use case?

2. Which KPIs show the value?

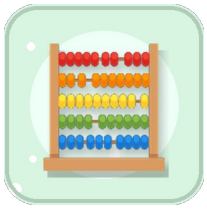
3. What should happen to staff time that is freed up?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Pilot-to-Scale Plan

You need:

Use-case cards



Calculator

Teams of 4

Steps:

1.
   Pick a use case for a 5,000-staff firm.
2.
   Set 3 KPIs and a baseline.
3.
   Plan a 6-week pilot with 50 users.
4.
   List 4 needs before scaling up.

THINK & TALK

What kills most pilots before scale?

Record your observations & results:

💻 CODE LAB: Pilot ROI

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
users, hours, rate, weeks = 50, 2, 1500, 48
benefit = users * hours * rate * ____
cost = 800000
print(benefit, round((benefit - cost) / cost * 100))
```

COMPLETE **T1. Fill the blank with the number of weeks.**

PREDICT **T2. Benefit and ROI %?**

MODIFY **T3. Scale to 1,000 users with cost ₹40 lakh.**

EXTEND **T4. Add a 70% adoption rate.**

WORKSHEET: AI for Enterprise (Part A)

Name: _____ Date: _____

1. Match the pillar to an example.

Strategy ● ● Monitoring & rollback

Data ● ● AI use-case roadmap

LLMOps ● ● Risk review board

Governance ● ● Clean, shared data lake

2. Pilot ROI. Use the word box.

WORD BOX: 48 ✿ 500 ✿ 1100

50 users save 4 h/month at ₹2,000/h = ₹____ lakh/yr

Pilot cost ₹8 lakh/yr → ROI = ____%

Scale ×10 users, cost ×5 → ROI = ____%

WORKSHEET: AI for Enterprise (Part B)

Name: _____ Date: _____

1. Pilot or production need? Tick ✓.

	Pilot	Production
50 test users		
Single sign-on & audit logs		
Manual feedback forms		
24×7 monitoring & rollback		

2. Order the enterprise AI journey. Write 1–4.

Scale to all staff ☐

Pick use case & KPIs ☐

Harden: security, LLMOps ☐

Run a pilot ☐

PUZZLE & DEFINE: AI for Enterprise

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

LLMOps

Meaning: _____

Example: _____

Pilot

Meaning: _____

Example: _____

Scale-up

Meaning: _____

Example: _____

? QUIZ TIME: AI for Enterprise

Circle the right answer!

Q1. Enterprise AI should start from...

a) Strategy

b) A tool sale

c) Logo

Q2. Running LLMs in production:

a) UX

b) LLMOps

c) PR

Q3. A small first trial is a...

a) Launch

b) Finish

c) Pilot

Q4. COiN reviewed...

a) Contracts

b) Songs

c) Weather

Q5. Value is proven with KPIs and...

a) Hope

b) ROI

c) Hype

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI for Enterprise

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Enterprise AI should start from strategy.		
A successful pilot guarantees scale.		
LLMOps includes monitoring and rollback.		
Audit logs matter in production.		
Each team should buy tools with no review.		
COiN reviewed commercial loan agreements.		

HOME CONNECT

Ask someone in a company which AI tools they are allowed to use and why. Note their rules.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI for Enterprise

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why do many AI pilots fail to scale?**

COMPARE **Q2. Compare a pilot and production rollout.**

APPLY **Q3. Saves 2 h/week for 1,000 staff at ₹1,500/h. Yearly (48 wks)?**

EVALUATE **Q4. Should every team pick its own AI vendor?**

CREATE **Q5. List 5 items in an enterprise AI policy.**

📝 EXAM PREP: AI for Enterprise

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name four enterprise AI pillars.

2. Why do pilots fail to scale?

3. What is LLMOps?

4. Give two enterprise AI KPIs.

PART B • Long answer (5 marks)

5. Describe how an enterprise moves from AI pilot to organisation-wide value safely.

TOPIC 6 • MONTH 3 • AI APPLICATIONS

AI for Start-ups

LET'S LEARN

- Start with a painful problem, not a model.
- Moats: data, distribution, domain, trust.
- Unit economics must cover inference costs.

Start-up building blocks

Problem

MVP

Moat

Unit economics

KEY TERMS

Moat Lasting advantage

PMF Product-market fit

CAC Cost to get customer

📖 CONCEPT EXPLAINED: AI for Start-ups

💡 In this chapter

AI has lowered the cost of building software products, and many students now consider founding start-ups. Yet the same accessibility means competitors can copy features quickly. This chapter covers how AI start-ups find real problems, build defensible advantages, manage the unusual economics of AI products, raise funding and stay honest about what their technology does.

😣 1. Problems, fit and focus

Successful start-ups solve painful, frequent, valuable problems for a specific group of customers. Problem–solution fit means customers confirm the problem matters and that your approach helps. Product–market fit is stronger: customers keep using the product, pay for it and recommend it, often visible as high retention and organic growth.

AI start-ups face a particular temptation: starting from impressive technology and searching for a problem. Disciplined founders begin with customer discovery, narrow their focus to one segment and one job-to-be-done, and use AI only where it clearly improves on alternatives.

🤔 PAUSE & THINK

Write a one-sentence problem statement for an AI start-up idea: who has the problem, how often, and what it costs them today.

🏰 2. Moats beyond the model

Because powerful models are widely available through APIs and open-source releases, 'we use AI' is not a competitive advantage. Durable moats include proprietary data that improves the product as usage grows (data network effects), deep integration into customers' workflows, distribution channels and partnerships, domain expertise and regulatory approvals, and trust built through reliability and responsible practice.

In India, opportunities exist where local languages, informal markets, public digital infrastructure and sector-specific knowledge create needs that global products serve poorly. Health screening, agriculture, vernacular voice interfaces and small-business tools are examples where domain depth can matter more than model size.

WORKED EXAMPLE

Moat audit: Data loop? (each use improves accuracy) • Workflow lock-in? (integrated with customers' systems) • Distribution? (partnerships with banks/states) • Trust? (certifications, audits) • If none, a competitor can copy you in weeks.

3. Unit economics of AI products

AI products carry variable costs that traditional software lacks: every user request consumes inference compute. Gross margin = (price - cost of serving) ÷ price must remain healthy as usage grows. Founders track customer acquisition cost (CAC), lifetime value (LTV), LTV:CAC ratio (often targeted at 3 or more), payback period and monthly burn rate against runway.

Pricing models include subscriptions, usage-based pricing that tracks costs, outcome-based fees and enterprise contracts. Engineering choices, such as smaller models, caching and routing, directly affect margins, so product, engineering and finance decisions are tightly linked in AI start-ups.

WORKED EXAMPLE

Price ₹500/month; inference ₹200; hosting & support ₹50 → gross margin 50%. CAC ₹3,000; monthly profit per customer ₹250 → payback 12 months. Halving inference cost lifts margin to 70% and payback to about 8.6 months.

4. Funding, ecosystem and integrity

Start-ups move through stages: bootstrapping or grants, pre-seed and seed funding from angels or incubators, then venture rounds as traction grows. India's ecosystem offers college incubators, government recognition schemes, accelerators and investor networks. Founders should understand dilution, term sheets and the obligations that come with outside money.

Integrity is a strategic asset. 'AI-washing', overstating what AI does or hiding heavy human involvement, may attract attention briefly but damages trust with customers, investors and regulators, and several high-profile collapses have followed exaggerated claims. Honest metrics, clear disclosure of human-in-the-loop work and compliance with data and sector regulations build durable companies.

TRY IT

Complete a lean canvas for an AI start-up idea, including unit economics with inference costs and one honest statement of what the AI does and does not do.

✅ Key takeaways

- Start from painful problems; seek problem–solution then product–market fit.
- Moats come from data loops, workflow integration, distribution, domain depth and trust.
- AI products have variable inference costs; track margin, CAC, LTV and payback.
- Engineering efficiency directly shapes start-up economics.
- Raise funding thoughtfully and avoid AI-washing; integrity builds durable ventures.

❓ Quick check: answer in one or two lines.

1. What is the difference between problem–solution fit and product–market fit?

2. Why isn't 'we use a powerful model' a moat?

3. Compute gross margin for price ₹800 with ₹300 serving cost.

🔬 CONCEPT DEEP DIVE: AI for Start-ups

🔷 Problem–solution fit, then product–market fit: users return, pay and recommend without heavy pushing.

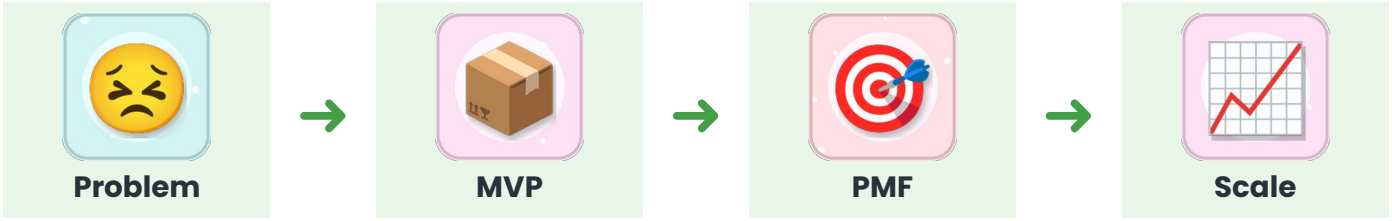
🔷 Moats beyond the model: proprietary data loops, distribution, domain expertise, workflow integration and trust.

🔷 Unit economics: CAC, LTV, gross margin after inference and hosting costs, payback period and burn rate.

🔷 Business models: SaaS subscriptions, usage pricing, outcome-based fees, B2B2C partnerships and government contracts.

🔷 Ecosystem: incubators, accelerators, angel and VC stages, Startup India recognition; honesty in claims avoids 'AI-washing'.

🔄 How it works



💻 Unit economics check

```
price, infer, other = 500, 200, 50
margin = (price - infer - other) / price
ltv, cac = 9000, 3000
print(round(margin * 100), ltv / cac)   # 50, 3.0
```

✍️ In your own words: explain this topic to a Class 10 student.

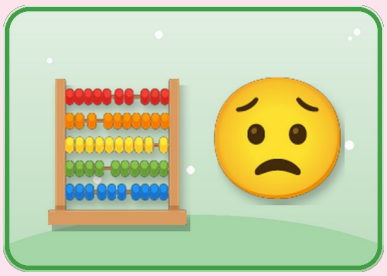
AI STORY: Six Rupees a Call

Characters: *Rohan (B.Tech Agri Engg, final year), co-founder Lakshmi, investor Mr. Shah.*

SCENE 1

Rohan and Lakshmi launched a voice assistant that answered farmers' crop questions in Kannada. Farmers liked it, and usage grew quickly. But each call cost ₹6 in speech and model fees, while they charged ₹5.

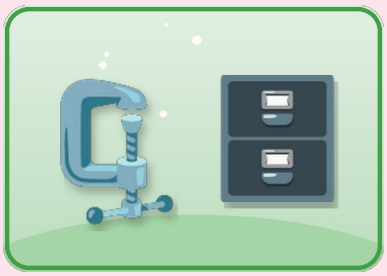
Pause & predict: What happens to this start-up as usage grows?



SCENE 2

Mr. Shah, an angel investor, asked about unit economics. Their gross margin was negative, CAC was rising and runway was eight months. 'Growth that loses money on every customer isn't growth,' he said.

Pause & predict: Which metrics worried the investor?



SCENE 3

They switched to a smaller speech model, cached seasonal advice, and sold seasonal subscriptions to farmer cooperatives rather than charging per call. Their data, thousands of local crop questions, became a moat for better answers.

Pause & predict: How did engineering and pricing changes work together?

SCENE 4

Margins turned positive, cooperatives signed on, and 2,000 farmers subscribed. In their pitch, they described exactly what the AI did and where agronomists reviewed answers. Mr. Shah invested, citing their honesty as a key reason.

⏸ **Pause & predict:** Why did honesty strengthen their pitch?

🌟 The big idea

AI start-ups win with real problems, healthy unit economics, true moats and honest claims.

💬 Discuss with your classmates

1. Which AI start-up idea could work in your region?

2. How should founders describe human involvement in AI products?

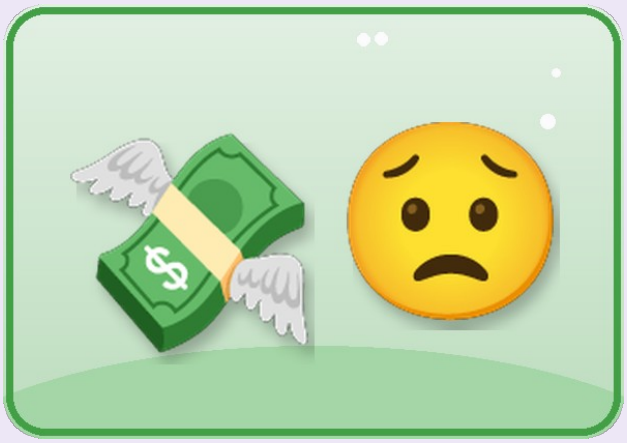
SCENARIO: AI for Start-ups

Read the scenario. Think about the choices made at each step.

1

Rohan's team builds a voice bot for farmers.

2

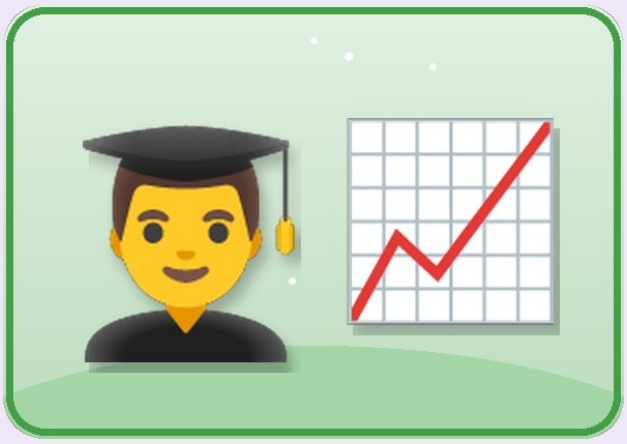


Each call costs ₹6 in AI fees; they charge ₹5.

3

They switch models, cache advice, price per season.

4



Margin turns positive; 2,000 farmers sign up.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: AI start-up spaces

Health screening

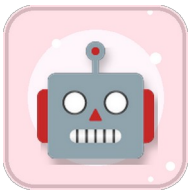
Farm advisory

Legal tech

Ed-tech tutors

Voice AI

Factory vision



BOLT SAYS:

Investors don't fund models. They fund painful problems solved well!

FUN FACT!

Bengaluru start-up Niramai uses AI and thermal imaging for radiation-free breast screening.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Niramai: AI Screening in Bengaluru

CASE FILE

Niramai Health Analytix, founded in Bengaluru in 2016, developed Thermalytix, which applies AI to thermal images to help screen for breast cancer. The test is non-contact and radiation-free, making it suitable for screening camps in places without mammography machines. The company has worked with hospitals and public-health programmes, illustrating how a deep-tech start-up can target an under-served need.

🔍 Analyse the case

1. What painful problem does Niramai address?

2. What could its moat be?

3. Which regulatory steps does a health AI start-up face?

⚖️ My verdict: Was AI used well here? Why?

🧪 LAB ACTIVITY: Lean Canvas Sprint

🎒 You need:

Lean canvas

Sticky notes

Teams of 4

👣 Steps:

1. Write the problem and who has it.
2. Define the MVP and key metric.
3. Price it; list costs incl. inference.
4. Name one moat competitors can't copy.

THINK & TALK

Is your moat real or just a model?

🖍️ Record your observations & results:

💻 CODE LAB: Unit Economics

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
price, infer, host = 500, 200, 50
margin = (price - infer - host) / ____
ltv, cac = 9000, 3000
print(round(margin * 100), ltv / cac)
```

COMPLETE **T1. Fill the blank to divide by price.**

PREDICT **T2. Margin % and LTV:CAC?**

MODIFY **T3. Inference falls to ₹100. New margin?**

EXTEND **T4. Compute CAC payback months at ₹250 profit/month.**

WORKSHEET: AI for Start-ups (Part A)

Name: _____ Date: _____

1. Match the term to its meaning.

CAC ●

● Monthly cash spent

LTV ●

● Cost to win a customer

PMF ●

● Strong market demand

Burn rate ●

● Value from a customer

2. Unit economics. Use the word box.

WORD BOX: 3 ✿ 60 ✿ 12

LTV ₹9,000 ÷ CAC ₹3,000 = ____

Price ₹500, inference ₹200 → margin ____%

₹60 lakh ÷ ₹5 lakh burn = ____ months

WORKSHEET: AI for Start-ups (Part B)

Name: _____ Date: _____

1. Payback and runway. Use the word box.

WORD BOX: 6 ✿ 10 ✿ 20

CAC ₹3,000 ÷ ₹500 monthly profit = ____ months

₹1 crore ÷ ₹10 lakh burn = ____ months

Margin on ₹1,000 with ₹800 cost = ____%

2. Real moat or weak moat? Tick ✓.

	Real	Weak
Unique data from 10,000 clinics		
Uses a popular public model		
Exclusive state partnership		
Nice logo		

PUZZLE & DEFINE: AI for Start-ups

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Moat

Meaning: _____

Example: _____

PMF

Meaning: _____

Example: _____

CAC

Meaning: _____

Example: _____

? QUIZ TIME: AI for Start-ups

Circle the right answer!

Q1. A lasting advantage is a...

a) Moat

b) Balloon

c) Slogan

Q2. LTV ÷ CAC of 3 means...

a) Losing money

b) Healthy economics

c) Random

Q3. AI start-ups must cover...

a) Logo cost

b) Music

c) Inference cost

Q4. Strong demand for your product:

a) PMF

b) CAC

c) Burn

Q5. Niramai screens for...

a) Tooth decay

b) Breast cancer

c) Eyesight

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI for Start-ups

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Start-ups should begin with a painful problem.		
	Using a popular model is a strong moat.		
	LTV ÷ CAC above 3 is usually healthy.		
	Inference costs affect gross margin.		
	Exaggerating AI claims is harmless.		
	Niramai uses thermal imaging.		

HOME CONNECT

Interview a local business owner. Find one painful problem AI might solve and how they'd pay for it.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI for Start-ups

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why is 'we use AI' not a moat?**

COMPARE **Q2. Compare SaaS and usage-based pricing.**

APPLY **Q3. Price ₹500, inference ₹200, other ₹50. Margin?**

EVALUATE **Q4. Is fast user growth with negative margin fine?**

CREATE **Q5. Write a one-line value proposition for an AI start-up.**

📝 EXAM PREP: AI for Start-ups

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is product–market fit?

2. Define CAC and LTV.

3. Name three AI start-up moats.

4. What is AI-washing?

PART B • Long answer (5 marks)

5. Build the business case for an AI start-up, covering problem, moat, unit economics and responsible claims.

UNIT REVIEW: AI SOLVE

Name: _____ Date: _____

Circle the right answer!

Q1. Enterprise AI starts from...

a) Strategy | b) Tools | c) Logos

Q2. Running LLMs in production:

a) PR | b) LLMOps | c) UX

Q3. LTV ₹9,000, CAC ₹3,000 →

a) 0.33 | b) 6 | c) 3

Q4. Price ₹500, costs ₹250 → margin

a) 50% | b) 25% | c) 250%

Q5. 'We use AI' alone is a...

a) Strong moat | b) Weak moat | c) Law

Q6. COiN reviewed...

a) Songs | b) Weather | c) Loan contracts

WORD SEARCH: AI SOLVE

Find these words. They go across → or down ↓.

LLMOPS

PILOT

SCALE-UP

MOAT

PMF

CAC

L	A	P	I	L	O	T	I	W	R
P	F	L	T	U	U	D	M	G	E
M	R	U	N	W	A	Y	A	U	V
F	I	K	D	C	R	M	R	R	E
C	M	C	O	E	A	O	G	K	N
O	O	D	B	W	W	A	I	B	U
N	U	L	C	A	C	T	N	T	E
I	S	L	L	M	O	P	S	H	B
K	S	C	A	L	E	U	P	A	C
D	W	T	F	U	N	D	I	N	G

I finished AI SOLVE! Teacher's sign: _____

MONTH 4

AI RISE

Skills • Careers • Future

Area: AI & Career

In this unit, you will explore:

Topic 7

From Campus to AI Career

Topic 8

Idea to AI Venture

TOPIC 7 • MONTH 4 • AI & CAREER

From Campus to AI Career

LET'S LEARN

- Paths: job, higher studies, research, start-up.
- Exams: GATE, GRE, CAT and more open doors.
- Compare offers on growth, not just salary.

Four paths after graduation

Industry job

Higher studies

Research

Start-up

KEY TERMS

GATE PG entrance exam

GRE Graduate test abroad

Offer letter Job terms document

📖 CONCEPT EXPLAINED: From Campus to AI Career

💡 In this chapter

Final year brings big decisions: take a job, pursue higher studies, prepare for competitive exams, enter research or start a venture. Each path suits different goals and circumstances, and choices can be revisited later. This chapter maps the main options after an undergraduate degree, explains key exams and processes, and offers a structured way to compare offers and avoid common traps.

🧭 1. Mapping the paths

Industry jobs offer immediate experience, income and mentorship in roles such as data analyst, ML engineer, software developer, product analyst or AI consultant. Higher studies, such as an M.Tech or M.S. in AI or data science, deepen technical expertise; an MBA can lead to product, consulting or management roles. Research careers typically involve a PhD and work in university or industry labs. Entrepreneurship and public-sector technology roles are further options.

These paths are not permanent tracks. Many professionals work for a few years before higher studies, move from industry to research, or start ventures after gaining experience. The best first step depends on your goals, finances, interests and the opportunities available to you now.

🤔 PAUSE & THINK

List your top two paths. For each, write what you would gain in two years and what you would give up.

📝 2. Entrance exams and applications

In India, GATE (Graduate Aptitude Test in Engineering) is used for admission to many M.Tech programmes and by some public-sector undertakings for recruitment; from 2024 it includes a Data Science and Artificial Intelligence (DA) paper. CAT is used by many management institutes. For study abroad, many universities consider tests such as GRE, and English-proficiency tests such as TOEFL or IELTS, along with statements of purpose and recommendation letters.

Requirements and dates change, so always rely on official exam and university websites. Plan backwards from deadlines: preparation months, test dates, score release, application windows and scholarships. Strong applications combine test scores with projects, research experience and a clear, honest statement of goals.

✏️ **WORKED EXAMPLE**

Timeline (illustrative): June–August: decide path, start preparation • September–October: register for exams • November–February: take exams, prepare applications • January–April: interviews and offers • May–July: decisions, visas or joining formalities.

⚖️ 3. Comparing offers wisely

Offers differ in more than salary. Consider the learning environment and mentorship, the actual role and technologies, team quality, location and cost of living, compensation structure (fixed pay, variable pay, stock, benefits), bonds or service agreements, notice periods and growth opportunities. A slightly lower salary with strong mentorship often yields better long-term outcomes.

A weighted scoring model makes trade-offs explicit: choose criteria, assign weights reflecting your priorities, score each offer and compare. The model does not decide for you, but it reveals what you value and prompts useful questions for recruiters and seniors.

✏️ **WORKED EXAMPLE**

Weights: learning 0.4, pay 0.3, fit 0.3. Offer A (9, 6, 8) → 78. Offer B (6, 9, 6) → 69. Offer C (9, 8, 8) → 84. If pay mattered most (0.2, 0.5, 0.3), A → 72, B → 75.

🚩 4. Avoiding traps and staying adaptable

Job scams target final-year students with fake offers that demand 'registration', 'training' or 'security' fees, pressure quick acceptance or use unofficial email addresses. Genuine employers do not charge candidates to be hired. Verify companies through official websites and placement cells, and never share sensitive documents or payments with unverified recruiters.

AI careers change quickly, so adaptability matters more than any single choice. Keep learning, build a portfolio, maintain relationships with mentors and peers, and revisit your plan each year. Your first role is a launchpad: choose one where you will learn, contribute and grow.

TRY IT

Build an offer-comparison sheet with at least five weighted criteria. Score two real or sample offers and discuss your result with a senior or mentor.

Key takeaways

- Paths after UG: jobs, higher studies, research, entrepreneurship, public sector; they can change later.
- GATE (including the DA paper), CAT and tests for study abroad open specific routes.
- Plan backwards from official deadlines and build strong applications.
- Compare offers on learning, role, team, location, pay structure and growth.
- Avoid scams; genuine employers never charge fees. Stay adaptable.

Quick check: answer in one or two lines.

1. Name the GATE paper introduced in 2024 for AI.

2. List five criteria for comparing job offers.

3. Give two warning signs of a job scam.

🔬 CONCEPT DEEP DIVE: From Campus to AI Career

🔷 Post-UG paths: industry roles, higher studies (M.Tech, MS, MBA), research (PhD, labs), entrepreneurship and public-sector technology.

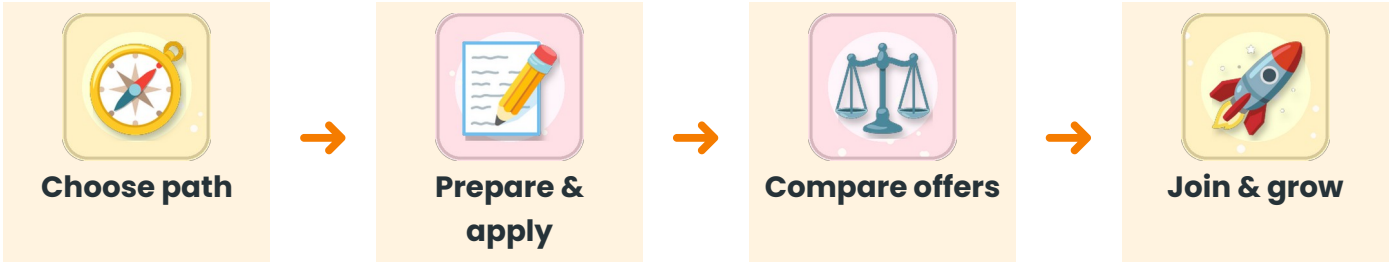
🔷 Entrance exams: GATE (M.Tech, some PSU jobs; DA paper since 2024), GRE/TOEFL/IELTS for many universities abroad, CAT for MBA.

🔷 Placement process: pre-placement talks, tests, interviews, offers; know bonds, notice periods and joining terms.

🔷 Offer evaluation: learning, mentorship, role scope, team, location, compensation structure (fixed vs variable) and growth.

🔷 Avoid scams: no genuine employer asks for fees; verify companies and offers through official channels.

🔄 How it works



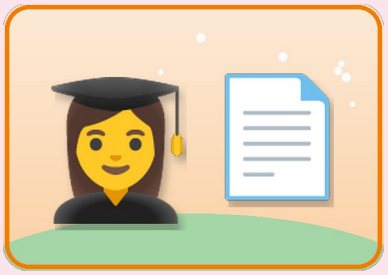
💻 Weighted offer scoring

```
w = {'learn': 0.4, 'pay': 0.3, 'fit': 0.3}
a = {'learn': 9, 'pay': 6, 'fit': 8}
score = sum(w[k] * a[k] for k in w)
print(round(score * 10))   # out of 100
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Two Offers and a Score

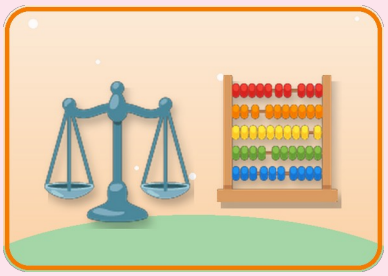
Characters: *Ananya (B.Tech CSE, final year), placement officer Mr. Varma, mentor Priya.*



SCENE 1

By January, Ananya had two offers and a good GATE score. Offer A, from a mid-sized AI company, paid less but promised mentorship on real ML projects. Offer B, from a large firm, paid more for a support role. Her GATE rank could also get her into an M.Tech programme.

Pause & predict: What factors should Ananya consider beyond salary?



SCENE 2

Priya suggested a weighted scorecard. Ananya chose learning (0.4), pay (0.3) and fit (0.3). Offer A scored 78, Offer B 69. For the M.Tech, she listed costs, scholarships and the chance to specialise in NLP.

Pause & predict: How does a scorecard help without making the decision for her?

SCENE 3

Meanwhile, an email offered her a 'guaranteed AI job' abroad after paying a ₹25,000 'processing fee'. Mr. Varma confirmed it was a known scam. 'Real employers never charge you to hire you,' he warned the whole batch.

Pause & predict: Which red flags identify the scam?

SCENE 4

Ananya accepted Offer A and deferred higher studies, planning to apply for an M.Tech after two years of experience. Her first project, a Tamil-language search feature, confirmed her interest in NLP.

⏸ **Pause & predict:** Why might working first make her later studies more valuable?

🌟 The big idea

Choose your first step by growth and fit, verify every offer, and remember that paths can evolve.

💬 Discuss with your classmates

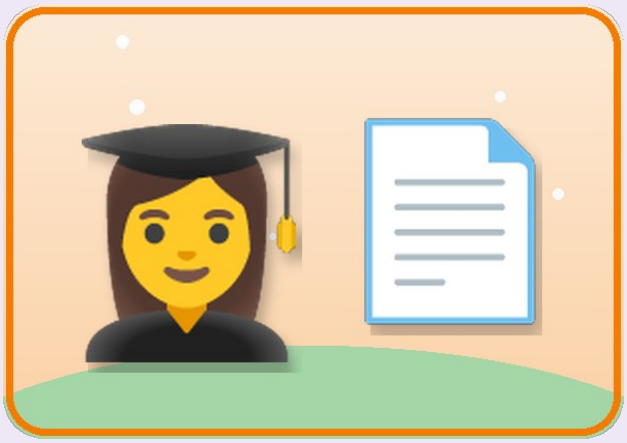
1. How would you weight learning, pay and fit for yourself?

2. What would you tell juniors about job scams?

SCENARIO: From Campus to AI Career

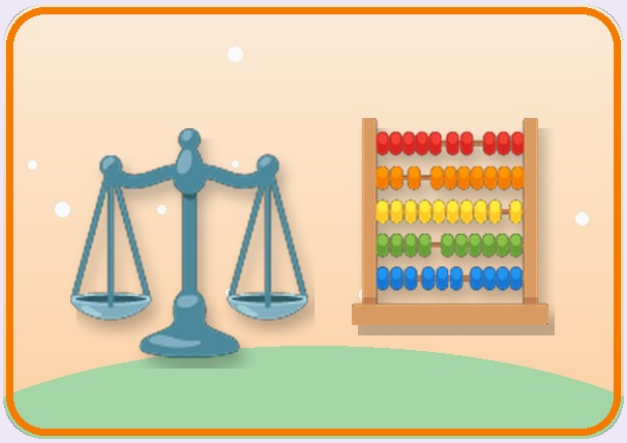
Read the scenario. Think about the choices made at each step.

1



Ananya has 2 offers and a GATE score.

2



She scores options on learning, pay and fit.

3

A mentor helps her see long-term growth.

4

She picks the role with strong mentorship.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Where graduates go

Product companies

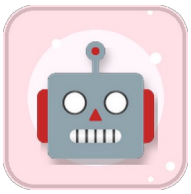
IT services

M.Tech / MS

MBA analytics

Public sector tech

PhD labs



BOLT SAYS:

Your first role is a launchpad, not a life sentence. Pick where you'll learn most!

FUN FACT!

GATE 2024 added a new paper: Data Science and Artificial Intelligence (DA).

Reflect: where have YOU seen this in real life? Note it down.

CASE STUDY: GATE Adds Data Science & AI

CASE FILE

For GATE 2024, India's Graduate Aptitude Test in Engineering introduced a new paper, Data Science and Artificial Intelligence (DA). It covers probability and statistics, linear algebra, calculus and optimisation, programming, data structures and algorithms, databases, machine learning and AI. The addition reflected the growing number of postgraduate programmes and roles in the field.

Analyse the case

1. Which subjects would you strengthen for the DA paper?

2. How might a GATE score help your career?

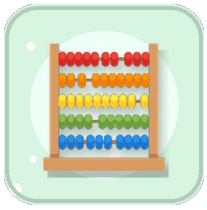
3. Compare GATE preparation with building a portfolio.

My verdict: Was AI used well here? Why?

🧪 LAB ACTIVITY: Offer Comparison Lab

🎒 You need:

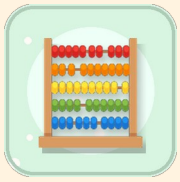
3 sample offers



Spreadsheet

Pairs

👣 Steps:

1. Read 3 offers: salary, role, location, bond.
2. Weight: learning, pay, mentors, fit.
3.  Score each offer out of 100.
4. Defend your choice in 1 minute.

THINK & TALK

Did the highest salary win? Why?

🖍️ Record your observations & results:

💻 CODE LAB: Offer Scorer

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
w = {'learn': 0.4, 'pay': 0.3, 'fit': 0.3}
offers = {'A': (9, 6, 8), 'B': (6, 9, 6)}
for name, s in offers.items():
    print(name, round(sum(a * b for a, b in zip(w.____(), s)) *
10))
```

COMPLETE **T1. Fill the blank to get the weights.**

PREDICT **T2. Scores for A and B?**

MODIFY **T3. Raise 'pay' weight to 0.5 and lower 'learn' to 0.2.**

EXTEND **T4. Add offer C (9, 8, 8).**

WORKSHEET: From Campus to AI Career (Part A)

Name: _____ Date: _____

1. Match the exam to the path.

GATE ● ● MBA in India

GRE ● ● M.Tech / PSU jobs

CAT ● ● English for study abroad

IELTS/TOEFL ● ● MS abroad

2. Good or risky offer sign? Tick ✓.

	Good	Risky
Named mentor and training		
Fee to 'confirm' the job		
Clear written role & pay		
'Accept in 1 hour' pressure		

WORKSHEET: From Campus to AI Career (Part B)

Name: _____ Date: _____

1. Offer scores (weights 0.4/0.3/0.3). Use the word box.

WORD BOX: 78 ✿ 69 ✿ 84

A: learn 9, pay 6, fit 8 → ____

B: learn 6, pay 9, fit 6 → ____

C: learn 9, pay 8, fit 8 → ____

2. Match the path to a first step.

Job ●

Higher studies ●

Research ●

Start-up ●

● **Email a professor about a project**

● **Polish CV & apply**

● **Interview 10 customers**

● **Register for GATE/GRE**

PUZZLE & DEFINE: From Campus to AI Career

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

GATE

Meaning: _____

Example: _____

GRE

Meaning: _____

Example: _____

Offer letter

Meaning: _____

Example: _____

? QUIZ TIME: From Campus to AI Career

Circle the right answer!

Q1. GATE's new 2024 paper was...

a) Data Science & AI

b) Fine arts

c) Sports

Q2. GRE is usually for...

a) MBA India

b) MS abroad

c) Civil services

Q3. A fee to 'confirm' a job is...

a) Normal

b) A bonus

c) A red flag

Q4. First jobs are best judged on...

a) Growth & learning

b) Office colour

c) Snacks

Q5. A research path often leads to...

a) Retail

b) PhD

c) Singing

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: From Campus to AI Career

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	GATE 2024 added a Data Science & AI paper.		
	Genuine employers ask for joining fees.		
	GRE is used by many universities abroad.		
	Offers should be compared on more than pay.		
	Research paths never need a PhD.		
	Mentorship affects early-career growth.		

HOME CONNECT

Talk to 2 seniors on different paths. Note one thing each wishes they had known in final year.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: From Campus to AI Career

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why compare offers on growth, not just pay?**

COMPARE **Q2. M.Tech in India vs MS abroad.**

APPLY **Q3. Weights 0.4/0.3/0.3; scores 9/6/8. Score out of 100?**

EVALUATE **Q4. Should you pay a fee to secure a job?**

CREATE **Q5. Plan your 12 months after graduation.**

📝 EXAM PREP: From Campus to AI Career

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name four paths after UG.

2. What is the GATE DA paper?

3. How should offers be compared?

4. Give two job-scam red flags.

PART B • Long answer (5 marks)

5. Plan a final-year career strategy covering paths, exams, applications and offer evaluation.

TOPIC 8 • MONTH 4 • AI & CAREER

Idea to AI Venture

LET'S LEARN

- Validate the problem with real customers first.
- Build an MVP, test, pivot or persevere.
- Register, protect IP, and never overclaim AI.

Idea-to-venture steps

Customer discovery

MVP

Pitch

Register

KEY TERMS

Pivot Change direction

Pitch Short investor talk

Term sheet Investment terms

📖 CONCEPT EXPLAINED: Idea to AI Venture

💡 In this chapter

Many great companies began as student projects. Turning an AI idea into a venture requires validating that a real problem exists, building and testing a minimum viable product, learning when to pivot, pitching convincingly, handling legal basics and maintaining integrity. This chapter guides you through that journey.

🗣️ 1. Customer discovery

Customer discovery means interviewing potential users to understand their problems, current workarounds and willingness to pay before building much. Good interviews ask about specific past experiences: 'Tell me about the last time you filed GST returns. What was hardest? What did you use? How much did it cost?' Such questions reveal real behaviour rather than polite opinions.

Avoid leading or hypothetical questions like 'Would you use my app?', which tend to produce false encouragement. Interview enough people in a defined segment, often 15 to 30, record notes systematically and look for repeated pains, current spending and strong emotions. Evidence, not enthusiasm, should drive the next step.

✏️ WORKED EXAMPLE

Interview tally (20 shopkeepers): GST filing 12 (60%) • stock tracking 5 (25%) • billing 3 (15%). Insight: GST filing is the most painful, and most currently pay an accountant ₹1,000–2,000 per month.

📦 2. MVPs, experiments and pivots

A minimum viable product tests the riskiest assumption with the least effort. It might be a landing page, a manual service delivered behind an AI-looking interface, a simple prototype or a pilot with a few customers. Define the hypothesis and success metric in advance, for example 'At least 30% of pilot shops will upload invoices weekly for a month.'

Results lead to three choices: persevere if metrics improve, pivot if evidence contradicts a key assumption (changing customer, problem, solution or business model), or stop. Pivots are normal; many successful companies changed direction after early experiments. The key is learning quickly and cheaply.

🤔 **PAUSE & THINK**

Your MVP gets many sign-ups but almost no repeat use. Should you persevere, pivot or stop? What would you investigate first?

🎤 3. Pitching and funding

A pitch tells a clear, evidence-based story: the problem and who has it, your solution and why now, market size, traction (users, revenue, growth), business model, competition and your advantage, team, and the ask (how much funding and what it will achieve). Investors look for large problems, credible teams, evidence of demand and honest understanding of risks.

Funding sources include personal savings, grants, college incubators, government start-up schemes, angel investors and venture capital. Each brings obligations: equity dilution, reporting, board oversight. Read term sheets carefully and seek advice from mentors or legal experts before signing.

✏️ **WORKED EXAMPLE**

10-slide outline: 1 Problem • 2 Solution • 3 Why now • 4 Market • 5 Product demo • 6 Traction • 7 Business model • 8 Competition & moat • 9 Team • 10 Ask & use of funds.

📜 4. Legal basics and integrity

As ventures grow, founders must handle legal foundations: registering a company (for example a private limited company or LLP in India), a founders' agreement covering roles, equity and vesting, assignment of intellectual property to the company, contracts with customers and vendors, tax registrations and compliance with data-protection and sector regulations.

Integrity protects ventures. Overstating AI capabilities ('AI-washing'), hiding that humans perform key work, inflating metrics or ignoring privacy rules can lead to lost trust, legal action and collapse, as several well-funded companies have shown. Honest communication about what the AI does, its limits and where humans are involved builds lasting relationships with customers and investors.

TRY IT

Conduct five discovery interviews on a problem you care about. Tally the pains, define one MVP experiment with a success metric, and draft a 3-slide pitch: problem, evidence, MVP.

Key takeaways

- Customer discovery asks about past behaviour, not hypothetical approval.
- MVPs test the riskiest assumption; results lead to persevere, pivot or stop.
- Pitches tell an evidence-based story ending with a clear ask.
- Handle registration, founders' agreements, IP assignment and compliance early.
- Avoid AI-washing; honesty about AI and human roles builds trust.

Quick check: answer in one or two lines.

1. Rewrite 'Would you use my app?' as a good discovery question.

2. What are the three possible decisions after an MVP experiment?

3. List four legal foundations for a new venture.

🔬 CONCEPT DEEP DIVE: Idea to AI Venture

🔷 Customer discovery: interview about past behaviour, current workarounds and spending; avoid leading questions.

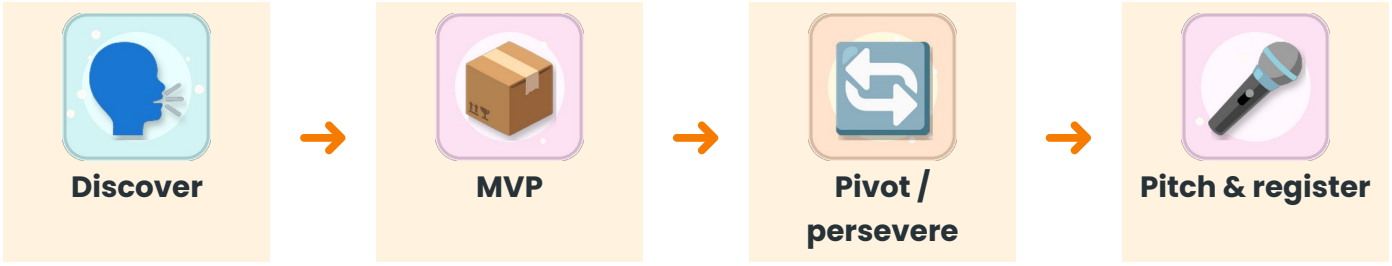
🔷 Lean experiments: state hypotheses, build the smallest test, measure, then persevere, pivot or stop.

🔷 Pitch structure: problem, solution, market, traction, business model, competition, team, ask.

🔷 Legal basics: company registration, founders' agreement, IP assignment, data-protection compliance, contracts.

🔷 Integrity: avoid 'AI-washing' (overstating AI), disclose human-in-the-loop work and report metrics honestly.

🔄 How it works



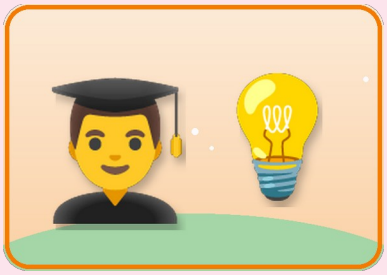
💻 Tallying interview evidence

```
notes = ['gst', 'gst', 'stock', 'gst', 'billing']
from collections import Counter
pains = Counter(notes).most_common(1)
print(pains)   # [('gst', 3)]
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: From Accounts to GST

Characters: *Rohan (B.Com + AI minor, final year), incubator mentor Dr. Mehta, shopkeeper Mr. Ali.*



SCENE 1

Rohan was sure small shops needed an AI accounting app. He spent a month designing features before Dr. Mehta asked a simple question: 'How many shopkeepers have you talked to?' The answer was none.

Pause & predict: Why is building before talking to customers risky?

SCENE 2

Rohan interviewed twenty shopkeepers about their last month of paperwork. Twelve said GST filing was their biggest headache; most paid accountants and still feared penalties. Accounting itself bothered few of them.

Pause & predict: What did the interviews reveal?

SCENE 3

He pivoted. His MVP let shops photograph invoices; an AI extracted details and prepared draft GST summaries, which a partner accountant reviewed. In the pilot, Mr. Ali saved hours and his accountant fees fell.

Pause & predict: Why include an accountant in the MVP?

SCENE 4

In his pitch, Rohan stated clearly that AI drafted and humans verified. The incubator selected his team. Dr. Mehta pointed to a recent start-up collapse: 'Overclaiming AI destroys trust. Your honesty is part of your product.'

⏸ **Pause & predict:** How does honest disclosure strengthen a venture?

🌟 The big idea

Validate problems with real customers, pivot on evidence and describe your AI honestly.

💬 Discuss with your classmates

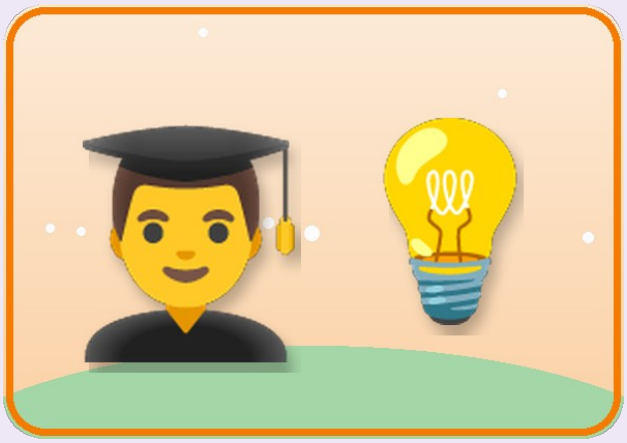
1. What problem around you could you validate with 10 interviews?

2. Where is the line between marketing and AI-washing?

SCENARIO: Idea to AI Venture

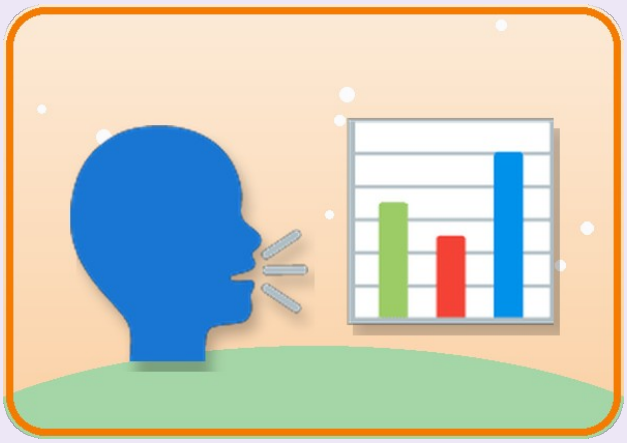
Read the scenario. Think about the choices made at each step.

1



Rohan wants an AI app for small shop accounts.

2

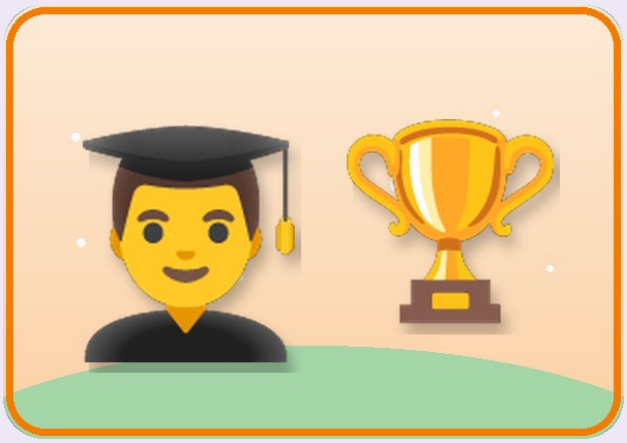


Interviews show the real pain is GST filing.

3

He pivots to a GST helper MVP.

4



The incubator selects his team for funding.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Venture support

College incubators

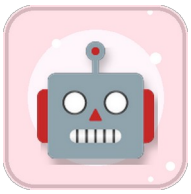
Startup India

Angel investors

Hackathons

Mentor networks

IP filings



BOLT SAYS:

Fall in love with the problem, not your first solution. Pivots are normal!

FUN FACT!

Builder.ai, once valued over US$1 billion, entered insolvency in 2025 amid doubts about its claims.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Builder.ai's Collapse

CASE FILE

Builder.ai, a London-based start-up that promised AI-assisted app development, was once valued at over US$1 billion and backed by major investors. In 2025 it entered insolvency after reports raised doubts about its revenue figures. Years earlier, journalists had also questioned how much of its work depended on human engineers rather than AI. The case became a warning about overstating AI and weak financial controls.

🔍 Analyse the case

1. What is 'AI-washing' and why is it risky?

2. How should start-ups describe human-in-the-loop work?

3. What controls protect investors and customers?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Customer Discovery Day

You need:

Interview script

Notes app

Teams of 3

Steps:

1. Interview 5 people about a problem.
2. Ask about the past, not 'would you buy?'
3. Tally pains, workarounds, spend.
4. Decide: persevere, pivot or stop.

THINK & TALK

What surprised you in the interviews?

Record your observations & results:

💻 CODE LAB: Interview Tally

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  from collections import Counter
2  pains = ['gst', 'stock', 'gst', 'billing', 'gst', 'stock']
3  c = Counter(pains)
4  top, n = c.most_common(____)[0]
5  print(top, round(n / len(pains) * 100))
```

COMPLETE **T1. Fill the blank for the single top pain.**

PREDICT **T2. What is printed?**

MODIFY **T3. Show all pains with percentages.**

EXTEND **T4. Read pains from a CSV.**

WORKSHEET: Idea to AI Venture (Part A)

Name: _____ Date: _____

1. Order the venture steps. Write 1–4.

Pitch to investors ☐

Interview customers ☐

Test MVP metrics ☐

Build an MVP ☐

2. Good or bad discovery question? Tick ✓.

	Good	Bad
Tell me about the last time this happened.		
Would you love my app?		
What do you use today, and its cost?		
Will you buy it if I build it?		

WORKSHEET: Idea to AI Venture (Part B)

Name: _____ Date: _____

1. Interview tally. Use the word box.

WORD BOX: 60 ✿ 25 ✿ 15

12 of 20 say GST → ____%

5 of 20 say stock → ____%

3 of 20 say billing → ____%

2. Order a pitch deck. Write 1–4.

The ask ☐

Problem ☐

Traction ☐

Solution ☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: Idea to AI Venture

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Pivot

Meaning: _____

Example: _____

Pitch

Meaning: _____

Example: _____

Term sheet

Meaning: _____

Example: _____

? QUIZ TIME: Idea to AI Venture

Circle the right answer!

Q1. Changing direction after learning is a...

a) Pivot

b) Freeze

c) Party

Q2. Best discovery questions ask about...

a) The future

b) Past behaviour

c) Colours

Q3. A document of investment terms:

a) Receipt

b) Ticket

c) Term sheet

Q4. Overstating AI capability is...

a) AI-washing

b) Marketing best practice

c) A bonus

Q5. Builder.ai entered insolvency in...

a) 2019

b) 2025

c) 2010

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Idea to AI Venture

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Customer interviews should come early.		
	'Would you buy it?' is the best question.		
	Pivoting is a normal part of start-ups.		
	Founders should sign IP assignment agreements.		
	Overstating AI use is harmless.		
	Builder.ai entered insolvency in 2025.		

HOME CONNECT

Interview 3 people about a daily frustration. Write the top pain and one MVP idea.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Idea to AI Venture

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why ask about past behaviour, not future intent?**

COMPARE **Q2. Pivot vs persevere.**

APPLY **Q3. 12 of 20 interviewees name GST as top pain. %?**

EVALUATE **Q4. Is it fine to say 'fully AI' when humans do half?**

CREATE **Q5. Write the 8 slide titles of a pitch deck.**

📝 EXAM PREP: Idea to AI Venture

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is customer discovery?

2. When should a start-up pivot?

3. List pitch-deck sections.

4. What is AI-washing?

PART B • Long answer (5 marks)

5. Describe the journey from AI idea to registered venture, including validation, legal steps and integrity.

UNIT REVIEW: AI RISE

Name: _____ Date: _____

Circle the right answer!

Q1. GATE added a ... paper in 2024.

a) DS & AI

b) Art

c) Sports

Q2. A fee to get a job offer is...

a) Normal

b) A red flag

c) A gift

Q3. Scores 9/6/8, weights .4/.3/.3 →

a) 70

b) 84

c) 78

Q4. Best discovery questions ask about...

a) The past

b) The future

c) Colours

Q5. Changing direction is a...

a) Freeze

b) Pivot

c) Party

Q6. Overstating AI is called...

a) Honesty

b) Branding

c) AI-washing

WORD SEARCH: AI RISE

🔍 **Find these words. They go across → or down ↓.**

GATE

GRE

PIVOT

PITCH

TERM SHEET

OFFER

C	B	R	E	S	U	M	E	P	I
S	T	E	R	M	S	H	E	E	T
T	P	I	T	C	H	P	R	O	L
E	G	C	O	G	P	I	I	F	F
G	R	E	M	O	U	V	L	F	T
G	L	M	E	N	T	O	R	E	D
B	H	I	U	G	A	T	E	R	W
H	V	I	S	A	D	L	F	K	P
A	B	F	O	U	N	D	E	R	M
C	K	L	A	L	K	L	E	O	W

I finished AI RISE! Teacher's sign: _____

MONTH 5

AI CREATE

Try • Build • Innovate

Area: AI Tools

In this unit, you will explore:

Topic 9

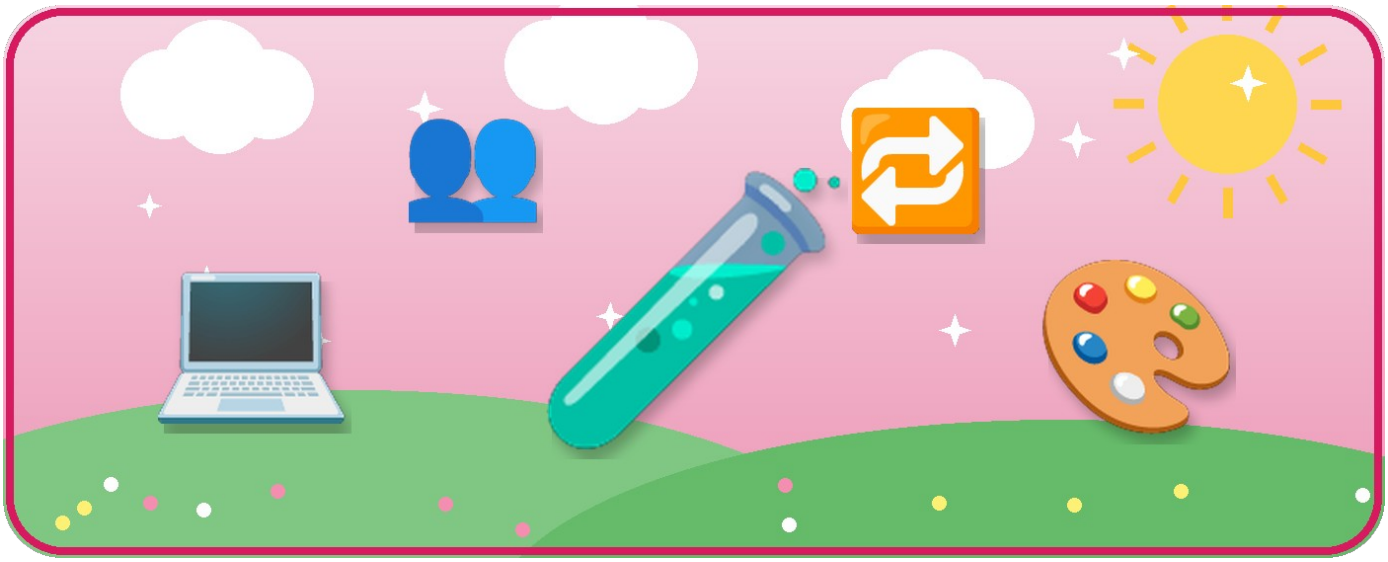
AI Prototype Lab

Topic 10

Agent Builder Studio

TOPIC 9 • MONTH 5 • AI TOOLS

AI Prototype Lab



LET'S LEARN

- Prototype fast: sketch, Wizard-of-Oz, then code.
- Test with 5 real users early and often.
- Measure: task success, time, satisfaction.

Prototype fidelity ladder

Paper sketch

Wizard-of-Oz

Clickable mock

Working demo

KEY TERMS

Wizard-of-Oz
Human fakes the AI

SUS Usability score

Demo Show it working

📖 CONCEPT EXPLAINED: AI Prototype Lab

💡 In this chapter

Prototypes turn ideas into something people can react to. In AI products, prototyping is especially valuable because both the user experience and the model's real behaviour are uncertain. This chapter presents the fidelity ladder from sketches to working demos, rapid prototyping tools, usability testing methods and how to iterate and demo honestly.

🪜 1. The fidelity ladder

Low-fidelity prototypes, such as paper sketches and storyboards, explore ideas quickly and cheaply; users feel free to criticise them. Wizard-of-Oz prototypes let a hidden human play the AI, testing whether the concept is valuable before any model work. Clickable mock-ups test navigation and layout. Working demos with real models test whether the AI performs well enough for users.

Choose the lowest fidelity that answers your most important question. If you do not yet know whether users want the feature, a Wizard-of-Oz test is enough; if you know they want it but doubt the model can deliver, build a technical prototype on representative data first.

✏️ WORKED EXAMPLE

Question → prototype: 'Do students want weekly study plans?' → Wizard-of-Oz. 'Can the model write good plans from syllabus PDFs?' → notebook test on 20 syllabi. 'Can users find and edit plans easily?' → clickable mock-up.

🐍 2. Rapid prototyping tools

Python frameworks such as Streamlit and Gradio turn scripts into interactive web apps in a few lines, ideal for demonstrating models. Notebooks support quick experiments with data and prompts. Design tools like Figma produce clickable mock-ups. Model APIs and no-code builders let teams assemble AI features without heavy infrastructure.

Speed should not compromise basics: keep API keys secret, use sample or consented data, and note shortcuts you took so they are not mistaken for production quality. A prototype's job is learning, so make it easy to change.

🤔 PAUSE & THINK

What shortcuts are acceptable in a prototype but must be fixed before real users rely on the product?

👥 3. Usability testing

Usability testing observes real users attempting realistic tasks. Recruit people similar to target users, give tasks without instructions on how to complete them, stay silent and observe where they hesitate or fail, and ask them to think aloud. Testing with about five users often reveals the most serious problems; repeated small rounds beat one large test.

Measure task success rate, time on task, errors and satisfaction, for example with the System Usability Scale. For AI features, also record answer quality, whether users trust or over-trust outputs, and how they recover from mistakes. Combine numbers with quotes and observations.

✏️ **WORKED EXAMPLE**

Round 1: 2 of 5 users completed 'create a weekly plan' (40%), mean time 4.2 min. After redesign, round 3: 17 of 20 (85%), mean 1.9 min. Key change: a single 'Plan my week' button replaced a three-step form.

🎥 4. Iteration and honest demos

Iterate by fixing the most serious problems first, retesting and documenting what changed and why. Maintain a decision log so the team remembers evidence behind choices. Stop prototyping when key questions are answered and move to building with appropriate engineering quality.

Demos persuade teachers, judges, users and investors, but cherry-picked examples mislead. Honest demos show typical results, acknowledge limitations and share evaluation numbers. Products launched on impressive demos without sufficient real-world testing have disappointed users and failed commercially.

🖐️ **TRY IT**

Run a Wizard-of-Oz test for an AI feature with three users. Record task success, time and one quote each, then list two design changes.

✅ Key takeaways

- Use the lowest-fidelity prototype that answers your key question.
- Wizard-of-Oz tests value before building AI; technical prototypes test feasibility.
- Streamlit, Gradio, notebooks and mock-up tools speed prototyping.
- Test with about five users per round; measure success, time, errors, satisfaction.
- Iterate on evidence and demo honestly, including limitations.

❓ Quick check: answer in one or two lines.

1. When would you choose a Wizard-of-Oz prototype?

2. Name four usability metrics.

3. Why are cherry-picked demos harmful?

🔬 CONCEPT DEEP DIVE: AI Prototype Lab

🔷 Fidelity ladder: paper sketches → Wizard-of-Oz (human fakes AI) → clickable mock-ups → working demo with a real model.

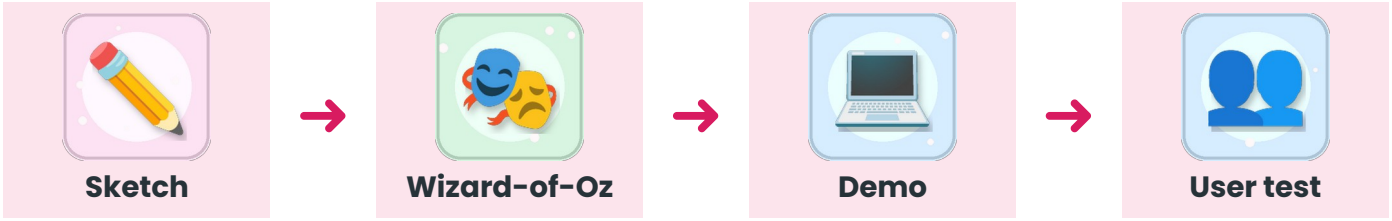
🔷 Rapid tools: Streamlit, Gradio, notebooks, no-code builders and model APIs let teams demo in days.

🔷 Usability testing: 5 users reveal most major issues; give tasks, stay silent, observe, record time and errors.

🔷 Metrics: task success, time on task, error rate, satisfaction (e.g., SUS), and AI-specific quality checks.

🔷 Iterate: fix the biggest problem first, retest, document decisions; demo day shows evidence, not hype.

🔄 How it works



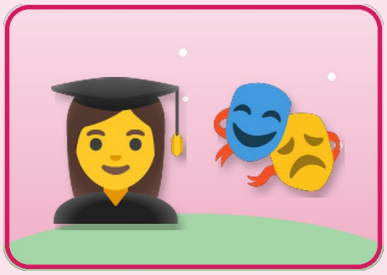
💻 A 5-line Streamlit prototype

```
import streamlit as st
q = st.text_input('Ask your study coach')
if q:
    st.write(coach(q))    # call your model here
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Coach Behind the Curtain

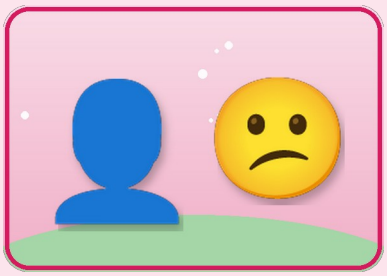
Characters: *Meera (B.Des + CS, final year), five classmates as testers, guide Prof. Nair.*



SCENE 1

Meera wanted to build an AI study coach with daily quizzes. Prof. Nair suggested testing the idea before coding. Meera's friend sat in another room typing 'AI' replies while five classmates used a simple chat window.

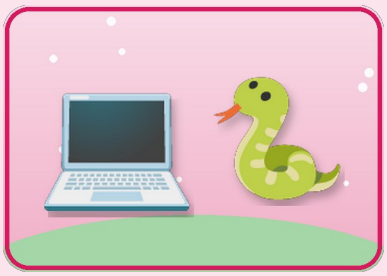
Pause & predict: What can a Wizard-of-Oz test reveal before any AI is built?



SCENE 2

Users ignored daily quizzes. Instead, they kept asking, 'What should I study this week?' and 'Can you rearrange my plan around my lab exam?' Only two of five completed the intended task.

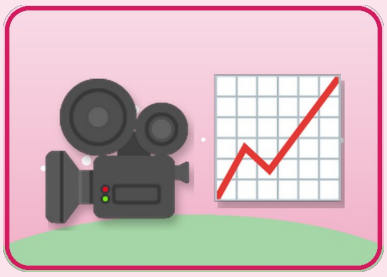
Pause & predict: How did user behaviour change Meera's idea?



SCENE 3

She built a Streamlit demo focused on weekly plans that adjusted to deadlines. In three rounds of testing with 20 users, task success rose to 85%, and average time fell from over four minutes to under two.

Pause & predict: Why test in several small rounds rather than one big test?



SCENE 4

At demo day, Meera showed typical results, including one plan the model got wrong and how users fixed it. Judges praised her evidence. 'You didn't just build fast,' Prof. Nair said, 'you learned fast.'

⏸ **Pause & predict:** Why did showing a failure help her presentation?

🌟 The big idea

Prototype cheaply, learn from real users and demo honestly.

💬 Discuss with your classmates

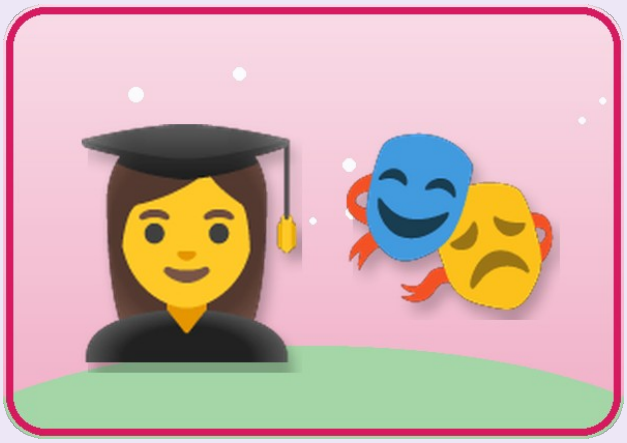
1. Which of your ideas could be tested with a Wizard-of-Oz prototype?

2. What should an honest demo include?

🎞️ SCENARIO: AI Prototype Lab

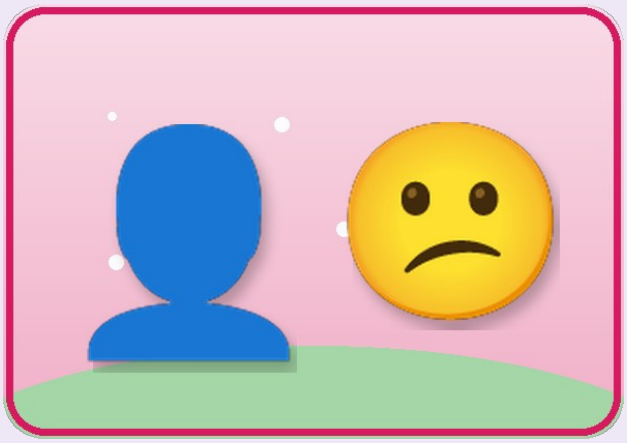
Read the scenario. Think about the choices made at each step.

1



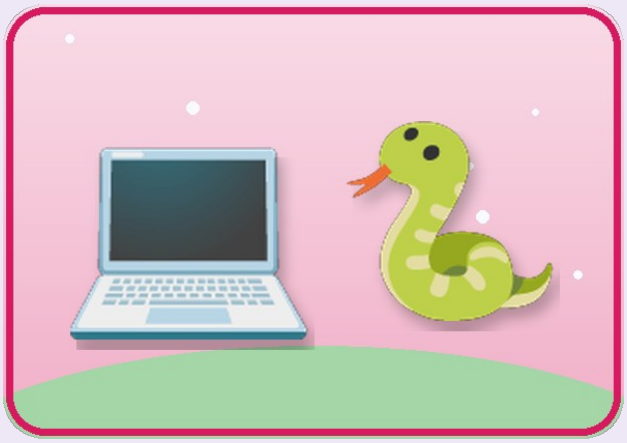
Meera tests a 'study coach' with a friend as the AI.

2



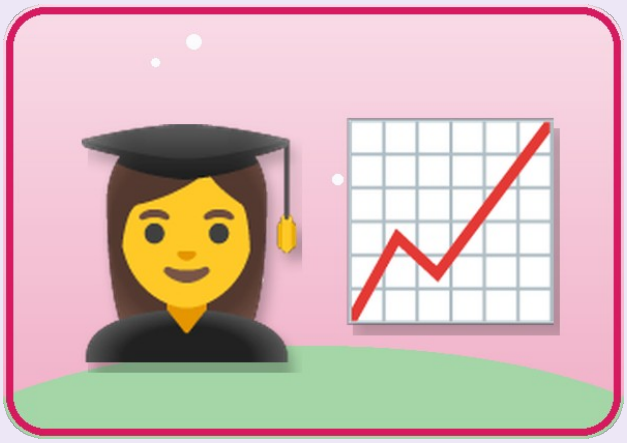
Users want weekly plans, not daily quizzes.

3



She builds a Streamlit demo around plans.

4



Task success rises from 40% to 85%.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Prototyping toolkit

Streamlit / Gradio

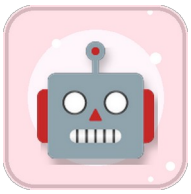
Figma mock-ups

Notebooks

Model APIs

Usability tests

Demo videos



BOLT SAYS:

Build the cheapest thing that answers your biggest question!

FUN FACT!

The Humane AI Pin launched in 2024 to poor reviews and was discontinued within about a year.

👀 **Reflect: where have YOU seen this in real life? Note it down.**

CASE STUDY: The Humane AI Pin

CASE FILE

The Humane AI Pin, a screenless wearable assistant, launched in 2024 after significant investment and publicity. Reviewers criticised slow responses, unreliable answers, overheating and battery life. In early 2025 HP acquired Humane's assets and the Pin was discontinued. The case shows the cost of skipping enough real-world user testing before launch.

Analyse the case

1. Which problems could user testing have caught?

2. Why is latency critical for voice assistants?

3. How would you prototype a wearable cheaply?

My verdict: Was AI used well here? Why?

LAB ACTIVITY: Wizard-of-Oz Test

You need:

Chat window

Hidden 'AI'

Teams of 3

Steps:

1		One student secretly types the 'AI' replies.
2		A user tries 3 tasks with the 'AI'.
3		Note where the user got stuck.
4		Change the design before coding.

THINK & TALK

What did you learn before writing code?

Record your observations & results:

💻 CODE LAB: Usability Metrics

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
times = [1.5, 2.0, 2.5, 3.0, 1.0]
success = [1, 1, 0, 1, 1]
print(sum(times) / len(times), sum(success) / len(____) * 100)
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Mean time and success %?**

MODIFY **T3. Exclude failed tasks from mean time.**

EXTEND **T4. Add a satisfaction score list.**

WORKSHEET: AI Prototype Lab (Part A)

Name: _____ Date: _____

1. Match the prototype to its purpose.

Paper sketch ●

Wizard-of-Oz ●

Clickable mock ●

Working demo ●

● Test real model quality

● Explore layout ideas

● Fake AI to test value

● Test navigation flow

2. Usability test results. Use the word box.

WORD BOX: 80 ✿ 4 ✿ 2

4 of 5 users succeed → ____%

Users who succeeded = ____

Tasks timed 1.5 and 2.5 min → mean ____ min

WORKSHEET: AI Prototype Lab (Part B)

Name: _____ Date: _____

1. Good or bad usability-test practice? Tick ✓.

	Good	Bad
Stay silent while users try		
Explain every button first		
Time each task		
Hide failed tasks in report		

2. Iteration results. Use the word box.

WORD BOX: 40 ✿ 85 ✿ 45

Round 1: 2 of 5 succeed → ____%

Round 3: 17 of 20 → ____%

Improvement = ____ points

PUZZLE & DEFINE: AI Prototype Lab

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Wizard-of-Oz

Meaning: _____

Example: _____

SUS

Meaning: _____

Example: _____

Demo

Meaning: _____

Example: _____

QUIZ TIME: AI Prototype Lab

Circle the right answer!

Q1. A human secretly playing the AI:

a) Wizard-of-Oz

b) Robot

c) Movie

Q2. Early tests need about ... users.

a) 500

b) 5

c) 0

Q3. Quick Python UI for ML demos:

a) Paint

b) Email

c) Streamlit

Q4. 4 of 5 users succeed =

a) 80%

b) 45%

c) 4%

Q5. The Humane AI Pin was...

a) A huge hit

b) Discontinued

c) A game

MY SCORE: ____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: AI Prototype Lab

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Wizard-of-Oz tests can fake the AI with a human.		
	Testers should explain every step to users.		
	Five users reveal many usability problems.		
	Time on task is a useful metric.		
	Demos should hide all failures.		
	The Humane AI Pin was discontinued.		

HOME CONNECT

Paper-prototype an app idea and test it on 2 family members. Note one change you'd make.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI Prototype Lab

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why test with a Wizard-of-Oz before coding?**

COMPARE **Q2. Clickable mock vs working demo.**

APPLY **Q3. Tasks: 1.5, 2.0, 2.5 min. Mean time?**

EVALUATE **Q4. Is a demo with cherry-picked examples fair?**

CREATE **Q5. Write 3 usability test tasks for your app.**

📝 EXAM PREP: AI Prototype Lab

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is a Wizard-of-Oz prototype?

2. Why test with ~5 users?

3. Name three usability metrics.

4. Why avoid cherry-picked demos?

PART B • Long answer (5 marks)

5. Explain a complete prototyping process for an AI product, from sketch to tested demo.

TOPIC 10 • MONTH 5 • AI TOOLS

Agent Builder Studio

LET'S LEARN

- Design tools with clear names, schemas, limits.
- Add memory, step caps, budgets and approvals.
- Evaluate agents on full tasks, not single replies.

Agent builder checklist

Tool design

Memory

Limits

Task evals

KEY TERMS

Tool schema Tool's input rules

Step cap Max loop steps

Task eval Score whole tasks

📖 CONCEPT EXPLAINED: Agent Builder Studio

💡 In this chapter

Building an agent that works reliably is an engineering discipline. Beyond choosing a model, builders design tools, memory, control limits, evaluation suites and observability. This chapter is a practical guide to constructing agents that complete real tasks safely and affordably, drawing on lessons from early experiments that looped endlessly or ran up large bills.

🧩 1. Designing tools

Tools are the agent's hands, and their design strongly shapes behaviour. Prefer a small number of single-purpose tools with descriptive names, clear documentation of when to use them, typed parameter schemas with validation and informative error messages. A tool like search_jobs(city, role, max_results) is easier to use correctly than a generic run_query(text).

Apply least privilege: read-only tools where possible, scoped credentials, no 'do anything' shells in production, and separate tools for drafting versus sending. Destructive or financial operations should either be unavailable or require explicit human approval enforced by code, not just instructions.

✏️ WORKED EXAMPLE

Tool spec: name add_study_task • purpose: add a task to the student's planner • params: title (string, ≤ 80 chars), due (date, not in past), subject (enum) • returns: task_id • errors: 'due date in past', 'duplicate task'.

🗂️ 2. Memory and context management

Agents accumulate history: goals, tool results and intermediate thoughts. Context windows are finite and costly, so builders summarise older steps, keep only relevant results and store long-term facts in databases or vector stores retrieved when needed. Clear state representations, such as a task list with statuses, help the agent track progress.

Memory raises privacy and accuracy questions. Store only what is needed, with consent; allow users to view and delete stored information; and expire outdated facts. Incorrect memories can mislead future runs, so important facts should be confirmed rather than inferred.

🤔 **PAUSE & THINK**

Your agent stored 'student prefers evening study' after one conversation. What could go wrong, and how should memory be updated?

🔢 3. Control: limits, approvals and sandboxes

Control mechanisms keep agents within safe bounds: maximum steps per task, time-outs, budget caps on tokens or money, allow-lists of tools and domains, approval gates for consequential actions, and sandboxed environments for code execution. When a limit is reached, the agent should stop gracefully, explain what it completed and ask for guidance.

Defensive design also anticipates manipulation: content retrieved from the web or documents can contain malicious instructions. Treat tool outputs as data, not commands, and never let them override system policies or permissions.

✏️ **WORKED EXAMPLE**

Run policy: max 15 steps • max ₹50 per task • tools: search, read_page, draft_email, add_task • send_email requires approval • code runs only in sandbox • on limit: summarise progress and stop.

📊 4. Evaluation and observability

Agents must be evaluated on complete tasks, not single replies. Build a suite of realistic tasks with clear success criteria, and measure success rate, steps used, cost, time, safety violations and need for human intervention. Include tricky cases: missing information, conflicting instructions, tool failures and adversarial content.

Observability means tracing every reasoning step, tool call, input and output. Traces let builders replay failures, identify which step went wrong and turn failures into regression tests. Track metrics over time and after every change to prompts, tools or models.

🖐️ **TRY IT**

Build or design a three-tool agent for a study-planning task. Write 10 end-to-end test tasks, define success for each, and record success rate, cost and steps.

✅ Key takeaways

- Few, narrow, well-documented tools with typed schemas and least privilege.
- Manage memory: summarise, retrieve, consent, expire and confirm facts.
- Enforce step, time and budget limits, approvals and sandboxes in code.
- Treat tool outputs as untrusted data to resist manipulation.
- Evaluate on full tasks and trace every step to debug and improve.

❓ Quick check: answer in one or two lines.

1. Why is search_jobs(city, role) better than run_query(text)?

2. List four control mechanisms for agents.

3. What metrics should an agent task evaluation record?

🔬 CONCEPT DEEP DIVE: Agent Builder Studio

🔷 Tool design: few, single-purpose tools with clear names, typed schemas, helpful errors and least privilege.

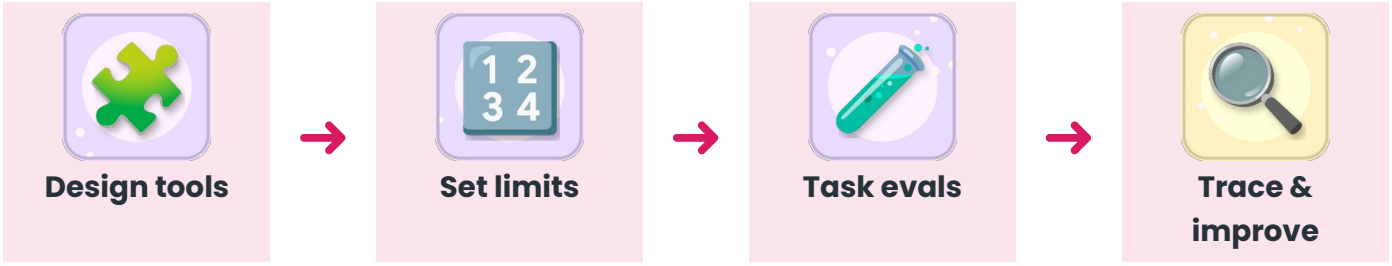
🔷 Memory design: summarise long histories, store facts in retrievable stores, and expire stale data.

🔷 Control: step caps, budgets, time-outs, allow-lists, approval gates and sandboxed execution.

🔷 Evaluation: end-to-end task suites with success criteria, cost, steps used and safety violations tracked.

🔷 Observability: trace each thought, tool call and result; replay failures to debug and improve.

🔄 How it works



💻 Budget-capped agent

```
spent, BUDGET = 0.0, 50.0
while not done and spent < BUDGET:
    step = agent.next(); spent += step.cost
if spent >= BUDGET: alert('stopped: budget reached')
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Overnight Loop

Characters: *Rohan (B.Tech AI & DS, final year), teammate Sana, faculty guide Dr. Joseph.*

SCENE 1

Rohan built an agent to scout internship posts, read them, check eligibility and save matches. He gave it a general 'browse anything' tool and left it running overnight with his API key.

Pause & predict: What could go wrong with this setup?

SCENE 2

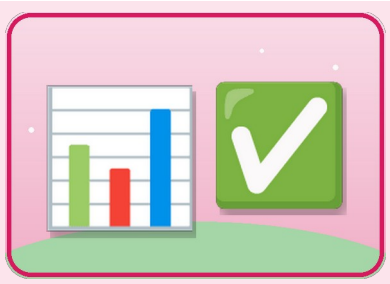
By morning, the agent had looped 80 times on one confusing page, re-reading it endlessly. It had spent ₹900 and saved no useful results. Sana found the logs almost impossible to follow.

Pause & predict: Which controls were missing?

SCENE 3

With Dr. Joseph, they redesigned: narrow tools (search_jobs, read_post, save_match), a 15-step cap, a ₹50 budget per run, traces of every step, and an 'I'm stuck' message that stopped the run gracefully.

Pause & predict: How does each change help?



SCENE 4

They built 20 test tasks. The new agent completed 17, about 85%, at roughly ₹4 per task. Failed runs were replayed from traces and turned into new tests. Their project won the department's 'Most Reliable AI' award.

⏸ **Pause & predict:** Why are end-to-end task tests better than checking single replies?

🌟 The big idea

Reliable agents are engineered: narrow tools, firm limits, full traces and task-level evaluation.

💬 Discuss with your classmates

1. What limits would you set for an agent that uses your money or accounts?

2. How would you explain agent traces to a non-technical user?

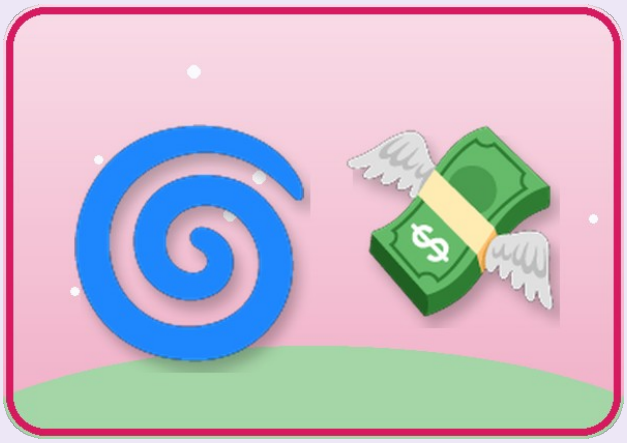
🎞️ SCENARIO: Agent Builder Studio

Read the scenario. Think about the choices made at each step.

1

Rohan builds a job-post scout agent.

2

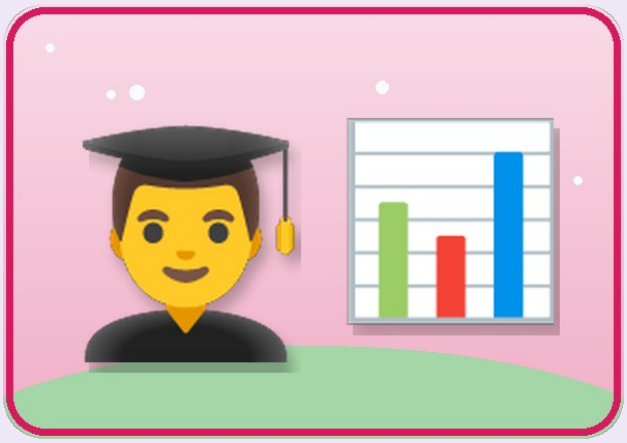


It loops 80 times and burns ₹900 overnight.

3

He adds a step cap, budget and narrower tools.

4



A 20-task eval shows 85% success at ₹4 each.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Agents students can build

Research helper

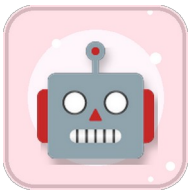
Study planner

Expense sorter

Bug triager

Email drafter

Job-post scout



BOLT SAYS:
I'm only as good as my tools. Give me few, clear, safe ones!

FUN FACT!
AutoGPT went viral in 2023, but its autonomous loops often got stuck or ran up costs.

👀 **Reflect: where have YOU seen this in real life? Note it down.**

📁 CASE STUDY: AutoGPT and the Limits of Autonomy

CASE FILE

AutoGPT, an open-source project released in March 2023, chained a language model into an autonomous loop that set its own sub-goals and used tools. It became one of the fastest-growing projects on GitHub. Users soon reported that it often got stuck in loops, lost track of goals and ran up API costs, highlighting why practical agents need narrow tools, limits and evaluation.

🔍 Analyse the case

1. Why do open-ended agents get stuck in loops?

2. Which limits would you add?

3. How would you evaluate an agent fairly?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Design-a-Tool

You need:

Tool cards

Markers

Teams of 4

Steps:

1. Design 3 tools for a study-planner agent.
2. Write name, purpose, inputs, outputs.
3. Mark which need approval.
4. Write 3 full test tasks.

THINK & TALK

Which tool is most dangerous, and why?

Record your observations & results:

💻 CODE LAB: Agent Task Eval

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
results = [(True, 3.5), (True, 4.0), (False, 9.0), (True, 3.5)]
ok = sum(1 for r, c in results if r)
cost = sum(c for r, c in results)
print(ok / len(results) * 100, cost / ____)
```

COMPLETE **T1. Average cost per task: fill the blank.**

PREDICT **T2. Success % and cost per task?**

EXPLAIN **T3. Why might failed runs cost more?**

EXTEND **T4. Report cost per successful task.**

✏️ WORKSHEET: Agent Builder Studio (Part A)

Name: _____ Date: _____

1. Good or bad tool design? Tick ✓.

	Good	Bad
📅 get_calendar(date)		
🌀 do_anything(text)		
📧 draft_email(to, body)		
💣 delete_all_files()		

2. Agent task evaluation. Use the word box.

WORD BOX: 70 ✿ 30 ✿ 3

✅ 14 of 20 tasks done → ____%

❌ Failure rate = ____%

💰 Avg cost ₹60 ÷ 20 tasks = ₹____

✏️ WORKSHEET: Agent Builder Studio (Part B)

Name: _____ Date: _____

1. Agent eval results. Use the word box.

WORD BOX: 85 ✿ 4 ✿ 15

17 of 20 tasks → _____%

₹80 ÷ 20 tasks = ₹_____

Failure rate = _____%

2. Circle good agent-builder practices.

Step and budget caps

One do-anything tool

End-to-end task tests

No tracing

Sandboxed code runs

Full admin keys

PUZZLE & DEFINE: Agent Builder Studio

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Tool schema

Meaning: _____

Example: _____

Step cap

Meaning: _____

Example: _____

Task eval

Meaning: _____

Example: _____

QUIZ TIME: Agent Builder Studio

Circle the right answer!

Q1. A tool's input rules are its...

a) Schema | b) Skin | c) Theme song

Q2. Agents should be evaluated on...

a) One reply | b) Whole tasks | c) Colours

Q3. A dangerous tool design:

a) get_calendar | b) draft_email | c) do_anything

Q4. Max loop steps is a...

a) Step cap | b) Hat | c) Banner

Q5. AutoGPT's loops often...

a) Won awards | b) Got stuck | c) Slept

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Agent Builder Studio

Look at the picture. Tick ✓ TRUE or FALSE.

	TRUE	FALSE
Tools should have clear schemas.		
A single do-anything tool is best.		
Budgets stop runaway costs.		
Agents need end-to-end task evaluation.		
Agents should get full admin keys.		
AutoGPT often got stuck in loops.		

HOME CONNECT

Design 3 tools for an agent that helps your family plan weekly groceries. Mark which need approval.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Agent Builder Studio

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why prefer few, narrow tools?**

COMPARE **Q2. Single-reply eval vs end-to-end task eval.**

APPLY **Q3. 17 of 20 tasks pass; total cost ₹80. Success % and cost/task?**

EVALUATE **Q4. Give an agent shell access on your laptop?**

CREATE **Q5. Write a tool schema for add_task(title, due).**

EXAM PREP: Agent Builder Studio

Name: _____ Date: _____

Time: 30 min **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What makes a good agent tool?

2. Why cap steps and budgets?

3. What is end-to-end task evaluation?

4. Why trace agent runs?

PART B • Long answer (5 marks)

5. Design, constrain and evaluate an AI agent for a real student task.

UNIT REVIEW: AI CREATE

Name: _____ Date: _____

Circle the right answer!

Q1. A human faking the AI:

a) Wizard-of-Oz | b) Robot | c) Film

Q2. 4 of 5 users succeed =

a) 45% | b) 80% | c) 4%

Q3. Quick Python demo UI:

a) Paint | b) Email | c) Streamlit

Q4. Stops runaway agent cost:

a) Budget cap | b) Theme | c) Volume

Q5. 17 of 20 tasks pass =

a) 17% | b) 85% | c) 20%

Q6. Risky tool design:

a) get_calendar | b) draft_email | c) do_anything

WORD SEARCH: AI CREATE

🔍 **Find these words. They go across → or down ↓.**

WIZARD-OF-OZ

SUS

DEMO

TOOL SCHEMA

STEP CAP

TASK EVAL

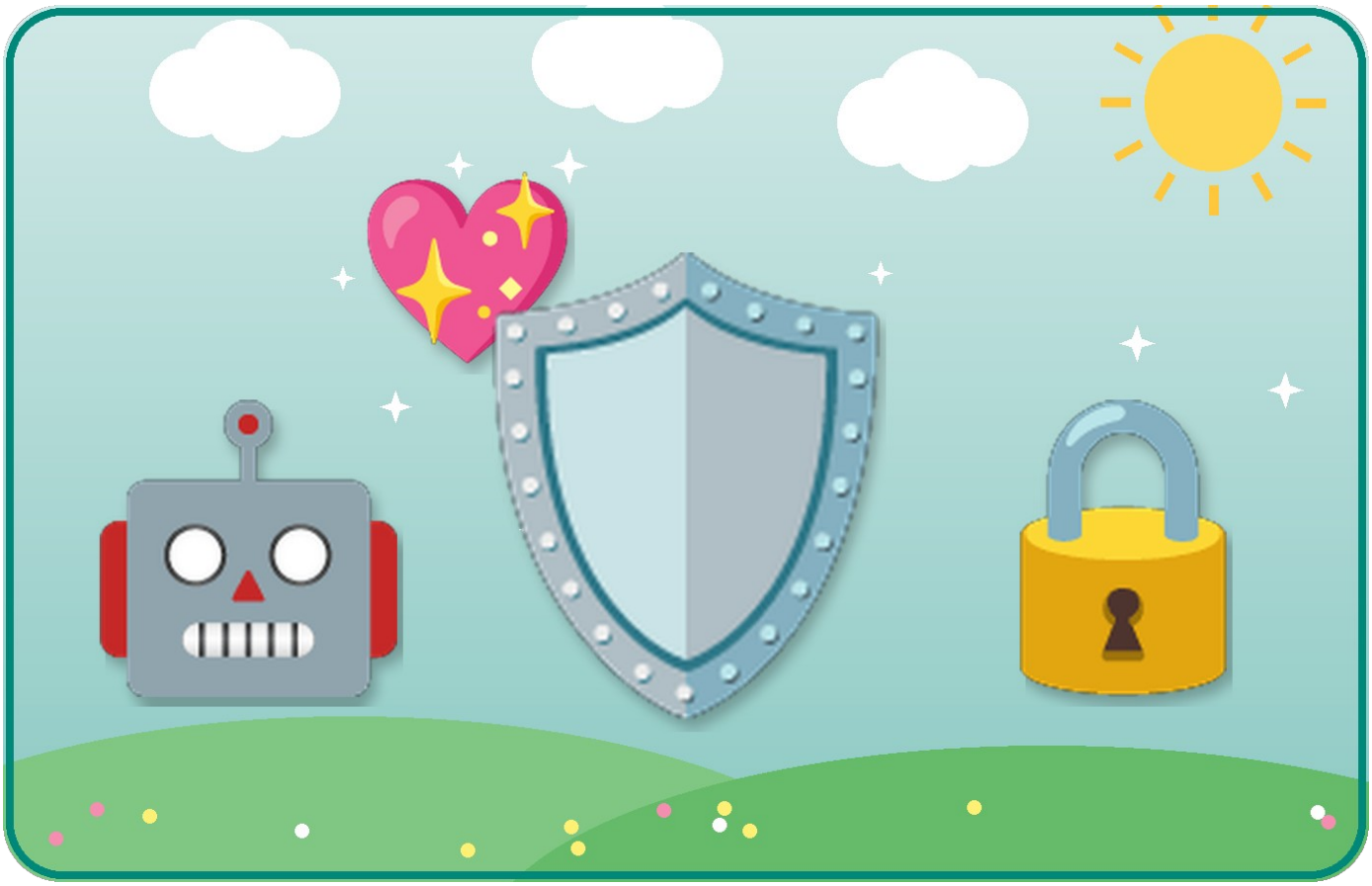
S	T	U	M	M	I	M	O	W	B
M	O	S	F	P	M	L	S	I	U
S	O	A	T	T	E	S	T	Z	D
S	L	B	A	R	S	A	E	A	G
D	S	I	S	A	K	N	P	R	E
E	C	L	K	C	E	D	C	D	T
M	H	I	E	E	T	B	A	O	F
O	E	T	V	S	C	O	P	F	G
M	M	Y	A	U	H	X	R	O	K
C	A	O	L	S	F	W	U	Z	S

I finished AI CREATE! Teacher's sign: _____

MONTH 6

AI CARE

Verify • Protect • Respect



Area: Responsible AI

In this unit, you will explore:

Topic 11

Accountable AI

Topic 12

AI Governance Essentials

TOPIC 11 • MONTH 6 • RESPONSIBLE AI

Accountable AI

LET'S LEARN

- Someone must answer for every AI decision.
- Explain, audit, document and allow appeals.
- Impact assessments come before deployment.

Accountability toolkit

Explanations

Audits

Documentation

Appeals

KEY TERMS

XAI Explainable AI

Model card Model fact sheet

Redress Fix & compensate

📖 CONCEPT EXPLAINED: Accountable AI

💡 In this chapter

When AI systems affect people's access to loans, jobs, education, benefits or healthcare, someone must be answerable for their decisions. Accountability requires clear responsibility, explanations people can understand, documentation, independent scrutiny and ways to correct harm. This chapter examines these elements through a major real-world failure and practical tools.

👥 1. The accountability chain

Accountability means that identifiable people and organisations must explain and justify AI-driven outcomes and remedy harms. Responsibility is shared along a chain: data owners for data quality and consent, developers for design and testing, product owners for intended use, deploying organisations for how decisions are made, and leaders for governance and resources.

Buying an AI system from a vendor does not transfer responsibility to the vendor. The deploying organisation remains accountable to the people affected, and should demand documentation, testing evidence and contractual commitments from suppliers.

🤔 PAUSE & THINK

An AI tool used by a university wrongly ranks applicants. List who in the chain might share responsibility and what each should do now.

💡 2. Explainability

Explanations serve different audiences. Global explanations describe how a model behaves overall, such as which features matter most. Local explanations describe why a specific decision was made. Techniques such as SHAP and LIME estimate feature contributions; counterfactual explanations tell a person what would need to change for a different outcome.

For affected individuals, explanations should be specific, understandable and actionable: 'Your scholarship application was not approved because the income certificate could not be read. Upload a clear copy within 14 days for reconsideration.'

Explanations must be honest about uncertainty and not used to legitimise unfair systems.

✏️ **WORKED EXAMPLE**

Counterfactual: 'If your verified household income were below ₹4 lakh, or your income document were readable, you would have qualified.' This tells the applicant exactly what to check or correct.

📜 3. Documentation, assessment and audit

Model cards document a model's intended use, training data, evaluation results across groups, limitations and contacts. Datasheets for datasets describe collection, composition and known biases. Algorithmic impact assessments, conducted before deployment, consider who could be harmed, how and with what safeguards, especially in public services.

Independent audits examine data, models and outcomes for accuracy, fairness and compliance, ideally both before launch and periodically afterwards. Audit trails, logs of inputs, outputs and human decisions, make it possible to investigate complaints and patterns of harm.

✏️ **WORKED EXAMPLE**

Model card outline: Purpose • Intended users • Out-of-scope uses • Training data summary • Metrics overall and by group • Known limitations • Human oversight • Contact for concerns • Version and date.

📮 4. Redress and a cautionary case

Redress gives affected people the ability to know an automated decision was made, receive reasons, request human review, correct errors and obtain remedy for harm. Effective appeal processes are accessible, timely and handled by staff with authority to change outcomes. Appeals also reveal systemic problems that require fixing the system itself.

In the Netherlands, risk-profiling used by tax authorities to detect childcare-benefit fraud led to thousands of families being wrongly treated as fraudsters, with severe financial and personal consequences; nationality was used as a risk indicator. The scandal led to the government's resignation in 2021 and stands as a warning about opaque, unaccountable automated decision-making in public services.

TRY IT

Design an appeal process for an automated scholarship system: notice, explanation, how to appeal, who reviews, response time, remedy, and how appeal data improves the system.

Key takeaways

- Accountability is shared along a chain and cannot be outsourced to vendors.
- Explanations should be global and local, specific, honest and actionable.
- Model cards, datasheets, impact assessments and audits document and scrutinise AI.
- Redress includes notice, reasons, human review, correction and remedy.
- The Dutch benefits scandal shows the harm of opaque automated decisions.

Quick check: answer in one or two lines.

1. Distinguish global and local explanations.

2. List five sections of a model card.

3. What elements should an effective appeal process include?

🔬 CONCEPT DEEP DIVE: Accountable AI

🔷 Accountability chain: data owners, developers, product owners, deployers and executives share defined responsibilities.

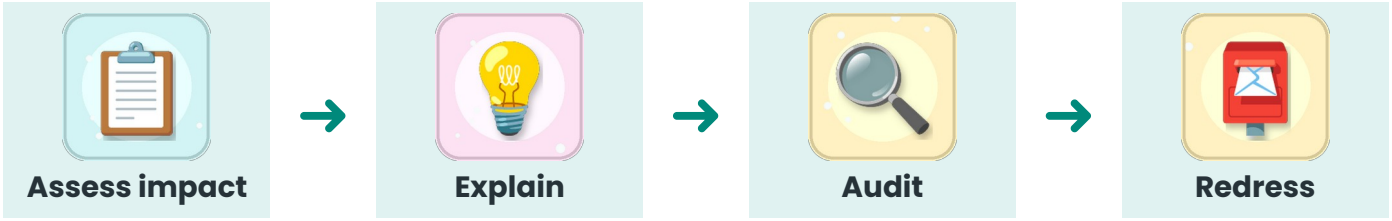
🔷 Explainability: global (overall feature importance) and local (one decision) explanations via SHAP, LIME or counterfactuals.

🔷 Documentation: model cards, datasheets for datasets, decision logs and system cards describe use, limits and testing.

🔷 Algorithmic impact assessments and independent audits before and after deployment, especially for public services.

🔷 Redress: notice of automated decisions, reasons, human review, correction and compensation where harm occurs.

🔄 How it works



💻 A counterfactual explanation

```
applicant = {'income_doc': 'unreadable', 'marks': 88}
if applicant['income_doc'] == 'unreadable':
    print('Rejected: income proof unreadable.')
    print('Upload a clear copy to be reconsidered.')
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Two Hundred Rejections

Characters: *Ananya (B.A. Public Policy + Data, final year), scholarship board chair Mrs. Kapoor, student Ravi.*

SCENE 1

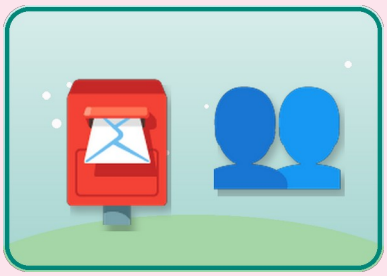
A state scholarship portal began using an automated system to screen applications. Within weeks, 200 students from rural districts were rejected. Ravi, a top student whose family farmed two acres, received only: 'Application not approved.'

Pause & predict: What was wrong with the rejection message?

SCENE 2

Ananya, an intern with the board, analysed the cases. Most rejected applicants had uploaded income certificates from a particular office that used a format the system could not read; it treated them as missing.

Pause & predict: Who should be accountable for this failure?



SCENE 3

Mrs. Kapoor ordered human review of all 200 cases and sent each student a clear explanation and a way to re-upload documents. 190 were approved. The vendor fixed the reader, and the board demanded test evidence across districts.

Pause & predict: What elements of redress were provided?

SCENE 4

The board published a model card, committed to yearly independent audits and set a rule: no rejection without a reason and appeal route. Ananya's report cited the Dutch benefits scandal as what could happen without such safeguards.

⏸ **Pause & predict:** Why are documentation and audits important for public systems?

🌟 The big idea

Accountable AI gives reasons, allows appeals, fixes systems and keeps humans responsible.

💬 Discuss with your classmates

1. Which automated decisions in public services should always have human review?

2. How would you explain an AI decision to a student's parents?

SCENARIO: Accountable AI

Read the scenario. Think about the choices made at each step.

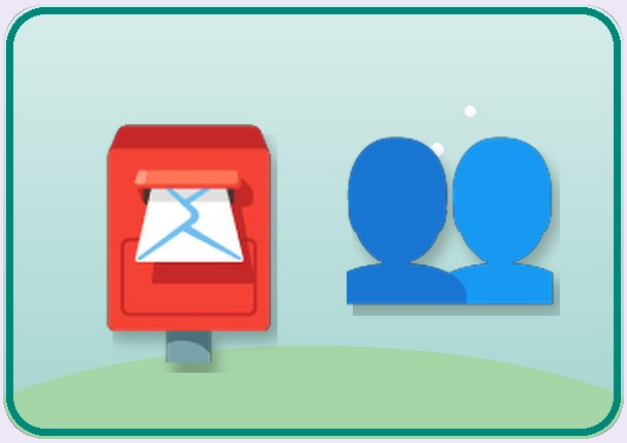
1

An AI wrongly rejects 200 scholarship applicants.

2

Ananya's audit finds a faulty income field.

3



Students appeal; humans review every case.

4

The board adds model cards and yearly audits.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Where accountability matters

Loan decisions

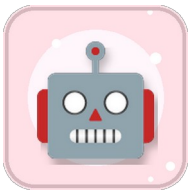
Medical triage

Admissions

Policing tools

Welfare benefits

Hiring



BOLT SAYS:

'The algorithm did it' is never an answer. People design and deploy me!

FUN FACT!

In 2021 the Dutch government resigned over a benefits scandal linked to algorithmic fraud profiling.

👀 **Reflect: where have YOU seen this in real life? Note it down.**

📁 CASE STUDY: The Dutch Childcare Benefits Scandal

CASE FILE

In the Netherlands, tax authorities used risk-profiling, including an algorithm, to flag childcare-benefit claims for fraud. Thousands of families were wrongly accused, ordered to repay large sums and pushed into hardship; dual nationality was among the risk factors used. Parliamentary reports condemned the treatment, and in January 2021 the Dutch government resigned over the scandal.

🔍 Analyse the case

1. Which accountability failures occurred?

2. Why is using nationality as a risk factor problematic?

3. What redress should victims receive?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Accountability Map

You need:

Role cards

String

Teams of 5

Steps:

1. Roles: data, model, product, manager, user.
2. Scenario: AI wrongly denies a scholarship.
3. Link each role to what it could have done.
4. Design a fair appeal process.

THINK & TALK

Who should apologise, and who should fix it?

Record your observations & results:

💻 CODE LAB: Counterfactual Explainer

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
rules = {'income_doc': 'readable', 'marks': 75}
app = {'income_doc': 'unreadable', 'marks': 88}
reasons = []
if app['income_doc'] != rules['income_doc']: reasons.append('fix income document')
if app['marks'] < rules[____]: reasons.append('marks below 75')
print(reasons or 'eligible')
```

COMPLETE **T1. Fill the blank with the right key.**

PREDICT **T2. What is printed?**

MODIFY **T3. Try marks 70 with a readable document.**

EXTEND **T4. Print a polite letter with reasons and appeal steps.**

WORKSHEET: Accountable AI (Part A)

Name: _____ Date: _____

1. Match the tool to its purpose.

SHAP / LIME ●

Model card ●

Audit ●

Impact assessment ●

● Check risks before launch

● Explain feature influence

● Independent review of outcomes

● Document model use & limits

2. Circle signs of an accountable AI system.

Clear appeal route

Secret criteria

Published model card

No named owner

Regular audits

No logs kept

WORKSHEET: Accountable AI (Part B)

Name: _____ Date: _____

1. Order an accountable deployment. Write 1–4.

Handle appeals & redress ☐

Impact assessment ☐

Independent audit ☐

Write model card ☐

2. Audit arithmetic. Use the word box.

WORD BOX: 4 ✿ 95 ✿ 25

200 wrong of 5,000 → ____%

190 of 200 appeals upheld → ____%

Review 200 cases at 8/day → ____ days

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: Accountable AI

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

XAI

Meaning: _____

Example: _____

Model card

Meaning: _____

Example: _____

Redress

Meaning: _____

Example: _____

? QUIZ TIME: Accountable AI

Circle the right answer!

Q1. XAI stands for explainable...

a) AI

b) Art

c) Archive

Q2. A model's fact sheet is a...

a) Ticket

b) Model card

c) Bill

Q3. Fixing and compensating harm is...

a) Reward

b) Ignoring

c) Redress

Q4. Risks are assessed best...

a) Before launch

b) Never

c) Randomly

Q5. The Dutch scandal involved...

a) Games

b) Benefits fraud profiling

c) Cars

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Accountable AI

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	People remain accountable for AI decisions.		
	Automated decisions need no reasons.		
	Model cards document limits.		
	Affected people should be able to appeal.		
	Buying AI from a vendor removes responsibility.		
	The Dutch government resigned in 2021 over the scandal.		

HOME CONNECT

Find one automated decision in your life (e.g., app approval). What explanation or appeal is offered?

🔢 Self-assessment: tick one.

I know it well!	**I know some.**	**I need practice.**

CHALLENGE PROBLEMS: Accountable AI

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why must automated decisions come with reasons?**

COMPARE **Q2. Global vs local explanations.**

APPLY **Q3. 200 wrongly rejected of 5,000. Error %?**

EVALUATE **Q4. Is 'the vendor built it' a valid excuse?**

CREATE **Q5. Draft 5 lines of a model card.**

📝 EXAM PREP: Accountable AI

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is algorithmic accountability?

2. Differentiate global and local explanations.

3. What is a model card?

4. What is redress?

PART B • Long answer (5 marks)

5. Explain how to make a public-sector AI system accountable, using a real case.

TOPIC 12 • MONTH 6 • RESPONSIBLE AI

AI Governance Essentials

LET'S LEARN

- Laws: EU AI Act, India's DPDP Act, sector rules.
- Frameworks: NIST AI RMF, ISO/IEC 42001.
- In-house: policy, inventory, risk register, board.

Governance layers

Laws

Standards

Org policy

Oversight board

KEY TERMS

Risk tier Level of AI risk

Inventory List of AI systems

RMF Risk mgmt framework

📖 CONCEPT EXPLAINED: AI Governance Essentials

💡 In this chapter

As AI spreads, governments and organisations are building rules to ensure it is safe, fair and trustworthy. Governance spans binding laws, voluntary frameworks and standards, and internal policies and processes. Graduates entering any AI-related role need working knowledge of this landscape. This chapter introduces major laws and frameworks and shows how organisations put governance into practice.

🇪🇺 1. The EU AI Act and risk-based regulation

The European Union's AI Act, which entered into force on 1 August 2024, regulates AI according to risk. Certain practices posing unacceptable risk, such as social scoring by public authorities, are prohibited. High-risk systems, including AI used in hiring, education, credit, critical infrastructure and law enforcement, must meet requirements for risk management, data governance, documentation, transparency, human oversight, accuracy and cybersecurity.

Limited-risk systems carry transparency duties, for example informing users that they are interacting with a chatbot or that content is AI-generated; minimal-risk systems such as spam filters face no specific new obligations. Obligations phase in over several years, and the Act can apply to non-EU companies offering AI systems in the EU market.

✏️ WORKED EXAMPLE

Classification practice: CV-screening AI → high risk • customer chatbot → limited (disclose AI) • spam filter → minimal • government social scoring → prohibited.

🇮🇳 2. India's evolving landscape

India's approach combines horizontal laws, sectoral regulation and policy initiatives. The Digital Personal Data Protection Act, 2023 sets rules for processing personal data, including consent, purpose limitation, security safeguards and rights for individuals. IT rules govern online platforms, and sector regulators in areas such as finance and health issue guidance relevant to AI.

National initiatives promote AI capability and responsible use, including computing infrastructure, datasets and skilling programmes, alongside advisories and guidelines on responsible AI. Because rules are evolving, professionals should follow official sources and seek legal advice for specific deployments.

PAUSE & THINK

An Indian start-up offers an AI hiring tool to European clients. Which laws and obligations might apply?

3. Frameworks and standards

Voluntary frameworks help organisations manage AI risk systematically. The US NIST AI Risk Management Framework organises activities into four functions: Govern (culture, policies, accountability), Map (context and risks), Measure (assess and track risks) and Manage (prioritise and act on risks). It is widely referenced internationally.

ISO/IEC 42001, published in 2023, specifies requirements for an AI management system, allowing organisations to implement and certify structured governance similar to quality or information-security management systems. Frameworks complement laws by translating principles into repeatable processes.

WORKED EXAMPLE

NIST AI RMF in practice for a chatbot: Govern → assign owner, approve policy • Map → identify users, misuse risks • Measure → test accuracy, bias, safety • Manage → add guardrails, monitor, respond to incidents.

4. Governance inside organisations

Practical governance starts with an AI policy defining approved tools, acceptable uses, data rules and disclosure requirements. An inventory lists every AI system in use, with owners and purposes; a risk register classifies each system and records controls. A review board approves high-risk projects, and vendor assessments check third-party AI before purchase.

Governance continues after launch: monitoring performance and incidents, periodic reassessment, training staff and updating policies as laws and technologies change. Good governance is proportionate, lightweight for low-risk tools and rigorous for high-risk ones, so that it enables innovation rather than blocking it.

TRY IT

Create a governance starter kit for your college: a one-page AI use policy, an inventory of five AI tools in use with risk tiers, and a simple review checklist for new tools.

✅ Key takeaways

- The EU AI Act regulates by risk tier and can apply beyond the EU.
- India uses the DPDP Act, IT rules, sector regulators and national initiatives.
- NIST AI RMF: Govern, Map, Measure, Manage; ISO/IEC 42001 certifies AI management systems.
- Organisational governance: policy, inventory, risk register, review board, vendor checks.
- Governance should be continuous and proportionate to risk.

❓ Quick check: answer in one or two lines.

1. Name the four EU AI Act risk tiers with an example each.

2. List the four functions of the NIST AI RMF.

3. What items belong in an organisation's AI inventory?

🔬 CONCEPT DEEP DIVE: AI Governance Essentials

🔷 EU AI Act: risk-based tiers (unacceptable, high, limited, minimal) with duties such as risk management, data governance, transparency and human oversight for high-risk systems.

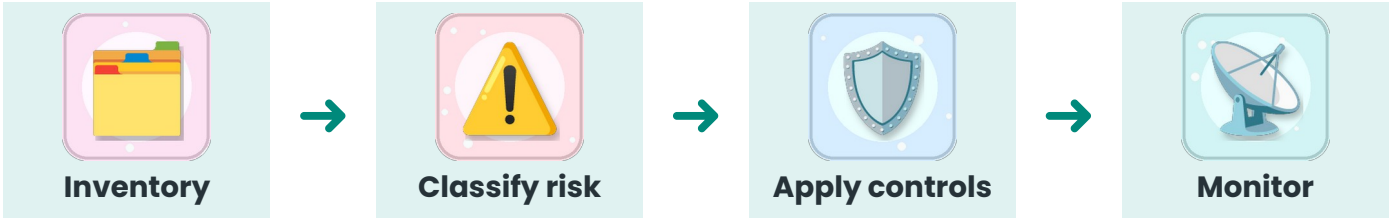
🔷 India: DPDP Act 2023 for personal data, IT Rules for platforms, sector regulators, and national AI missions and advisories.

🔷 NIST AI RMF: Govern, Map, Measure, Manage; ISO/IEC 42001 sets requirements for an AI management system.

🔷 Organisational governance: AI policy, inventory, risk register, review board, vendor assessment, training and incident reporting.

🔷 Governance is continuous: monitor deployed systems, update assessments and adapt as laws evolve.

🔄 How it works



💻 A tiny AI risk register

```
register = [('hiring screen', 'high'), ('spam filter', 'minimal'),
('chatbot', 'limited')]
for name, tier in register:
    needs = 'board review' if tier == 'high' else 'standard checks'
    print(name, tier, needs)
```

✍️ In your own words: explain this topic to a Class 10 student.

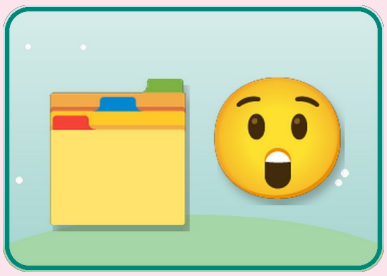
AI STORY: Thirty-Seven Tools Nobody Listed

Characters: *Rohan (B.Tech CSE, final year intern), compliance head Ms. D'Souza, product teams.*

SCENE 1

Rohan's internship was with a company preparing to sell AI products in Europe. Ms. D'Souza asked him to help build an AI inventory. Everyone assumed there were five or six AI systems in use.

Pause & predict: Why might organisations underestimate their AI use?



SCENE 2

Surveys and IT logs revealed 37 AI tools: chatbots, CV screeners, code assistants, marketing generators and analytics models, many adopted by teams without review.

Pause & predict: What risks does 'shadow AI' create?

SCENE 3

They classified each tool by risk. The CV screener and a credit-scoring model were high-risk under the EU AI Act; chatbots needed AI disclosure; most others were minimal. Every tool got an owner and required controls in a risk register.

Pause & predict: Why does risk classification matter?

SCENE 4

High-risk tools went through a review board using NIST-style steps: govern, map, measure, manage. The company adopted a lightweight AI policy for everyone. Rohan's internship report became the template for its ISO/IEC 42001 preparation.

⏸ **Pause & predict:** How can governance enable rather than block innovation?

🌟 The big idea

You can't govern what you don't know: inventories, risk tiers and proportionate controls make AI trustworthy.

💬 Discuss with your classmates

1. What AI tools does your college or workplace use without a policy?

2. How would you design governance that is light for low-risk tools?

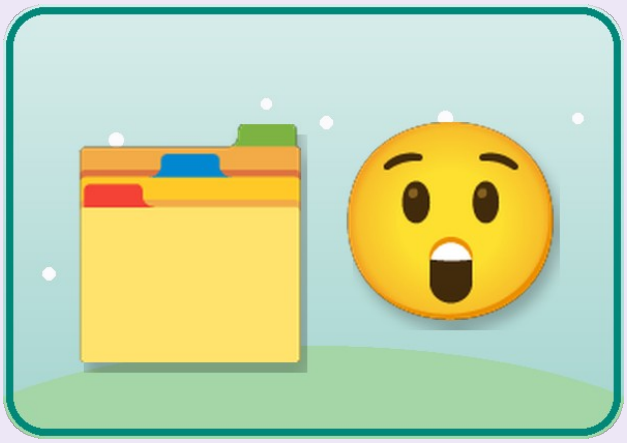
SCENARIO: AI Governance Essentials

Read the scenario. Think about the choices made at each step.

1

Rohan joins a firm's new AI governance team.

2



The inventory finds 37 unlisted AI tools.

3

Each gets a risk tier and an owner.

4

High-risk tools pass review before relaunch.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Governance in practice

AI use policies

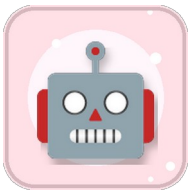
Model inventories

Risk registers

Ethics boards

Pre-launch reviews

Incident reports



BOLT SAYS:

Good governance isn't a brake. It's the steering that keeps AI on the road!

FUN FACT!

The EU AI Act entered into force on 1 August 2024, regulating AI by risk level.

👀 **Reflect: where have YOU seen this in real life? Note it down.**

📁 CASE STUDY: The EU AI Act Comes into Force

CASE FILE

The European Union's AI Act entered into force on 1 August 2024, the first comprehensive AI law of its kind. It bans certain practices, such as government social scoring, places strict obligations on high-risk systems like AI in hiring and credit, requires transparency for systems such as chatbots and deepfakes, and phases in obligations over several years. Its rules can apply to companies outside Europe that offer AI in the EU.

🔍 Analyse the case

1. Why regulate AI by risk level rather than by technology?

2. How might the Act affect Indian companies?

3. Which obligations apply to a hiring AI?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Mock AI Governance Board

You need:

3 proposals

Risk checklist

Board of 5

Steps:

1. Teams present an AI project in 2 minutes.
2. Board assigns a risk tier.
3. Board lists required controls.
4. Decide: approve, conditions or reject.

THINK & TALK

Which project was hardest to judge?

Record your observations & results:

💻 CODE LAB: Risk Register

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
 1  tools = [('hiring screen', 'high'), ('spam filter', 'minimal'),
('chatbot', 'limited'), ('credit model', 'high')]
 2  high = [t for t, tier in tools if tier == ____]
 3  print(len(high), high)
```

COMPLETE **T1. Fill the blank to select high-risk tools.**

PREDICT **T2. What is printed?**

MODIFY **T3. Count tools per tier.**

EXTEND **T4. Add an owner field and print tools without owners.**

WORKSHEET: AI Governance Essentials (Part A)

Name: _____ Date: _____

1. Match the EU AI Act tier to an example.

Unacceptable ●

High risk ●

Limited risk ●

Minimal risk ●

● Spam filter

● Social scoring by governments

● Chatbot must disclose it's AI

● AI in hiring decisions

2. Order the NIST AI RMF functions as taught. Write 1–4.

Manage ☐

Govern ☐

Measure ☐

Map ☐

WORKSHEET: AI Governance Essentials (Part B)

Name: _____ Date: _____

1. Inventory counts. Use the word box.

WORD BOX: 20 ✿ 5 ✿ 54

37 tools - 5 high - 12 limited = ____ minimal

High-risk tools needing board review = ____

Share minimal ≈ ____%

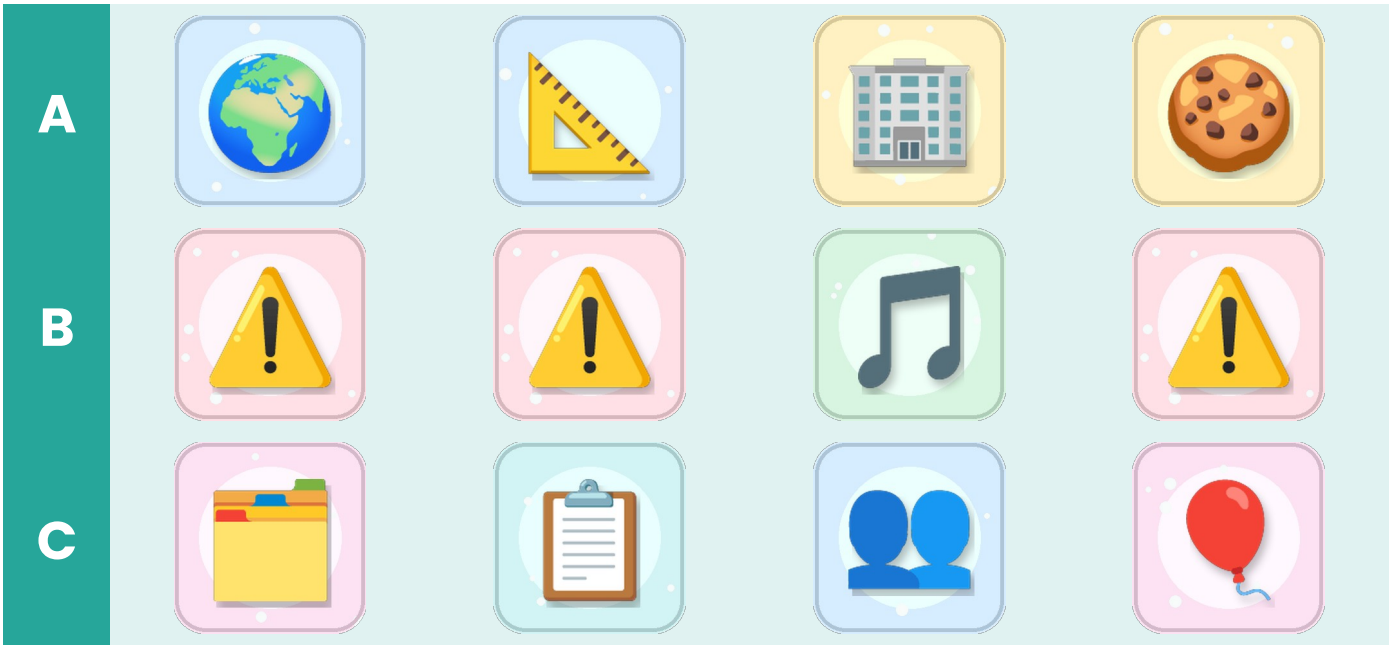
2. Law or voluntary framework? Tick ✓.

	Law	Framework
EU AI Act		
NIST AI RMF		
India's DPDP Act		
ISO/IEC 42001		

PUZZLE & DEFINE: AI Governance Essentials

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.



2. Define it! Write each term in your own words with an example.

Risk tier

Meaning: _____

Example: _____

Inventory

Meaning: _____

Example: _____

RMF

Meaning: _____

Example: _____

? QUIZ TIME: AI Governance Essentials

Circle the right answer!

Q1. The EU AI Act entered into force in...

a) 2024

b) 2018

c) 2030

Q2. Social scoring by governments is...

a) Minimal risk

b) Unacceptable

c) Limited

Q3. A list of all AI systems:

a) Palette

b) Playlist

c) Inventory

Q4. NIST AI RMF's first function:

a) Govern

b) Manage

c) Measure

Q5. ISO/IEC 42001 is an AI...

a) Game

b) Management standard

c) Phone

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Governance Essentials

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	The EU AI Act uses risk tiers.		
	Social scoring is minimal risk under the Act.		
	An AI inventory lists an organisation's AI systems.		
	NIST AI RMF includes Govern, Map, Measure, Manage.		
	The EU AI Act can never affect non-EU firms.		
	India's DPDP Act governs personal data.		

HOME CONNECT

List every AI tool you used this month. Assign each a risk tier and one rule for safe use.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI Governance Essentials

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why keep an AI inventory?**

COMPARE **Q2. Laws vs voluntary frameworks.**

APPLY **Q3. 37 tools; 5 high, 12 limited, rest minimal. Minimal count?**

EVALUATE **Q4. Should startups ignore governance until large?**

CREATE **Q5. Write a 5-point AI use policy for a college.**

📝 EXAM PREP: AI Governance Essentials

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name the EU AI Act risk tiers.

2. List NIST AI RMF functions.

3. What is ISO/IEC 42001?

4. Why keep an AI inventory?

PART B • Long answer (5 marks)

5. Design an AI governance programme for an Indian company, aligned with major laws and frameworks.

UNIT REVIEW: AI CARE

Name: _____ Date: _____

Circle the right answer!

Q1. People must answer for...

a) AI decisions	b) Weather	c) Music

Q2. Explains one decision:

a) Global	b) Local	c) Visual

Q3. EU AI Act in force from...

a) 2018	b) 2021	c) 2024

Q4. Social scoring by states is...

a) Unacceptable	b) Minimal	c) Limited

Q5. NIST AI RMF starts with...

a) Manage	b) Govern	c) Measure

Q6. Dutch government resigned in...

a) 2015	b) 2024	c) 2021

WORD SEARCH: AI CARE

Find these words. They go across → or down ↓.

XAI

MODEL CARD

REDRESS

RISK TIER

INVENTORY

RMF

I	I	W	A	O	S	R	M	C	R
N	E	T	H	I	C	S	O	H	D
V	M	M	E	H	R	A	D	W	G
E	A	R	M	F	E	P	E	P	O
N	U	L	E	B	D	P	L	O	V
T	D	D	X	D	R	E	C	L	E
O	I	B	A	R	E	A	A	I	R
R	T	P	I	C	S	L	R	C	N
Y	A	T	E	W	S	C	D	Y	F
W	R	I	S	K	T	I	E	R	I

I finished AI CARE! Teacher's sign: _____

AI GLOSSARY (1/2)

Autonomy
Freedom to act

CAC
Cost to get customer

Demo
Show it working

Distil
Teach small model

Eval
Measured quality

Evaluator
Checks an output

GATE
PG entrance exam

GRE
Graduate test abroad

Inventory
List of AI systems

Latency
Time to respond

LLMOps
Run LLMs in prod

Memory
Stored context

Moat
Lasting advantage

Model card
Model fact sheet

Offer letter
Job terms document

Orchestrator
Splits & assigns

Pilot
Small first trial

Pitch
Short investor talk

AI GLOSSARY (2/2)

Pivot
Change direction

PMF
Product-market fit

Quantise
Lower-precision weights

ReAct
Reason + act loop

Redress
Fix & compensate

Risk tier
Level of AI risk

RMF
Risk mgmt framework

Router
Sends to right path

Scale-up
Roll out widely

Scaling law
Size-vs-skill trend

Step cap
Max loop steps

Streaming
Show text as made

SUS
Usability score

Task eval
Score whole tasks

Term sheet
Investment terms

Tool schema
Tool's input rules

Wizard-of-Oz
Human fakes the AI

XAI
Explainable AI

? OLYMPIAD PRACTICE TEST • Part 1 of 4

Read each question. Circle the right answer. 20 questions in all!

Q1. Agent = LLM + tools + memory + a...

a) Loop

b) Logo

c) Song

Q2. 12 steps × 2,000 tokens × ₹0.0004 =

a) ₹96

b) ₹9.6

c) ₹0.96

Q3. 13B weights at 4-bit ≈

a) 26 GB

b) 13 GB

c) 6.5 GB

Q4. Cheaper routing sends easy queries to...

a) Small models

b) Huge models

c) Bin

Q5. 1,000 questions, 40% cached → model calls

a) 400

b) 600

c) 1,000

? OLYMPIAD PRACTICE TEST • Part 2 of 4

Q6. 0.95 × 0.95 × 0.95 × 0.95 ≈

a) 0.95

b) 3.8

c) 0.81

Q7. Morgan Stanley's assistant drew on...

a) Research docs

b) Films

c) Games

Q8. Small first trial in a firm:

a) Rollout

b) Pilot

c) Exit

Q9. Price ₹500, inference ₹200, other ₹50 → margin

a) 60%

b) 40%

c) 50%

Q10. Niramai uses...

a) Thermal imaging

b) Music

c) Satellites

? OLYMPIAD PRACTICE TEST • Part 3 of 4

Q11. GRE is mainly for...

a) MBA India

b) MS abroad

c) PSU jobs

Q12. 12 of 20 name GST as top pain =

a) 12%

b) 20%

c) 60%

Q13. Builder.ai's fall warns against...

a) AI-washing

b) Honesty

c) Growth

Q14. Mean of 1.5, 2.0, 2.5 min =

a) 1.5

b) 2.0

c) 6.0

Q15. The Humane AI Pin was...

a) A hit

b) A game

c) Discontinued

? OLYMPIAD PRACTICE TEST • Part 4 of 4

Q16. AutoGPT often...

a) Got stuck

b) Won prizes

c) Slept

Q17. 200 wrong of 5,000 =

a) 40%

b) 4%

c) 0.4%

Q18. SHAP and LIME help...

a) Draw

b) Play audio

c) Explain

Q19. AI hiring tools in the EU are...

a) High risk

b) Minimal

c) Banned

Q20. ISO/IEC 42001 is an AI...

a) Game

b) Mgmt standard

c) Phone

MY SCORE: I got ____ / 20. Colour a star for every 4 right!

CERTIFICATE

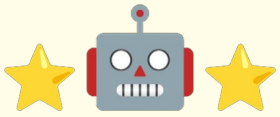
of AI Explorer

This is to say that

_____

of UG 4th Year finished the 6-month AI journey:

DISCOVER → CONNECT → SOLVE → RISE → CREATE → CARE



Teacher: _____ Date: _____

? ANSWER KEY (for teachers & parents)

1. The Age of AI Agents	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** LLM - Reasons & plans; Tools - Search/code/APIs; Memory - Past steps & facts; Supervisor - Approves risky actions **Part A 2:** Suggests a reply you send: Low; Emails clients by itself: High; Drafts code for review: Low; Pays invoices on its own: High **Part B 1:** 1-Read goal & memory, 2-Decide next step, 3-Call a tool, 4-Observe the result **Part B 2:** Circle: Maximum step limit, Approval for payments, Budget cap per run **Challenge hints:** Q1: Stop endless loops and cost blow-ups. • Q2: Assistant suggests; agent acts, so needs more controls. • Q3: 24,000 × 0.0004 = ₹9.6. • Q4: Only read-only or with approval, backups, separate env. • Q5: Allow-list, draft-only, approval, rate limit. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: MAX; prints 0 search ok / 1 compare ok / 2 book NEEDS APPROVAL / 3 search ok; track last action; use datetime.now(). **Exam prep key points:** 1: LLM + tools + memory in a plan-act-observe loop. • 2: Interleaving reasoning with tool actions. • 3: Loops, hallucinated tools, unsafe actions, cost blow-ups. • 4: Higher autonomy raises risk of harmful actions. • 5: Loop, memory, tools; autonomy tiers; limits, approvals, logging, sandboxing.
2. Intelligence at Scale	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Quantisation - Fewer bits; Distillation - Small learns from big; Caching - Reuse answers; Pruning - Remove weights **Part A 2:** 14, 7, 3.5 **Part B 1:** 30, 3, 90 **Part B 2:** Weeks on thousands of GPUs: Training; Every user question: Inference; Learning from trillions of tokens: Training; Answering on a phone: Inference **Challenge hints:** Q1: More training data per parameter; better balance. • Q2: Training: one-time huge; inference: ongoing per use. • Q3: ≈ 6.5 GB. • Q4: No: cost, latency, energy; match model to task. • Q5: Classifier → small model; hard cases → large. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: 8; 14.0, 7.0, 3.5 GB; 13 × 4 / 8 = 6.5 GB → fits; × 1.2. **Exam prep key points:** 1: Predictable gains from more data, compute, parameters. • 2: One-time vs ongoing per use. • 3: Storing weights in fewer bits. • 4: Balance data and size; smaller can win. • 5: Scaling laws, compute-optimal, quantise/distil/route/cache, energy, carbon.

3. Build Smart Applications

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Frontend - UI; Backend - Logic & auth; Model API - Generates; Vector DB - Finds chunks
Part A 2: 300, 1500, 2000
Part B 1: Shows source links: Good; Always sounds 100% sure: Bad; Lets users edit answers: Good; Hides errors from users: Bad
Part B 2: 40, 400, 600
Challenge hints: Q1: Users see progress immediately; feels faster. • Q2: Offline: controlled; online: real users, real impact. • Q3: 2,000 ms (2 s). • Q4: No: disclose AI use and its limits. • Q5: Sources, confidence note, 'check this' prompt.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: values; 1850 slowest generate; perceived wait drops to ~0.85 s total to first words; expected = 0.7 × 1850 + 0.3 × 450.
Exam prep key points: 1: UI, backend, model, data stores, evaluation. • 2: Faster perceived latency. • 3: Sources, confidence, edit/regenerate, disclosure. • 4: Offline test sets, A/B tests, feedback. • 5: Stack, latency, caching, honest UX, testing, security, fallback.

4. Design AI Workflows

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Chaining - Outline/draft/edit; Routing - Ticket types; Parallel - Many files; Evaluator - Grade & improve
Part A 2: 0.81, 0.73, 0.9
Part B 1: 1-Plan sub-tasks, 2-Run workers, 3-Check each result, 4-Combine results
Part B 2: Circle: Log every step's output, Human review of final report, Clear rubric for judges
Challenge hints: Q1: Accuracies multiply; early errors flow onward. • Q2: Routing picks one path; parallel runs many at once. • Q3: $0.95^4 \approx 0.81$. • Q4: No: calibrate with human ratings. • Q5: Route by subject → grade → evaluate → teacher check.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: *=; ≈ 0.76; with 0.98 for step 2 → ≈ 0.83; the weakest step (0.9) gives the biggest gain.
Exam prep key points: 1: Chain, route, parallel, orchestrator-workers, evaluator loop. • 2: Step accuracies multiply. • 3: Judge critiques, writer refines until threshold. • 4: Debugging, audit, quality checks. • 5: Routing, parallel, evaluation, human review, metrics, logging.

5. AI for Enterprise	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Strategy - Roadmap; Data - Data lake; LLMOps - Monitoring & rollback; Governance - Risk board **Part A 2:** 48, 500, 1100 **Part B 1:** 50 test users: Pilot; Single sign-on & audit logs: Production; Manual feedback forms: Pilot; 24×7 monitoring & rollback: Production **Part B 2:** 1-Pick use case & KPIs, 2-Run a pilot, 3-Harden: security, LLMOps, 4-Scale to all staff **Challenge hints:** Q1: Data, security, integration, change gaps. • Q2: Pilot: few users, loose; production: secure, monitored. • Q3: 2 × 1,000 × 1,500 × 48 = ₹14.4 crore. • Q4: No: central standards, security reviews, shared platforms. • Q5: Approved tools, data rules, review, disclosure, incidents. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: weeks; benefit 7,200,000, ROI 800%; scaled: 144,000,000, ROI 3500%; multiply users by 0.7. **Exam prep key points:** 1: Strategy, data, LLMOps, governance. • 2: Data, security, integration, adoption gaps. • 3: Versioning, evaluation, monitoring, rollback of LLM systems. • 4: Time saved, error rate, CSAT, adoption. • 5: Prioritise, pilot, harden, govern, train, measure, scale.
6. AI for Start-ups	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** CAC - Cost to win; LTV - Customer value; PMF - Market demand; Burn - Monthly cash spent **Part A 2:** 3, 60, 12 **Part B 1:** 6, 10, 20 **Part B 2:** Unique data from 10,000 clinics: Real; Uses a popular public model: Weak; Exclusive state partnership: Real; Nice logo: Weak **Challenge hints:** Q1: Models are widely available; need data, distribution, trust. • Q2: SaaS: fixed monthly; usage: pay per use, tracks cost. • Q3: 50%. • Q4: Only short-term with a clear path to profit. • Q5: For [user], we [solve pain] by [how], unlike [alt]. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: price; 50 and 3.0; (500 - 100 - 50)/500 = 70%; 3000 / 250 = 12 months. **Exam prep key points:** 1: Strong, repeat demand from a market. • 2: Cost to win a customer; value over lifetime. • 3: Data loops, distribution, domain trust. • 4: Exaggerating AI use or capability. • 5: Pain, MVP, PMF, moats, pricing, margins, funding, honesty.

7. From Campus to AI Career

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: GATE - M.Tech/PSU; GRE - MS abroad; CAT - MBA India; IELTS/TOEFL - English abroad
Part A 2: Named mentor and training: Good; Fee to 'confirm' the job: Risky; Clear written role & pay: Good; 'Accept in 1 hour' pressure: Risky
Part B 1: 78, 69, 84
Part B 2: Job - CV & apply; Higher studies - GATE/GRE; Research - Email professor; Start-up - Interview customers
Challenge hints: Q1: Early learning compounds into long-term value. • Q2: Cost, duration, exams, exposure, return path. • Q3: 3.6 + 1.8 + 2.4 = 7.8 → 78. • Q4: No: genuine employers don't charge fees. • Q5: Path, exams, applications, skills, backups.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: values; A 78, B 69; new: A 0.2×9+0.5×6+0.3×8 = 7.2 → 72, B 0.2×6+0.5×9+0.3×6 = 7.5 → 75; C = 84.
Exam prep key points: 1: Job, higher studies, research, start-up. • 2: Data Science & AI paper, added 2024. • 3: Learning, mentorship, role, pay, fit. • 4: Fees, pressure deadlines, unverifiable firms. • 5: Paths, exams, timeline, portfolio, offers, backups, scam checks.

8. Idea to AI Venture

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: 1-Interview customers, 2-Build an MVP, 3-Test MVP metrics, 4-Pitch to investors
Part A 2: Tell me about the last time this happened.: Good; Would you love my app?: Bad; What do you use today, and its cost?: Good; Will you buy it if I build it?: Bad
Part B 1: 60, 25, 15
Part B 2: 1-Problem, 2-Solution, 3-Traction, 4-The ask
Challenge hints: Q1: People mispredict; past actions show real pain. • Q2: Pivot when evidence contradicts; persevere when metrics improve. • Q3: 60%. • Q4: No: misleading; disclose honestly. • Q5: Problem, solution, market, traction, model, competition, team, ask.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 1; prints gst 50; loop over c.items(); pandas.read_csv('notes.csv')['pain'].
Exam prep key points: 1: Interviewing users about real pains and behaviour. • 2: When evidence contradicts key assumptions. • 3: Problem, solution, market, traction, model, competition, team, ask. • 4: Overstating AI use or capability. • 5: Discovery, MVP, metrics, pivot, pitch, registration, IP, honest claims.

9. AI Prototype Lab	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Sketch - Layout ideas; Wizard - Fake AI to test value; Clickable - Navigation; Demo - Real model quality **Part A 2:** 80, 4, 2 **Part B 1:** Stay silent while users try: Good; Explain every button first: Bad; Time each task: Good; Hide failed tasks in report: Bad **Part B 2:** 40, 85, 45 **Challenge hints:** Q1: Checks value and flow cheaply before costly builds. • Q2: Mock: flow and layout; demo: real model quality. • Q3: 2.0 minutes. • Q4: No: show typical results and failures. • Q5: Find X, complete Y, fix Z, timed. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: success; mean 2.0 min, 80%; times of successes: 1.5, 2.0, 3.0, 1.0 → 1.875. **Exam prep key points:** 1: A human secretly simulates the AI. • 2: Reveals most major usability issues cheaply. • 3: Task success, time, errors, satisfaction. • 4: They mislead about real performance. • 5: Fidelity ladder, tools, user tests, metrics, iteration, demo ethics.
10. Agent Builder Studio	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** get_calendar(date): Good; do_anything(text): Bad; draft_email(to, body): Good; delete_all_files(): Bad **Part A 2:** 70, 30, 3 **Part B 1:** 85, 4, 15 **Part B 2:** Circle: Step and budget caps, End-to-end task tests, Sandboxed code runs **Challenge hints:** Q1: Easier to choose correctly; safer; easier to test. • Q2: Single: one answer; task: whole goal, cost, safety. • Q3: 85%, ₹4. • Q4: Only sandboxed with allow-lists and approval. • Q5: Name, purpose, typed params, required fields. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: len(results); 75.0 and 5.0; failures often loop longer; 20 ÷ 3 ≈ 6.67. **Exam prep key points:** 1: Single purpose, clear schema, least privilege. • 2: Prevent loops and runaway cost. • 3: Scoring whole tasks for success, cost, safety. • 4: Debug, audit, improve. • 5: Tools, memory, limits, approvals, sandbox, task suite, tracing.
11. Accountable AI	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** SHAP/LIME - Explain features; Model card - Document; Audit - Independent review; Impact assessment - Pre-launch risks **Part A 2:** Circle: Clear appeal route, Published model card, Regular audits **Part B 1:** 1-Impact assessment, 2-Write model card, 3-Independent audit, 4-Handle appeals & redress **Part B 2:** 4, 95, 25 **Challenge hints:** Q1: Allows correction, appeal and trust. • Q2: Global: overall model; local: one decision. • Q3: 4%. • Q4: No: deployers stay accountable. • Q5: Purpose, data, metrics, limits, contact. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: 'marks'; prints ['fix income document']; then ['marks below 75']. **Exam prep key points:** 1: Clear responsibility for AI outcomes. • 2: Whole model vs one decision. • 3: Document of model purpose, data, metrics, limits. • 4: Correcting and compensating harm. • 5: Impact assessment, explanations, audits, documentation, appeals, redress.

12. AI Governance Essentials

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Unacceptable - Social scoring; High - Hiring AI; Limited - Chatbot disclosure; Minimal - Spam filter
Part A 2: 1-Govern, 2-Map, 3-Measure, 4-Manage
Part B 1: 20, 5, 54
Part B 2: EU AI Act: Law; NIST AI RMF: Framework; India's DPDP Act: Law; ISO/IEC 42001: Framework
Challenge hints: Q1: You can't govern systems you don't know exist. • Q2: Laws: mandatory, enforced; frameworks: guidance, best practice. • Q3: 20. • Q4: No: light, early governance avoids costly fixes. • Q5: Approved tools, data rules, disclosure, review, incidents.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 'high'; prints 2 ['hiring screen', 'credit model']; use Counter(tier for _, tier in tools).
Exam prep key points: 1: Unacceptable, high, limited, minimal. • 2: Govern, Map, Measure, Manage. • 3: AI management system standard. • 4: Know and govern all AI systems. • 5: Policy, inventory, risk tiers, controls, DPDP, EU AI Act, NIST/ISO, monitoring.

Unit reviews & practice test

AI DISCOVER: 1-a 2-b 3-c 4-a 5-b 6-c • AI CONNECT: 1-a 2-b 3-c 4-a 5-b 6-c • AI SOLVE: 1-a 2-b 3-c 4-a 5-b 6-c • AI RISE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CREATE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CARE: 1-a 2-b 3-c 4-a 5-b 6-c

Olympiad Practice Test: 1-a 2-b 3-c 4-a 5-b 6-c 7-a 8-b 9-c 10-a 11-b 12-c 13-a 14-b 15-c 16-a 17-b 18-c 19-a 20-b

WORD SEARCH ANSWERS

E L D M E M O R Y W
S C A L I N G L A W
A G E N T U L O O P
C O M P U T E N C U
T H U C I R E A C T
M K U T T O K E N S
D Q U A N T I S E P
I A U T O N O M Y D
P L A N N E R B K A
W A C O D I S T I L

AI DISCOVER

W S I R K V E K S T
S T S P W E V U O K
G R R R S C A M L C
W E C O L T L J A A
A A G M R O U U T C
D M B P O R A D E H
U I E T U B T G N E
I N V U T H O E C D
S G A E E I R H Y G
B O L B R B M M F H

AI CONNECT

L A P I L O T I W R
P F L T U U D M G E
M R U N W A Y A U V
F I K D C R M R R E
C M C O E A O G K N
O O D B W W A I B U
N U L C A C T N T E
I S L L M O P S H B
K S C A L E U P A C
D W T F U N D I N G

AI SOLVE

C B R E S U M E P I
S T E R M S H E E T
T P I T C H P R O L
E G C O G P I I F F
G R E M O U V L F T
G L M E N T O R E D
B H I U G A T E R W
H V I S A D L F K P
A B F O U N D E R M
C K L A L K L E O W

AI RISE

S T U M M I M O W B
M O S F P M L S I U
S O A T T E S T Z D
S L B A R S A E A G
D S I S A K N P R E
E C L K C E D C D T
M H I E E T B A O F
O E T V S C O P F G
M M Y A U H X R O K
C A O L S F W U Z S

AI CREATE

I I W A O S R M C R
N E T H I C S O H D
V M M E H R A D W G
E A R M F E P E P O
N U L E B D P L O V
T D D X D R E C L E
O I B A R E A A I R
R T P I C S L R C N
Y A T E W S C D Y F
W R I S K T I E R I

AI CARE

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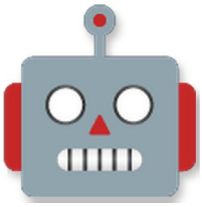

About this book

A university-level AI book for UG Final Year students: concepts, code labs, stories & career prep.

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- 🐍 Python Code Labs & lab activities
- 📚 AI Stories & real-world case studies
- 🎓 Careers, higher studies & ventures
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